Gaussian states, covariance matrices and the formal Fock space

Gaussian formalism expresses properties of bosonic quantum states in terms of phase-space data. For Gaussian states, the first moments and covariance matrix determine the state: the former give its mean position and momentum, and the latter describe its fluctuations. These notes develop the matrix algebra of covariances and connect it with explicit normalized Gaussian wave functions, laying foundations for the study of Gaussian matrix product states. For general background, see Weedbrook et al. [1, Sec. II]. Sections 6 and 7 construct the formal Fock space of all coefficient families over the number basis, with its ladder operators and the Bargmann picture, and place the bosonic Fock space inside it, where coefficient-finite maps define closed operators and closed forms. Section 8 second-quantizes complex matrices on the formal Fock space, and Section 9 factors every real symplectic matrix into orthogonal symplectic matrices and squeezing. Sections 10 and 11 construct the displacement and squeezing operators and the Gaussian unitaries on the bosonic Fock space, and Section 12 characterizes the pure Gaussian states by their characteristic functions.

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About these notes

These notes are the mathematical account of the Lean library BosonicLean. We assume finite-dimensional linear algebra, complex inner products, Lebesgue integration and elementary differential calculus; Section 6 needs only algebra and formal power series, and Section 7 assumes the elements of closed operators and forms on Hilbert space, as in Kato [8]; Sections 8 and 9 use the spectral theorem for normal and symmetric matrices, the positive square root, and, in Proposition 8.12, the functional calculus of direct sums of self-adjoint operators; Sections 10–12 use bounded and unitary operators on Hilbert space and the square root of a positive semidefinite matrix. The reading guide below relates our conventions to those of the literature, the roadmap after it shows which sections use which, and an index of terms precedes the references.

A statement marked Lean ✓ has a checked proof in BosonicLean whose correspondence with the statement here has been reviewed; the declaration index at the end names the checked declarations. Definitions, Remarks and paragraphs of motivation carry no marks.

Reading guide

Where our objects or conventions differ from the usual physics presentation, the table names the object used here, the one used in the literature, and where the two are related.

Here	**In the literature**	**Relation**
Covariance γ, twice the symmetrized covariance; vacuum I_2	Vacuum I_2 as here [1]; or $\sigma = \gamma/2$, vacuum $\frac{1}{2}I_2$; or $\hbar$-dependent units	Definition 2.3; units in Section 2
Coordinates $(q_1, \dots, q_n, p_1, \dots, p_n)$, form Ω_n	Mode by mode, $(q_1, p_1, \dots, q_n, p_n)$	A permutation of coordinates; admissibility is unchanged (proof of Theorem 1.5)
Admissibility $\gamma - i\Omega_n \succeq 0$	Uncertainty relation for the covariance, e.g. $V + i\Omega \succeq 0$ [12, Sec. IX]	Equivalent to $\gamma + i\Omega_n \succeq 0$ (Definition 1.1)
Formal Fock space of all coefficient families; e^T summed coefficientwise; Bargmann map into formal power series	$\Gamma_s(\mathfrak{h})$; operator exponential; Bargmann's space of entire functions	$\Gamma_s(\mathfrak{h})$ is the square-summable part; there the coefficientwise objects are the usual ones wherever these are defined (Section 6)
$\Omega_N = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$	Also $J = \left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right) = -\Omega_N$	The same symplectic group (Lemma 9.1)
$\Sigma(r) = \operatorname{diag}(e^{-r_1}, \dots, e^{r_N})$, positions first	Momenta first	The same squeezers have the matrix $\Sigma(-r)$ (after Definition 9.5)
$O_{\mathcal{U}} = \left(\begin{smallmatrix} X & -Y \\ Y & X \end{smallmatrix}\right)$ for $\mathcal{U} = X + iY$, positions first	Also $\left(\begin{smallmatrix} X & Y \\ -Y & X \end{smallmatrix}\right)$ for $X + iY$	The latter is $O_{\bar{\mathcal{U}}}$; ours gives $a \mapsto \mathcal{U}a$, the convention of [4] (after Theorem 9.4)
Weyl operator $W(u) = D(\alpha(u))$, $\alpha(u) = (q + ip)/\sqrt{2}$	$e^{i\xi^T \Omega \hat{R}}$ or $D(\xi)$, with varying signs and scalings	Formally $W(u) = e^{-i\sigma(u,\hat{R})}$, and $W(u)W(v) = e^{-i\sigma(u,v)/2}W(u+v)$ (Proposition 11.4)
Gaussian unitary: $U^* W(u) U = e^{-i\sigma(u,d)} W(M^{-1}u)$	$U^\dagger \hat{R} U = M\hat{R} + d$; or $e^{i\chi}e^{-iH}$, H quadratic	Ours implies $U^\dagger \hat{R} U = M\hat{R} + d$ (Corollary 11.7(c)) and includes products that are not single exponentials

Here	In the literature	Relation
Generator: $U_t = e^{-itA}$, A self-adjoint	Also $U_t = e^{itA}$	A changes sign; with ours, e^{-itH} is the Schrödinger evolution of the Hamiltonian H (Definition 13.1)

Roadmap

Section 1 characterizes covariance matrices by the matrix inequality (1.1) and proves that symplectic congruence, the selection of a subsystem and independent sums preserve it. Section 2 constructs normalized Gaussian wave functions and computes their covariances, which realize the equality case of the one-mode criterion of Proposition 1.6. Section 3 defines the contraction of covariance matrices, a Schur complement motivated by EPR measurements on virtual modes, proves that it preserves admissibility, and assembles from it the covariance of a periodic Gaussian matrix product state. Sections 4 and 5 treat two cases of that construction: chains with the same local covariance at every site, whose covariance splits into one 2×2 block per frequency, and products.

Section 6 uses none of Sections 1–5. From algebra and formal power series alone it builds the formal Fock space of all coefficient families over the number basis, its ladder operators, the exponentials of operators that raise the particle number, and the Bargmann picture; it is the foundation of the Gaussian-augmented bosonic matrix product states of [4]. Section 7 places the bosonic Fock space inside the formal one, realizes coefficient-finite maps as closed operators and their sums of squared norms as closed forms, and proves the invariance of the photon-added class. Section 8 defines the second quantization $\Gamma(g)$ of every complex matrix on the formal Fock space, shows that the unitary matrices give a strongly continuous unitary representation on the bosonic Fock space, and identifies $\Gamma(e^h)$ with the coefficientwise exponential of $d\Gamma(h)$. Section 9 uses only the symplectic matrices of Section 1: it identifies the orthogonal symplectic matrices with the unitary matrices and proves the Bloch–Messiah and polar decompositions. Section 10 constructs the displacement and squeezing operators on the bosonic Fock space from their normal-ordered forms, through the exponential vectors, and proves the irreducibility of the displacements. Section 11 defines Gaussian unitaries by their action on the Weyl operators and factors each into passive, displacement and squeezing unitaries, using the Bloch–Messiah decomposition. Section 12 proves that the pure Gaussian states, the vacuum vectors of the Gaussian unitaries, are exactly the vectors with a Gaussian characteristic function. Section 13 proves that e^{-itH} is a Gaussian unitary for every quadratic Hamiltonian $H = \frac{1}{2}\hat{R}^T K \hat{R} + f^T \hat{R}$, with symplectic matrix $e^{t\Omega K}$; it first develops the one-parameter unitary groups and their generators that the statement needs. Figure 1 shows which sections use which.

1 Covariance matrices

For $n \geq 0$, we use coordinates $(q_1, \ldots, q_n, p_1, \ldots, p_n)$ on $\mathbb{R}^{2n}$ and define

$$\Omega_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

For a complex matrix H, the notation $H \succeq 0$ means that $H = H^*$ and $z^* H z \geq 0$ for every complex column vector z. Here H^* is the conjugate transpose. We regard real matrices as complex matrices in this inequality.

Definition 1.1 (Admissible covariance)**.** A real $2n \times 2n$ matrix γ is called *admissible* if

$$\gamma^T = \gamma, \qquad \gamma - i\Omega_n \succeq 0. \tag{1.1}$$

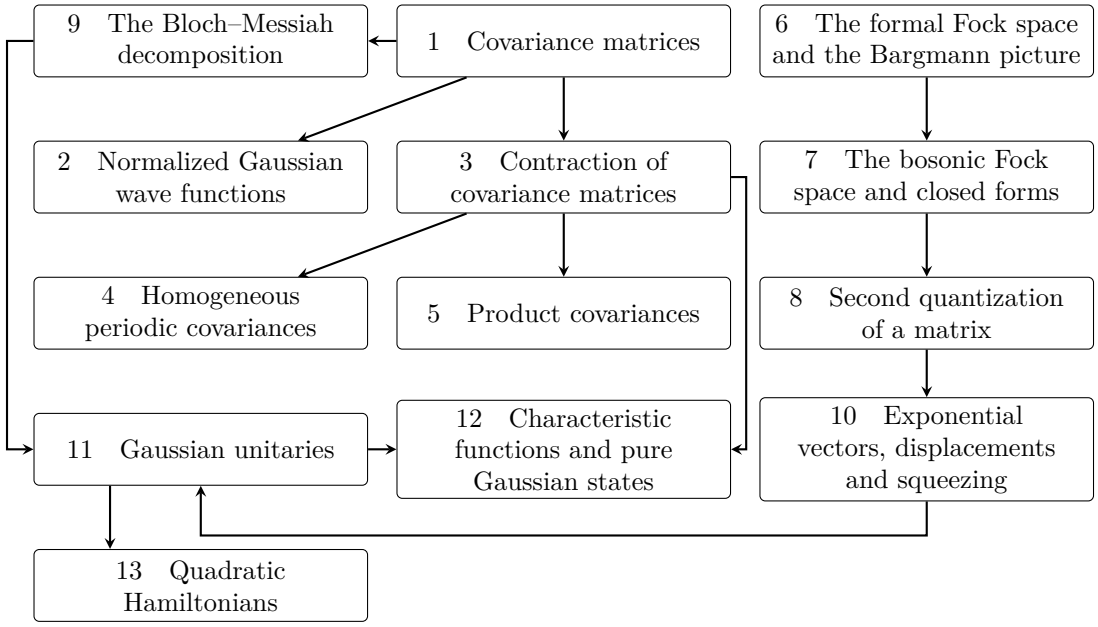


Figure 1: Dependence of the sections. An arrow means that the later section uses the earlier one; arrows implied by a chain of arrows are omitted. Section 2 enters Sections 3 and 5 only in remarks, and Section 6 uses no other section; Section 7 uses only Section 6, and Section 9 refers to Section 8 only in remarks; Section 12 uses Section 3 only for the positivity of admissible matrices.

These coordinates are dimensionless. For a real $2n\times 2n$ matrix γ we call $\gamma - i\Omega_n$ its *uncertainty matrix*. For one mode the condition concerns the Hermitian matrix

$$\begin{pmatrix} a & c-i \\ c+i & b \end{pmatrix}, \qquad \gamma = \begin{pmatrix} a & c \\ c & b \end{pmatrix}.$$

The imaginary part records the canonical pairing between position and momentum. The empty matrix is admissible when $n = 0$.

1.1 Changes of coordinates

A real matrix S is *symplectic* in these coordinates when $S\Omega_n S^T = \Omega_n$. Transforming a covariance in both indices gives $\gamma \mapsto S\gamma S^T$. This is a *congruence transformation*. Similarity has the form $A \mapsto SAS^{-1}$; the two formulas agree for orthogonal S, but symplecticity alone does not give $S^T = S^{-1}$.

Theorem 1.2. Lean ✓ *Let $n \geq 0$ and let S, γ be real $2n \times 2n$ matrices. If γ is admissible and $S\Omega_n S^T = \Omega_n$, then $S\gamma S^T$ is admissible.*

Proof. Symmetry follows from $(S\gamma S^T)^T = S\gamma^T S^T = S\gamma S^T$. Put $H = \gamma - i\Omega_n$. Since S is real,

$$S\gamma S^T - i\Omega_n = S(\gamma - i\Omega_n)S^T = SHS^*.$$

This matrix is Hermitian, and for every $z \in \mathbb{C}^{2n}$,

$$z^* SHS^* z = (S^*z)^* H (S^*z) \geq 0.$$

The same identities hold for empty matrices. □

1.2 Subsystems and independent sums

To select a subsystem, a position coordinate and its matching momentum coordinate must be selected together. A *one-to-one map*, also called an *injective map* or an *injection*, is a map e such that $e(i) = e(j)$ implies $i = j$. Thus a one-to-one map $e : \{1, \dots, m\} \to \{1, \dots, n\}$ selects distinct modes without repetitions, in the order

$$(q_{e(1)}, \dots, q_{e(m)}, p_{e(1)}, \dots, p_{e(m)}).$$

Write $E(q_j) = q_{e(j)}$ and $E(p_j) = p_{e(j)}$. The selected covariance has entries $\gamma'_{ab} = \gamma_{E(a),E(b)}$.

Theorem 1.3. Lean ✓ *For all $m, n \geq 0$, every such selection from an admissible n-mode covariance is an admissible m-mode covariance. The empty selection is included.*

Proof. Symmetry is inherited by restriction. The qq and pp blocks of Ω_n vanish, and its qp and pq entries are δ_{ij} and $-\delta_{ij}$. Injectivity gives $\delta_{e(i),e(j)} = \delta_{ij}$, so restriction along E sends Ω_n to Ω_m.

Thus $H' = \gamma' - i\Omega_m$ is the selected principal submatrix of $H = \gamma - i\Omega_n$. It is Hermitian. Extending a vector $z \in \mathbb{C}^{2m}$ by zero on the unselected coordinates gives $\widetilde{z} \in \mathbb{C}^{2n}$ and

$$z^* H' z = \widetilde{z}^* H \widetilde{z} \geq 0.$$

□

Conversely, we can combine independent covariance blocks without cross terms.

Definition 1.4 (Independent sum)**.** For arbitrary real matrices A and B of sizes $2n \times 2n$ and $2m \times 2m$, write their position/momentum blocks as $A_{qq}, A_{qp}, A_{pq}, A_{pp}$ and similarly for B. Define

$$A \boxplus B = \begin{pmatrix} A_{qq} & 0 & A_{qp} & 0 \\ 0 & B_{qq} & 0 & B_{qp} \\ A_{pq} & 0 & A_{pp} & 0 \\ 0 & B_{pq} & 0 & B_{pp} \end{pmatrix} \tag{1.2}$$

in the order (q_A, q_B, p_A, p_B). This is $\mathrm{diag}(A, B)$ reordered from (q_A, p_A, q_B, p_B); the definition makes no symmetry assumption.

Theorem 1.5. Lean ✓ *For all $n, m \geq 0$ and all real matrices A, B of the stated sizes, $A \boxplus B$ is admissible if and only if both A and B are admissible.*

Proof. The same coordinate permutation sends $\mathrm{diag}(\Omega_n, \Omega_m)$ to Ω_{n+m}. Hence the uncertainty matrix of $A \boxplus B$ is the corresponding permutation of $\mathrm{diag}(A - i\Omega_n, B - i\Omega_m)$. When both blocks are admissible, symmetry is preserved and the quadratic form is the sum of two nonnegative quadratic forms. Hermiticity is inherited from the blocks. Conversely, restriction to either collection of matching position/momentum coordinates recovers that block and its uncertainty matrix. Theorem 1.3 applies. Empty blocks cause no exception. □

1.3 One-mode diagonal matrices

The diagonal case gives explicit examples and fixes the scale of the uncertainty bound. Put $D(a, b) = \mathrm{diag}(a, b)$ in the order (q, p).

Proposition 1.6. Lean ✓ *For all $a, b \in \mathbb{R}$, the matrix $D(a, b)$ is admissible if and only if $a > 0$ and $ab \geq 1$.*

Proof. The uncertainty matrix is $H = \left(\begin{smallmatrix} a & -i \\ i & b \end{smallmatrix}\right)$. If $H \succeq 0$, the coordinate vectors give $a, b \geq 0$. If $a = 0$, the vector $(b + 1, -i)^T$ gives the negative value $-b - 2$; hence $a > 0$. The vector $(i, a)^T$ then gives $0 \leq a(ab - 1)$, and therefore $ab \geq 1$.

For the converse, suppose $a > 0$ and $ab \geq 1$. Write $z = x + iy$, $w = u + iv$. The quadratic form is real and satisfies

$$Q = (\overline{z}, \overline{w})H(z,w)^T = a(x^2+y^2) + b(u^2+v^2) + 2(xv - yu),$$
$$aQ = (ax+v)^2 + (ay-u)^2 + (ab-1)(u^2+v^2) \geq 0.$$

Thus $Q \geq 0$; Hermiticity of H is immediate. □

In particular, $I_2 = D(1,1)$ is admissible. More generally, $D(t,t^{-1})$ is admissible for every $t > 0$. Its two variances in our covariance convention are $t/2$ and $1/(2t)$; their product is $1/4$. By contrast, cI_2 with $0 \leq c < 1$ is a positive semidefinite real matrix but is not admissible, since $c^2 < 1$.

Notes and references

The admissibility condition (1.1) is the uncertainty principle for the covariance matrix in the form of Robertson and Schrödinger, and Theorem 1.2 is its invariance under the real symplectic group. Arvind, Dutta, Mukunda and Simon [12, Secs. VIII–IX] state both for the ordinary covariance matrix, which is half of ours; Simon, Mukunda and Dutta [18] study the multimode condition and its normal forms.

We now construct wave functions whose covariances give the one-mode equality case.

2 Normalized Gaussian wave functions

Definition 2.1 (Schwartz space)**.** Let $\mathcal{D} = \mathcal{S}(\mathbb{R};\mathbb{C})$ be the Schwartz space: a smooth function f belongs to $\mathcal{D}$ when

$$\sup_{x\in\mathbb{R}} |x|^r |f^{(k)}(x)| < \infty \qquad (r,k \in \mathbb{N}_0).$$

We use Lebesgue measure and put $\mathcal{H} = L^2(\mathbb{R};\mathbb{C})$. Elements of $\mathcal{H}$ are equivalence classes of functions: two functions represent the same element if they agree almost everywhere. Schwartz decay makes every $f \in \mathcal{D}$ square-integrable. The usual inclusion

$$J : \mathcal{D} \longrightarrow \mathcal{H}, \qquad Jf = [f],$$

sends a Schwartz function to its class. It is injective: continuous functions that agree almost everywhere agree everywhere. We henceforth identify f with Jf when taking an L^2 norm or inner product, writing

$$\|f\|_2 = \|Jf\|_{\mathcal{H}}, \qquad \langle f, g\rangle = \int_{\mathbb{R}} \overline{f(x)} g(x)\, dx.$$

The inner product is conjugate-linear in its first argument.

Definition 2.2 (Normalized representative)**.** A normalized state representative is a Schwartz function $f \in \mathcal{D}$ with $\|f\|_2 = 1$.

For $\hbar > 0$, position and momentum act on the common invariant domain $\mathcal{D}$ by

$$(Xf)(x) = xf(x), \qquad (P_\hbar f)(x) = -i\hbar f'(x).$$

Multiplication by x and differentiation preserve Schwartz space. Their images therefore have well-defined L^2 norms and inner products.

Definition 2.3 (Means and covariance)**.** For an operator $A : \mathcal{D} \to \mathcal{D}$ and a normalized f, define

$$\mu_f(A) = \operatorname{Re}\langle f, Af\rangle, \qquad u_f(A) = Af - \mu_f(A) f.$$

For the ordered pair $(R_1, R_2) = (X, P_\hbar)$, the covariance is

$$(\gamma_f)_{jk} = 2\operatorname{Re}\langle u_f(R_j), u_f(R_k)\rangle. \tag{2.1}$$

In particular, $(\gamma_f)_{jj} = 2\|u_f(R_j)\|_2^2$ is twice the variance.

2.1 Existence and normalization

For a width $s > 0$, we consider the positive real function

$$\psi_s(x) = \frac{1}{\sqrt{\sqrt{\pi}\, s}} \exp\left(-\frac{x^2}{2s^2}\right). \tag{2.2}$$

Theorem 2.4. Lean ✓ *For every $s > 0$, the function ψ_s belongs to $\mathcal{D}$ and satisfies $\|\psi_s\|_2 = 1$. It determines a unique normalized state representative with the pointwise values in* (2.2).

Proof. Put $\phi(x) = e^{-x^2/2}$. We define the *Hermite polynomials*, in the probabilists' normalization, by

$$\mathrm{He}_0(x) = 1, \qquad \mathrm{He}_{k+1}(x) = x\,\mathrm{He}_k(x) - \mathrm{He}_k'(x) \quad (k \geq 0).$$

The *Hermite derivative identity* is

$$\phi^{(k)}(x) = (-1)^k \mathrm{He}_k(x)\phi(x).$$

It follows by induction from $\phi' = -x\phi$ and the product rule. For every $r \geq 0$, $|x|^r\phi(x) \to 0$ as $|x| \to \infty$. Continuity bounds this function globally. Expanding He_k into monomials then bounds $|x|^r|\phi^{(k)}(x)|$ for each r, k. Dilation by $x \mapsto x/s$, inclusion of real values into $\mathbb{C}$, and multiplication by a constant preserve Schwartz space, so $\psi_s \in \mathcal{D}$.

The Gaussian integral gives

$$\|\psi_s\|_2^2 = \frac{1}{\sqrt{\pi}\, s} \int_{\mathbb{R}} e^{-x^2/s^2}\, dx = \frac{\sqrt{\pi}\, s}{\sqrt{\pi}\, s} = 1.$$

A function with the prescribed pointwise values is equal to ψ_s. □

The pointwise formula (2.2) fixes the overall phase of ψ_s.

2.2 Action and moments

Theorem 2.5. Lean ✓ *For every $s, \hbar > 0$,*

$$P_\hbar\psi_s = \frac{i\hbar}{s^2} X\psi_s, \qquad \mu_{\psi_s}(X) = \mu_{\psi_s}(P_\hbar) = 0, \tag{2.3}$$

$$\|X\psi_s\|_2^2 = \frac{s^2}{2}, \qquad \|P_\hbar\psi_s\|_2^2 = \frac{\hbar^2}{2s^2}, \tag{2.4}$$

$$\langle X\psi_s, P_\hbar\psi_s\rangle = \frac{i\hbar}{2}. \tag{2.5}$$

The first identity holds as an equality of Schwartz functions.

Proof. Differentiating (2.2) yields $\psi_s' = -(x/s^2)\psi_s$ and hence the momentum action. The integrand in $\langle\psi_s, X\psi_s\rangle$ is odd and integrable, so this complex inner product vanishes.

Set $A = \|X\psi_s\|_2^2$. With $\langle\cdot,\cdot\rangle_{\mathbb{R}}$ denoting the real part of the complex inner product, integration by parts gives

$$1 - \frac{A}{s^2} = \langle\psi_s, (X\psi_s)'\rangle_{\mathbb{R}} = -\langle\psi_s', X\psi_s\rangle_{\mathbb{R}} = \frac{A}{s^2}. \tag{2.6}$$

The first equality uses $(X\psi_s)' = \psi_s + x\psi_s'$ and normalization. Schwartz decay removes the boundary term, and the derivative formula gives the last equality. Thus $A = s^2/2$.

Complex linearity in the second argument now gives

$$\langle\psi_s, P_\hbar\psi_s\rangle = 0, \qquad \langle X\psi_s, P_\hbar\psi_s\rangle = \frac{i\hbar}{s^2}A = \frac{i\hbar}{2}.$$

Finally, $\|P_\hbar\psi_s\|_2^2 = (\hbar^2/s^4)A = \hbar^2/(2s^2)$. □

2.3 Covariance and units

Theorem 2.6. Lean ✓ *For every $s, \hbar > 0$, the covariance of ψ_s is*

$$\gamma_{\psi_s} = \begin{pmatrix} s^2 & 0 \\ 0 & \hbar^2/s^2 \end{pmatrix}.$$

If $T = \operatorname{diag}(s^{-1}, s/\hbar)$ and $\widetilde{\gamma} = T\gamma_{\psi_s}T^T$, then $\widetilde{\gamma} = I_2$ and $\widetilde{\gamma}$ is admissible in the sense of (1.1).

Proof. The means vanish, so the centered vectors in (2.1) are $X\psi_s$ and $P_\hbar\psi_s$. Equations (2.4) give the diagonal entries. The cross moment (2.5) is purely imaginary, as is its complex conjugate, so both off-diagonal entries vanish. Diagonal multiplication gives $\widetilde{\gamma} = I_2$. Proposition 1.6, with $a = b = 1$, proves admissibility. □

The matrix T converts position and momentum to the dimensionless quantities X/s and $sP_\hbar/\hbar$. The covariance before this change of units retains $\hbar$; we express the admissibility criterion of Section 1 in dimensionless coordinates. With $\hbar = 1$ and $t = s^2$, the original covariance is $D(t, t^{-1})$, realizing the diagonal equality cases above.

Corollary 2.7. Lean ✓ *Let $m, \omega, \hbar > 0$ and set $s = \sqrt{\hbar/(m\omega)}$. Then*

$$\gamma_{\psi_s} = \operatorname{diag}\left(\frac{\hbar}{m\omega}, \hbar m\omega\right).$$

Proof. Positivity gives $s > 0$ and $s^2 = \hbar/(m\omega)$. Substitution into Theorem 2.6 gives the formula. □

Notes and references

We use the Schwartz space as in Reed and Simon [9, Sec. V.3]. The polynomials He_k in the proof of Theorem 2.4 are the probabilists' Hermite polynomials of the NIST Digital Library of Mathematical Functions [19, Sec. 18.3]; the physicists' H_k, orthogonal for the weight e^{-x^2}, are $H_k(x) = 2^{k/2}\,\mathrm{He}_k(\sqrt{2}\,x)$.

3 Contraction of covariance matrices

We now turn to constructing Gaussian matrix product states from local Gaussian resources [3, Secs. 3 and 6]. At each site, we distinguish *physical modes*, which form the final system, from auxiliary *virtual modes*. Neighbouring sites are joined by EPR measurements on their virtual modes, and we seek the covariance of the physical modes conditioned on the measurement outcomes. At the covariance level, this construction leads to a Schur complement. We first establish the general matrix result, then explain EPR measurements and apply it to a periodic chain.

Theorem 3.1. Lean ✓ *Every admissible $2n\times 2n$ covariance matrix is positive definite: $x^T\gamma x > 0$ for every nonzero $x \in \mathbb{R}^{2n}$.*

Proof. Put $H = \gamma_{\mathbb{C}} - i\Omega_n$, where the subscript denotes entrywise inclusion in $\mathbb{C}$. For real x, skew-symmetry gives $x^T\Omega_n x = 0$, and hence

$$x^* H x = x^T \gamma x \geq 0.$$

For a positive semidefinite Hermitian matrix, a vector has zero quadratic form precisely when it lies in the kernel. Thus, if $x^T\gamma x = 0$, then $Hx = 0$. Taking imaginary parts gives $\Omega_n x = 0$. Since $\Omega_n^2 = -I$, we obtain $x = 0$. The quadratic form is therefore strictly positive away from zero. When $n = 0$, there is no nonzero vector and the assertion is vacuous. □

Definition 3.2 (Contracted covariance)**.** Let W be a real $(k + 2m) \times 2n$ matrix. For a real symmetric $2n \times 2n$ matrix γ, write

$$V = W\gamma W^T = \begin{pmatrix} E & F \\ F^T & G \end{pmatrix}, \qquad E \in \mathbb{R}^{k\times k}.$$

The first k output coordinates are to be eliminated; the last $2m$ are retained in position-then-momentum order. When E is invertible, define

$$\gamma_{\text{out}} = G - F^T E^{-1} F.$$

This is the *Schur complement* of E in V. For $k = 0$, it reduces to G.

Schuch, Wolf and Cirac [3, Sec. 3] give the covariance of the physical modes after the EPR measurements by this formula. The eliminated coordinates need not form a canonical subsystem. The next condition makes their symplectic pairings vanish, both among themselves and with the retained coordinates.

Theorem 3.3. Lean ✓ *Let $n, m, k \geq 0$ and let γ be admissible. Suppose W^T is one-to-one and*

$$W\Omega_n W^T = \begin{pmatrix} 0_{k\times k} & 0 \\ 0 & \Omega_m \end{pmatrix}.$$

Then E is positive definite, hence invertible, and γ_{out} is admissible.

Proof. Theorem 3.1 and injectivity of W^T give, for $y \neq 0$,

$$y^T V y = (W^T y)^T \gamma (W^T y) > 0.$$

Thus V and its principal block E are positive definite. Congruence of the input uncertainty matrix yields

$$Q = W_{\mathbb{C}}(\gamma_{\mathbb{C}} - i\Omega_n)W_{\mathbb{C}}^* = \begin{pmatrix} E_{\mathbb{C}} & F_{\mathbb{C}} \\ F_{\mathbb{C}}^* & G_{\mathbb{C}} - i\Omega_m \end{pmatrix} \succeq 0.$$

For any complex vector z, choose the eliminated component to cancel the first block row. Direct block multiplication gives

$$\begin{pmatrix} -E_{\mathbb{C}}^{-1}F_{\mathbb{C}}z \\ z \end{pmatrix}^{*} Q \begin{pmatrix} -E_{\mathbb{C}}^{-1}F_{\mathbb{C}}z \\ z \end{pmatrix} = z^{*}((G - F^{T}E^{-1}F)_{\mathbb{C}} - i\Omega_m)z \geq 0.$$

Here complexification commutes with the inverse of the invertible real matrix E. The output is real symmetric because E and G are symmetric and $(E^{-1})^{T} = E^{-1}$. Its uncertainty matrix is consequently Hermitian, so the quadratic-form inequality proves admissibility. The argument includes empty eliminated or retained coordinate sets. □

3.1 EPR measurements and covariance contraction

Two pairs of canonical quadratures admit the commuting combinations

$$X_{-} = q_A - q_B, \qquad P_{+} = p_A + p_B, \qquad [X_{-}, P_{+}] = [q_A, p_A] - [q_B, p_B] = 0.$$

An *EPR measurement* is the joint measurement of these two observables, the pair of Einstein, Podolsky and Rosen [22]. Their exact joint eigenvectors are generalized eigenvectors: they do not belong to the two-mode Hilbert space. In particular, the ideal EPR vector is not a normalizable state. Finite squeezing gives normalizable approximations, but we must not treat the infinitely squeezed vector as a density operator of trace one. Giedke and Cirac [2, Secs. III–IV] develop the covariance description of Gaussian measurements.

Reversing both quadratures of B is a canonical phase rotation. It replaces the preceding pair by

$$Y_q = q_A + q_B, \qquad Y_p = p_A - p_B. \tag{3.1}$$

We use this convention for the virtual pairs below. Put $\Theta = \mathrm{diag}(I_M, -I_M)$ for M such pairs. If we denote the remaining modes by C, the matrix

$$L = \begin{pmatrix} I & \Theta & 0 \\ 0 & 0 & I_C \end{pmatrix}$$

maps the quadrature vector in A, B, C order to (Y, C). The identity $\Theta\Omega_M\Theta^{T} = -\Omega_M$ gives $L\Omega L^{T} = \mathrm{diag}(0, \Omega_C)$, as required in Theorem 3.3.

We now compute the corresponding characteristic function, including its normalization. For a centered Gaussian with covariance γ, write

$$\chi(t) = \langle e^{it^{T}R} \rangle = \exp(-t^{T}\gamma t/4).$$

The factor $1/4$ follows from our convention that γ is twice the ordinary covariance. Thus the joint characteristic function of Y and the retained quadratures is $\chi(L^{T}(u, z))$. In A, B, C coordinates, $L^{T}(u, z) = (u, \Theta u, z)$: this explains the momentum sign in the contraction of characteristic-function variables. Write

$$L\gamma L^{T} = \begin{pmatrix} E & F \\ F^{T} & G \end{pmatrix}.$$

Conditioning on the zero outcome integrates out the Fourier variable u. The quadratic expression in the exponent is

$$\begin{aligned} u^{T}Eu + 2u^{T}Fz + z^{T}Gz = (u + E^{-1}Fz)^{T}E(u + E^{-1}Fz) \\ + z^{T}(G - F^{T}E^{-1}F)z. \end{aligned}$$

Since E is positive definite, the Gaussian integral is finite. Dividing by its value at $z = 0$ leaves the characteristic function with covariance $G - F^{T}E^{-1}F$. More generally, for means d_Y, d_C and outcome a, the usual Gaussian conditioning formula [21, Sec. 2.5] is

$$d_{C|a} = d_C + F^{T}E^{-1}(a - d_Y), \qquad \gamma_{C|a} = G - F^{T}E^{-1}F. \tag{3.2}$$

The outcome changes the mean, but not the conditional covariance. We understand conditioning here through the outcome density, not as an event of positive probability at one exact value. These formulas give the measurement interpretation; Theorem 3.3 establishes the covariance inequality directly.

In the Choi–Jamiołkowski construction [23, 24], the resource associated with a map is obtained formally by applying the map to one half of an ideal EPR pair; for Gaussian maps see [2]. EPR contraction with the other half represents feeding an input to the map. This turns the construction of Gaussian matrix product states into contractions of local resources, but still involves ideal EPR measurements. Moreover, a channel representative obtained from an infinitely squeezed pair need not be a finite covariance matrix: the identity channel already illustrates the difficulty. The results below use finite admissible local matrices. Schuch, Wolf and Cirac describe the resource construction in Sections 3 and 6 of [3]; their Section 5 replaces infinitely entangled bonds by finite ones while moving the infinite entanglement into the local maps.

3.2 A periodic chain

At each site $i \in \mathbb{Z}/N\mathbb{Z}$, we introduce M left virtual modes A_i, M right virtual modes B_i, and one retained mode C_i. The pair on each edge joins B_{i-1} to A_i. We assume $N \geq 3$ and $M \geq 1$. The integer M counts the *virtual modes per bond*. Each bosonic mode has an infinite-dimensional Hilbert space, so even $M = 1$ can support entanglement. With no virtual modes, $M = 0$, the virtual Hilbert space is one-dimensional; this is the counterpart of bond dimension $D = 1$ for a qudit MPS. We can also represent product covariances with $M \geq 1$ by adding unused virtual modes. A bound on M alone does not bound entanglement entropy: the squeezing must also be controlled.

Definition 3.4 (Periodic covariance)**.** Put $d = 2M + 1$ and let Γ_i be real $2d \times 2d$ matrices. The local mode order is $(A_{i,1}, \dots, A_{i,M}, B_{i,1}, \dots, B_{i,M}, C_i)$, with positions preceding momenta. Define γ_{loc} on Nd modes by

$$(\gamma_{\mathrm{loc}})_{(u,i,a),(v,j,b)} = \begin{cases} (\Gamma_i)_{(u,a),(v,b)}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad u, v \in \{q, p\}.$$

Globally, positions precede momenta; within either group, sites precede local modes. The real linear map $W : \mathbb{R}^{2Nd} \to \mathbb{R}^{2NM} \oplus \mathbb{R}^{2N}$ has eliminated coordinates

$$(Wx)_{q,i,a} = x_{q,A_i,a} + x_{q,B_{i-1},a}, \qquad (Wx)_{p,i,a} = x_{p,A_i,a} - x_{p,B_{i-1},a},$$

and retained coordinates $(Wx)_{q,C_i} = x_{q,C_i}$ and $(Wx)_{p,C_i} = x_{p,C_i}$. Each output group is in position-then-momentum order. When its eliminated block is invertible, the *periodic covariance* is

$$\gamma_{\mathrm{per}} = \mathrm{SC}_{\mathbb{R}^{2NM}}(W\gamma_{\mathrm{loc}}W^T),$$

where SC denotes the Schur complement of Definition 3.2.

For comparison with [3], we group all A coordinates together, then all B and all C coordinates. If R shifts the site index to the right, $(Rx)_i = x_{i-1}$, the same map is

$$W = \begin{pmatrix} I_A & R\Theta_B & 0 \\ 0 & 0 & I_C \end{pmatrix}.$$

The shift acts identically on positions and momenta and preserves each mode label. This is the matrix in Section 6, Eq. (3) of [3].

Theorem 3.5. Lean ✓ *Suppose every Γ_i is admissible. Then W^T is injective,*

$$W\Omega_{Nd}W^T = \mathrm{diag}(0_{2NM}, \Omega_N),$$

the eliminated block of $W\gamma_{\mathrm{loc}}W^T$ is positive definite, and γ_{per} is admissible.

Proof. The independent sum of the local covariances is admissible by Theorem 1.5, iterated over the sites. Reordering its positions and momenta together gives γ_{loc} and preserves the uncertainty inequality. For an output vector (y_q, y_p, z_q, z_p), the A components of its image under W^T are y_q, y_p, and the C components are z_q, z_p. Hence W^T is injective. Each eliminated position–momentum pairing has an A contribution $+1$ and a B contribution -1. They cancel, including after the common cyclic shift. Distinct mode pairs and all pairings with C vanish. The retained pairings form Ω_N. Apply Theorem 3.3. □

3.3 Three correlated sites

Example 3.6. Lean ✓ Take $N = 3$, $M = 1$, and at each site let

$$K = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} I_3 & 0 \\ K & I_3 \end{pmatrix}, \qquad \Gamma_i = SS^T.$$

Then S is symplectic, Γ_i is admissible, and the periodic covariance is

$$\gamma_{\mathrm{per}} = \mathrm{diag}(Q, P), \qquad Q = \begin{pmatrix} 3/5 & 1/5 & 1/5 \\ 1/5 & 3/5 & 1/5 \\ 1/5 & 1/5 & 3/5 \end{pmatrix}, \quad P = \begin{pmatrix} 2 & -1/2 & -1/2 \\ -1/2 & 2 & -1/2 \\ -1/2 & -1/2 & 2 \end{pmatrix}.$$

It is admissible. In particular, distinct sites have position–position covariance entry $1/5$ and momentum–momentum covariance entry $-1/2$.

Proof. Since $K = K^T$, direct multiplication gives $S\Omega_3 S^T = \Omega_3$. The covariance I_6 is an independent sum of three one-mode vacua, so Theorem 1.2 gives admissibility of SS^T. Let R be the three-site cyclic right shift and U the 3×3 all-ones matrix. Multiplying the local covariance blocks gives

$$E = \begin{pmatrix} 2I & 0 \\ 0 & 4I - R - R^T \end{pmatrix}, \quad F = \begin{pmatrix} 0 & I + R \\ I - R & 0 \end{pmatrix}, \quad G = \begin{pmatrix} I & 0 \\ 0 & 3I \end{pmatrix}.$$

Here $R + R^T = U - I$ and $U^2 = 3U$. Consequently

$$(4I - R - R^T)^{-1} = (5I - U)^{-1} = \tfrac{1}{5}I + \tfrac{1}{10}U,$$

as multiplication checks. Since $U(I - R) = 0$, the two diagonal blocks of the Schur complement are

$$\begin{aligned} Q &= I - (I - R)^T(\tfrac{1}{5}I + \tfrac{1}{10}U)(I - R) = \tfrac{2}{5}I + \tfrac{1}{5}U, \\ P &= 3I - \tfrac{1}{2}(I + R)^T(I + R) = \tfrac{5}{2}I - \tfrac{1}{2}U. \end{aligned}$$

The off-diagonal blocks vanish. Theorem 3.5 proves admissibility. □

For this example, $QP = I_3$, so the associated Gaussian state is pure. Its single-site covariance is $\mathrm{diag}(3/5, 2)$, with determinant $6/5 > 1$; the single-site reduced state is therefore mixed. Thus the state is entangled across every partition of one site from the other two. These deductions use the Gaussian purity criterion [1], which is cited, not proved here; Example 3.6 establishes the displayed covariance and its admissibility.

Notes and references

Theorem 3.1 is a standard consequence of the uncertainty principle; see [12, Sec. IX]. The proof of Theorem 3.3 applies to the complex uncertainty matrix the positive semidefiniteness of a Schur complement in a positive semidefinite block matrix [20, Sec. 7.7]. Example 3.6 is worked out for these notes; we have not found it in the literature we checked.

4 Homogeneous periodic covariances

Suppose that every site in the periodic construction uses the same local covariance Γ. A cyclic shift then leaves the construction unchanged. The finite Fourier transform separates its spatial frequencies: at each frequency, we obtain the covariance by eliminating only the local virtual coordinates. This gives the finite-chain form of the Fourier calculation in [3, Sec. 6].

Fix $N \geq 3$ and $M \geq 1$, with one physical mode at each site. We retain the ordering and predecessor wiring of Section 3. For the local block calculations, we group the coordinates as (a, b, c), where $a = (q_A, p_A)$, $b = (q_B, p_B)$ and $c = (q_C, p_C)$. Write

$$\Gamma = \begin{pmatrix} \Gamma_{vv} & \Gamma_{vC} \\ \Gamma_{Cv} & \Gamma_{CC} \end{pmatrix}, \qquad v = (a, b), \qquad \Theta = \operatorname{diag}(I_M, -I_M).$$

Thus the virtual block has size $4M \times 4M$, the cross block has size $4M \times 2$, and the physical block has size 2×2. All blocks are taken in the order (a, b, c).

Definition 4.1 (Fourier blocks)**.** Label sites and frequencies by $0, \ldots, N-1$, with indices interpreted modulo N. Set

$$\phi_k = \frac{2\pi k}{N}, \qquad \widehat{f}_k = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} e^{-\mathrm{i}\phi_k j} f_j, \qquad Z_k = \begin{pmatrix} I_{2M} & e^{-\mathrm{i}\phi_k}\Theta \end{pmatrix}.$$

Define

$$E_k = Z_k \Gamma_{vv} Z_k^*, \qquad F_k = Z_k \Gamma_{vC}, \qquad S_k = \Gamma_{CC} - F_k^* E_k^{-1} F_k, \tag{4.1}$$

where $*$ denotes conjugate transpose. Invertibility of E_k follows from the next theorem.

The eliminated coordinates are $y_j = a_j + \Theta b_{j-1}$. Substituting $j = r + 1$ in the Fourier sum gives

$$\widehat{(b_{j-1})}_k = e^{-\mathrm{i}\phi_k} \widehat{b}_k, \qquad \widehat{y}_k = Z_k \begin{pmatrix} \widehat{a}_k \\ \widehat{b}_k \end{pmatrix}.$$

Theorem 4.2 (Fourier contraction)**.** Lean ✓ *Let Γ be admissible and let γ_{per} be its homogeneous periodic covariance. Each E_k is Hermitian positive definite, and*

$$(\gamma_{\mathrm{per}})_{\alpha,i+1;\,\beta,j+1} = (\gamma_{\mathrm{per}})_{\alpha i,\beta j} \qquad (\alpha, \beta \in \{q, p\}).$$

Its Fourier transform is block diagonal:

$$\frac{1}{N} \sum_{i,j=0}^{N-1} e^{-\mathrm{i}\phi_k i} (\gamma_{\mathrm{per}})_{\alpha i,\beta j} e^{\mathrm{i}\phi_l j} = \begin{cases} (S_k)_{\alpha\beta}, & k = l, \\ 0, & k \neq l. \end{cases} \tag{4.2}$$

Proof. For r coordinates at each site, define $T_k^{(r)} : \mathbb{C}^r \to \mathbb{C}^{Nr}$ by $(T_k^{(r)} v)_j = e^{\mathrm{i}\phi_k j} v$. Character orthogonality gives

$$(T_l^{(r)})^* T_k^{(r)} = N \delta_{lk} I_r, \qquad \frac{1}{N} \sum_k T_k^{(r)} (T_k^{(r)})^* = I_{Nr}.$$

The independent local covariance acts as Γ on each such subspace. The shift calculation above shows that the wiring acts there as $L_k = \mathrm{diag}(Z_k, I_2)$. Its transpose acts as L_k^*, since the contribution from a B coordinate uses the successor site. Consequently the wired covariance has local matrix

$$L_k \Gamma L_k^* = \begin{pmatrix} E_k & F_k \\ F_k^* & \Gamma_{CC} \end{pmatrix}.$$

Write its global blocks as E, F, G, as in the contraction theorem. In particular,

$$E T_k^{(2M)} = T_k^{(2M)} E_k, \qquad F T_k^{(2)} = T_k^{(2M)} F_k, \qquad G T_k^{(2)} = T_k^{(2)} \Gamma_{CC}.$$

Theorem 3.1 gives $\Gamma \succ 0$ over the reals. For complex vectors, put $Q = \Gamma - \mathrm{i}\Omega$, with the symplectic matrix Ω expressed in the same coordinate order. Both Q and Q^T are positive semidefinite, so their average Γ is positive semidefinite as well. Since Γ is nonsingular, it is positive definite. Its virtual principal block is therefore positive definite. The map Z_k^* is injective, since its first block is the identity, and hence $E_k = Z_k \Gamma_{vv} Z_k^* \succ 0$. By Theorem 3.5, the global block E is also positive definite. The first mode identity therefore implies $E^{-1} T_k^{(2M)} = T_k^{(2M)} E_k^{-1}$. Taking the Schur complement yields

$$\gamma_{\mathrm{per}} T_k^{(2)} = T_k^{(2)} S_k.$$

Multiplication by $(T_l^{(2)})^*$ and orthogonality prove (4.2). A cyclic shift acts by a scalar on each Fourier subspace, so γ_{per} commutes with cyclic shifts and is translation invariant. □

Theorem 4.3 (Reconstruction and uncertainty)**.** Lean ✓ *Under the same assumptions,*

$$(\gamma_{\mathrm{per}})_{\alpha i, \beta j} = \frac{1}{N} \sum_{k=0}^{N-1} e^{\mathrm{i}\phi_k (i-j)} (S_k)_{\alpha\beta}. \tag{4.3}$$

Moreover,

$$S_{-k} = \overline{S_k}, \qquad S_k - \mathrm{i}\Omega_1 \succeq 0, \qquad \Omega_1 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

where $-k$ is interpreted modulo N and the bar is entrywise conjugation.

Proof. Insert the Fourier resolution of the identity into $\gamma_{\mathrm{per}} T_k^{(2)} = T_k^{(2)} S_k$ to obtain (4.3). Since γ_{per} is real, conjugating its Fourier transform replaces each frequency by its negative. The diagonal blocks in (4.2) therefore satisfy $S_{-k} = \overline{S_k}$. Finally, Theorem 3.5 gives $\gamma_{\mathrm{per}} - \mathrm{i}\Omega_N \succeq 0$. Its pullback along $T_k^{(2)}$ is $N(S_k - \mathrm{i}\Omega_1)$: the position and momentum coordinates use the same Fourier vectors, whose squared norm is N. Dividing by $N > 0$ proves the inequality. □

The matrices S_k can have complex entries. Conjugate frequencies together reconstruct the real covariance; the uncertainty inequality for an individual frequency is an inequality of complex Hermitian matrices.

Notes and references

For translation-invariant chains Schuch, Wolf and Cirac [3, Sec. 6, Eq. (4)] compute the covariance by a Schur complement in Fourier space; Theorem 4.2 is its form for a finite periodic chain. We have not found the frequency-wise inequality $S_k - \mathrm{i}\Omega_1 \succeq 0$ of Theorem 4.3 in the literature we checked; the proof above derives it from Theorem 3.5.

5 Product covariances

The periodic construction includes products: if the physical mode at each site is independent of the virtual modes, contraction leaves its covariance unchanged. We make this explicit, allowing the physical covariance to vary from site to site. We retain $N \geq 3$, $M \geq 1$ and the ordering of Section 3.

Definition 5.1. Let η_i be a real one-mode covariance at site i. Take the local covariance

$$\Gamma_i = \operatorname{diag}(I_{2M}, I_{2M}, \eta_i)$$

in grouped phase coordinates $(q_{A_i}, p_{A_i}, q_{B_i}, p_{B_i}, q_{C_i}, p_{C_i})$, then reorder it to the position-then-momentum coordinates used for periodic assembly. Define the physical product covariance by

$$(\gamma_{\mathrm{prod}})_{(u,i),(v,j)} = \begin{cases} (\eta_i)_{uv}, & i = j, \\ 0, & i \neq j, \end{cases} \qquad u, v \in \{q, p\}.$$

Each virtual mode is a vacuum and is independent of all other modes at its site. The physical mode has covariance η_i.

Theorem 5.2. Lean ✓ *If every η_i is admissible, then every Γ_i is admissible and*

$$W \gamma_{\mathrm{loc}} W^T = \begin{pmatrix} 2I_{2NM} & 0 \\ 0 & \gamma_{\mathrm{prod}} \end{pmatrix}.$$

Consequently the periodic contraction equals γ_{prod} and is admissible. In particular, one virtual mode per bond suffices for every $N \geq 3$.

Proof. The independent-sum rule of Theorem 1.5 applies to the local resources. In grouped phase coordinates its uncertainty matrix is

$$\Gamma_i - i \operatorname{diag}(\Omega_M, \Omega_M, \Omega_1) = (I_{2M} - i\Omega_M) \oplus (I_{2M} - i\Omega_M) \oplus (\eta_i - i\Omega_1) \succeq 0.$$

The two vacuum blocks and the prescribed physical block are positive semidefinite; reordering the coordinates preserves the inequality.

Each eliminated quadrature is the signed sum of two independent virtual vacuum coordinates. Its diagonal covariance entry is $1 + 1 = 2$. Distinct eliminated coordinates have zero covariance: distinct virtual modes are independent, and within each vacuum mode the position–momentum covariance vanishes. The virtual coordinates are also independent of every physical coordinate. Thus, in the notation of Definition 3.2,

$$E = 2I_{2NM}, \qquad F = 0, \qquad G = \gamma_{\mathrm{prod}}.$$

The Schur-complement formula therefore gives

$$\gamma_{\mathrm{per}} = G - F^T E^{-1} F = \gamma_{\mathrm{prod}}.$$

Admissibility follows from Theorem 3.5. □

Corollary 5.3. Lean ✓ *For $\eta_i = I_2$ at every site, the periodic contraction is I_{2N}. For arbitrary $s_i > 0$, the choices $\eta_i = \operatorname{diag}(s_i^2, s_i^{-2})$ give the corresponding product of squeezed one-mode covariances.*

Proof. The vacuum covariance is admissible. The diagonal criterion of Proposition 1.6 states

$$\operatorname{diag}(a, b) \text{ is admissible} \iff a > 0 \text{ and } ab \geq 1.$$

Apply it with $a = s_i^2$ and $b = s_i^{-2}$, whose product is one, and use Theorem 5.2. In the vacuum case, the entries of γ_{prod} are those of the identity. □

The one-mode Gaussians of Theorem 2.6, with $\hbar = 1$, have these squeezed covariances. The virtual vacua in this construction are unused; adding them allows products to fit the same $M \geq 1$ convention as the correlated examples.

6 The formal Fock space and the Bargmann picture

The number basis of a bosonic Fock space labels vectors by occupation numbers. Allowing all coefficient families, with no summability condition, gives a space on which the creation operators and the exponentials of operators that raise the particle number act exactly, one coefficient at a time. This section develops that formal Fock space, the canonical commutation relations on it, the exponential law for these operators, and the Bargmann picture, in which creation and annihilation become multiplication by and differentiation in formal variables. The exponentials of the Gaussian-augmented bosonic matrix product states of Tjoa and Cirac [4] are of this kind. Those states are a different construction from the Gaussian matrix product states whose covariance matrices are built by contraction in Sections 3–5; these notes do not yet relate the two.

Notation for this section.

ι	a set of modes
$n = (n_i)_{i\in\iota}$	a multi-index of occupation numbers, finitely many nonzero
m	an occupation number of a single mode
e_i	the multi-index with a single 1 at i
$\lvert n\rvert = \sum_i n_i$	total particle number of n
$n! = \prod_i n_i!$	factorial of a multi-index
$\lvert n\rangle$	number state (Definition 6.1)
$\mathcal{F}(\iota, R)$	formal Fock space with coefficients in R (Definition 6.1)
$\mathfrak{h}$, $\Gamma_s(\mathfrak{h})$	for finite ι, the one-particle space $\mathbb{C}^\iota$ and its Fock space
$a_i^\dagger$, a_i	creation and annihilation operators (Definition 6.2)
T, T', e^T	operators that raise the particle number and their exponential (Definition 6.6)
$R[[z]]$, ∂_j	formal power series in $(z_i)_{i\in\iota}$ over R; partial derivative
$[z^n]\, f$	coefficient of $z^n = \prod_i z_i^{n_i}$ in $f \in R[[z]]$
$\mathfrak{B}$	Bargmann map (Definition 6.9)

In this section R is a $\mathbb{C}$-algebra, $X \in R$, n a multi-index, W a word in the ladder operators, T an operator that raises the particle number, $p \geq 1$ an exponent, z the formal variables and Λ a bound on particle numbers; letters shared with Sections 1–5 do not carry their meanings from there.

6.1 The formal Fock space

Throughout, R is a $\mathbb{C}$-algebra: a ring together with a ring homomorphism from $\mathbb{C}$ into its centre. The two cases used later are $R = \mathbb{C}$ (states) and $R = M_\beta(\mathbb{C})$, the complex matrices indexed by a finite set β (vectors that carry a matrix index); for β empty, $M_\beta(\mathbb{C})$ is the zero ring, which is allowed.

Definition 6.1 (Formal Fock space)**.** $\mathcal{F}(\iota, R)$ is the set of all functions ψ from multi-indices to R, with pointwise addition and scalar multiplication; ψ is read as $\sum_n \psi(n)|n\rangle$. The number state $c|n\rangle$ has coefficient c at n and 0 elsewhere, and $|0\rangle$ is the vacuum. A vector ψ *vanishes below* k if $\psi(n) = 0$ whenever $|n| < k$.

No summability is imposed. For finite ι and the one-particle space $\mathfrak{h} = \mathbb{C}^\iota$ with its standard orthonormal basis, the k-particle space $\otimes_s^k \mathfrak{h}$ has as orthonormal basis the number states $|n\rangle$ with $|n| = k$, so that, sector by sector,

$$\Gamma_{\text{fin}}(\mathfrak{h}) = \bigoplus_k \otimes_s^k \mathfrak{h} \;\subset\; \Gamma_s(\mathfrak{h}) = \overline{\bigoplus_k} \otimes_s^k \mathfrak{h} \;\subset\; \prod_k \otimes_s^k \mathfrak{h} = \mathcal{F}(\iota, \mathbb{C}) :$$

the finite-particle space, the bosonic Fock space of [5, Ch. 3], [6, Sec. 5.2.1] (the Hilbert direct sum, that is, the square-summable families) and the full product. The symmetric-tensor construction

of $\Gamma_s(\mathfrak{h})$ is recalled from [5, 6]; Section 7 works with it as ℓ^2 of the multi-indices (Definition 7.1). Section 8 treats the second quantization of one-particle matrices. The Fock representation of the CCR algebra and the identification of $\Gamma_s(\mathfrak{h})$ with $L^2(\mathbb{R}^\iota)$ through the Hermite functions that would join this section to Sections 1–5 are not treated in these notes. For a single mode, the vector with coefficients $\sqrt{(2k)!}/k!$ at $|2k\rangle$ and 0 at odd occupations lies in $\mathcal{F}(\{1\},\mathbb{C})$ but not in Γ_s, since $(2k)!/(k!)^2 \geq 1$. After Definition 6.6 it is $e^{(a^\dagger)^2}|0\rangle$.

Definition 6.2 (Ladder operators and left multiplication)**.** For $i \in \iota$, the creation and annihilation operators are the $\mathbb{C}$-linear maps

$$(a_i^\dagger \psi)(n) = \begin{cases} \sqrt{n_i}\,\psi(n - e_i), & n_i \geq 1, \\ 0, & n_i = 0, \end{cases} \qquad (a_i\psi)(n) = \sqrt{n_i + 1}\,\psi(n + e_i).$$

For $X \in R$, $X\psi$ is the vector with coefficients $X\psi(n)$. For a word W in the ladder operators, $X \otimes W$ denotes the composite XW; the two factors commute, since a ladder word multiplies each coefficient by complex scalars and moves it, while X multiplies it on the left, and complex scalars are central in R.

On number states these are the formulas $a_i^\dagger|n\rangle = \sqrt{n_i + 1}|n + e_i\rangle$ and $a_i|n\rangle = \sqrt{n_i}|n - e_i\rangle$. On $\Gamma_s(\mathfrak{h})$ the creation and annihilation operators of [5] are $a^*(h) = \sum_i h_i a_i^\dagger$ and $a(h) = \sum_i \overline{h_i}\, a_i$ for $h \in \mathfrak{h}$. Each coefficient of $a_i^\dagger\psi$ or $a_i\psi$ involves a single coefficient of ψ, so both operators are defined on the whole formal Fock space.

Theorem 6.3 (Canonical commutation relations)**.** Lean ✓ *On all of $\mathcal{F}(\iota, R)$: $[a_i, a_j^\dagger] = \delta_{ij}$, $[a_i, a_j] = 0$ and $[a_i^\dagger, a_j^\dagger] = 0$.*

Proof. We check the identities coefficientwise.

(a) For $i = j$: $(a_i a_i^\dagger\psi)(n) = \sqrt{n_i + 1}\sqrt{n_i + 1}\,\psi(n)$ and $(a_i^\dagger a_i\psi)(n) = \sqrt{n_i}\sqrt{n_i}\,\psi(n)$, the second being 0 when $n_i = 0$. The difference is $\psi(n)$.

(b) For $i \neq j$, both orders give $\sqrt{n_j}\sqrt{n_i + 1}\,\psi(n + e_i - e_j)$ when $n_j \geq 1$, and 0 otherwise, because shifting n_i does not change n_j.

(c) For $i \neq j$, $a_i a_j$ and $a_j a_i$ both multiply $\psi(n + e_i + e_j)$ by $\sqrt{n_i + 1}\sqrt{n_j + 1}$; $a_i^\dagger a_j^\dagger$ and $a_j^\dagger a_i^\dagger$ both multiply $\psi(n - e_i - e_j)$ by $\sqrt{n_i}\sqrt{n_j}$ when $n_i, n_j \geq 1$, and give 0 otherwise. For $i = j$ there is nothing to prove. □

Since $a_i^\dagger(c|n\rangle) = \sqrt{n_i + 1}\, c|n + e_i\rangle$ and the creation operators commute, $\prod_i (a_i^\dagger)^{n_i}(c|0\rangle) = \sqrt{n!}\, c|n\rangle$, the product running over the finitely many i with $n_i \neq 0$. So $|n\rangle = \prod_i (a_i^\dagger)^{n_i}|0\rangle/\sqrt{n!}$, as usual.

For a single mode we write a and $a^\dagger$ for the ladder operators on $\mathcal{F}(\{1\},\mathbb{C})$ and $\hat{n} = a^\dagger a$ for the number operator. By Theorem 6.3, $aa^\dagger = a^\dagger a + 1$, and this relation moves every annihilation operator in a product of ladder operators to the left of every creation operator at the cost of shorter products. The result is the *anti-normal order*, the order in which Tjoa and Cirac compute the generalized transfer matrices of their bosonic matrix product states [4, Prop. 2]. The polynomials in a and $a^\dagger$ form the algebra generated by a and $a^\dagger$: the smallest subalgebra of the linear maps on $\mathcal{F}(\{1\},\mathbb{C})$ that contains 1, a and $a^\dagger$.

Theorem 6.4 (Anti-normal ordering)**.** Lean ✓ *For a single mode, the monomials $a^m(a^\dagger)^n$, $m, n \in \mathbb{N}$, span the algebra generated by a and $a^\dagger$.*

Proof. Let $\mathcal{M}$ be the span of the monomials. Each monomial lies in the algebra, so $\mathcal{M}$ does. Conversely, we show that $a\mathcal{M} \subseteq \mathcal{M}$ and $a^\dagger\mathcal{M} \subseteq \mathcal{M}$. Then the operators Y with $Y\mathcal{M} \subseteq \mathcal{M}$ form a subalgebra containing a and $a^\dagger$, hence the whole algebra, and every Y of the algebra is

$Y \cdot 1 \in \mathcal{M}$, since $1 = a^0(a^\dagger)^0 \in \mathcal{M}$. Left multiplication by a sends $a^m(a^\dagger)^n$ to $a^{m+1}(a^\dagger)^n$. For $a^\dagger$, induction on m from $aa^\dagger = a^\dagger a + 1$ gives

$$a^{m+1}a^\dagger = a^\dagger a^{m+1} + (m+1)\,a^m,$$

the step being $a^{m+2}a^\dagger = a(a^\dagger a^{m+1} + (m+1)a^m) = (a^\dagger a + 1)a^{m+1} + (m+1)a^{m+1}$. Hence $a^\dagger a^{m+1}(a^\dagger)^n = a^{m+1}(a^\dagger)^{n+1} - (m+1)\,a^m(a^\dagger)^n$ lies in $\mathcal{M}$, and so does $a^\dagger(a^\dagger)^n = (a^\dagger)^{n+1}$. □

For the square of the number operator the reordering takes two steps.

Proposition 6.5 (The square of the number operator)**.** Lean ✓ *For a single mode, $\hat{n}^2 = a^2(a^\dagger)^2 - 3\,aa^\dagger + 1$.*

Proof. By Theorem 6.3, $\hat{n} = a^\dagger a = aa^\dagger - 1$, so $\hat{n}^2 = aa^\dagger aa^\dagger - 2\,aa^\dagger + 1$. Inside the first term, $a^\dagger a = aa^\dagger - 1$ again gives $aa^\dagger aa^\dagger = a^2(a^\dagger)^2 - aa^\dagger$. □

Both sides give $j^2|j\rangle$ on a number state $|j\rangle$, since $(j+1)(j+2) - 3(j+1) + 1 = j^2$.

6.2 Operators that raise the particle number

Operators that raise the particle number have exponential series that are finite at each coefficient.

Definition 6.6 (Raising the particle number; the exponential)**.** A $\mathbb{C}$-linear map T on $\mathcal{F}(\iota, R)$ *raises the particle number* if, for every k, $T\psi$ vanishes below $k+1$ whenever ψ vanishes below k. For such T,

$$(e^T\psi)(n) = \sum_{r=0}^{|n|} \frac{(T^r\psi)(n)}{r!}.$$

The basic examples are the polynomials in the creation operators with zero constant term (Lemma 6.7(c)), such as $\kappa(a^\dagger)^2 + \ell a^\dagger$, whose exponential applied to the vacuum is, for $|\kappa| < \frac{1}{2}$, a squeezed coherent state up to normalization [4, (8)–(9)], and $K \otimes (a^\dagger)^2 + L \otimes a^\dagger$ with matrix coefficients.

Lemma 6.7. Lean ✓ *Let T and T' raise the particle number.*

(a) *If $T_1, \dots, T_r$ raise the particle number and ψ vanishes below k, then $T_1 \cdots T_r\psi$ vanishes below $k+r$. In particular $(T^r\psi)(n) = 0$ for $r > |n|$, and $(e^T\psi)(n) = \sum_{r=0}^{\Lambda}(T^r\psi)(n)/r!$ for every $\Lambda \geq |n|$.*

(b) *$(T\psi)(n)$ depends only on the coefficients $\psi(n')$ with $|n'| < |n|$.*

(c) *$T + T'$, TT' and cT ($c \in \mathbb{C}$) raise the particle number; so do $a_i^\dagger$ and, for $X \in R$ and $p \geq 1$, $X(a_i^\dagger)^p$. More generally, every word $Xa_{i_1}^\dagger \cdots a_{i_r}^\dagger$ with $r \geq 1$ raises the particle number, hence so does every polynomial in the creation operators with zero constant term and coefficients in R on the left.*

Proof. (a) Apply the definition to T_r, then T_{r-1}, and so on; each step raises the bound by one. Every ψ vanishes below 0.

(b) If ψ and ψ' agree in degrees below $|n|$, then $\psi - \psi'$ vanishes below $|n|$, so $T(\psi - \psi')$ vanishes below $|n| + 1$, in particular at n.

(c) The first three claims follow from the definition. $(a_i^\dagger\psi)(n)$ is a multiple of $\psi(n - e_i)$ and $|n - e_i| = |n| - 1$, so $a_i^\dagger$ raises the particle number; hence so does $(a_i^\dagger)^p$ for $p \geq 1$, and left multiplication by X preserves vanishing below k. By (a), a word $Xa_{i_1}^\dagger \cdots a_{i_r}^\dagger$ maps a vector that vanishes below k to one that vanishes below $k + r$, so it raises the particle number since $r \geq 1$, and sums of such maps raise it by the first claim. □

We define the exponential by the finite sum of Definition 6.6, not as a limit, so no topology on $\mathcal{F}(\iota, R)$ is needed. By Lemma 6.7(a) this is the operator series $\sum_r T^r/r!$ itself: at every coefficient the partial sums are constant from $r = |n|$ on. When the series converges in $\Gamma_s(\mathfrak{h})$, its limit has these coefficients, since convergence there implies convergence of each coefficient.

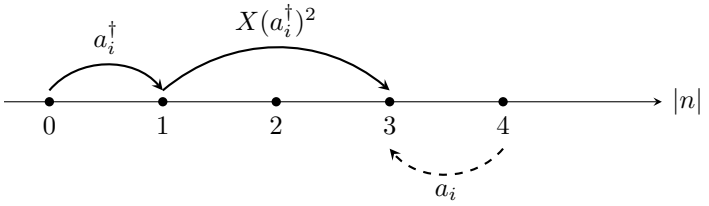


Figure 2: Maps that raise the particle number (solid, above the axis) move every coefficient to strictly higher particle number; the annihilation operator (dashed, below the axis) moves it down. Hence the coefficient of $e^T\psi$ at particle number k receives only the terms $T^r\psi$ with $r \le k$: the finite sum of Definition 6.6.

Theorem 6.8 (Exponential law). Lean ✓ *If T and T' raise the particle number and $TT' = T'T$, then $e^{T+T'} = e^T e^{T'}$.*

Proof. By Lemma 6.7(c), $T + T'$ raises the particle number, so both sides are defined; likewise so does T^r for $r \ge 1$, which is used below. The coefficient of $e^T e^{T'}\psi$ at n reduces to a finite double sum, which the binomial theorem collapses. Fix n and ψ, and put $\Lambda = |n|$. By Lemma 6.7(a), $e^{T'}\psi$ agrees in all degrees $\le \Lambda$ with $\sum_{t=0}^{\Lambda} T'^t\psi/t!$. By Lemma 6.7(b), applied to the map T^r for $r \ge 1$ (and trivially for $r = 0$),

$$(e^T e^{T'}\psi)(n) = \sum_{r=0}^{\Lambda}\sum_{t=0}^{\Lambda} \frac{(T^r T'^t\psi)(n)}{r!\,t!} = \sum_{k=0}^{\Lambda} \frac{1}{k!} \sum_{r+t=k} \binom{k}{r} (T^r T'^t\psi)(n) = \sum_{k=0}^{\Lambda} \frac{((T+T')^k\psi)(n)}{k!}.$$

We dropped the terms with $r + t > \Lambda$ because $T^r T'^t$ is a product of $r + t$ maps that raise the particle number and therefore, by Lemma 6.7(a), vanishes at n; the last step is the binomial theorem for commuting T and T'. The right side is $(e^{T+T'}\psi)(n)$. □

For matrices the same identity rests on absolute convergence of the double series. Here the grading by particle number replaces it. Without the hypothesis that T raises the particle number the finite sum of Definition 6.6 need not agree with the series $\sum_r (T^r\psi)(n)/r!$, and the series need not converge. For the annihilation operator a of a single mode and $\psi = |1\rangle$, the finite sum at $n = 0$ is $\psi(0) = 0$, while the series gives $(a\psi)(0) = 1$. For $\psi(m) = m!$, the series $\sum_k (a^k\psi)(0)/k! = \sum_k \sqrt{k!}$ diverges.

6.3 The Bargmann picture

$R[[z]]$ is the set of all functions f from multi-indices to R, written $f = \sum_n [z^n]\, f\, z^n$, with the Cauchy product $[z^n]\,(fg) = \sum_{n'+n''=n} [z^{n'}]\, f\, [z^{n''}]\, g$, a finite sum. For $f \in R[[z]]$, we define the partial derivative coefficientwise by $[z^n]\,(\partial_j f) = (n_j + 1)\,[z^{n+e_j}]\, f$; for $R = \mathbb{C}$ it is the usual derivative of formal power series.

Definition 6.9 (Bargmann map). $\mathfrak{B} : \mathcal{F}(\iota, R) \to R[[z]]$ is given by $[z^n]\,\mathfrak{B}\psi = \psi(n)/\sqrt{n!}$. It is a $\mathbb{C}$-linear bijection, with inverse $\psi(n) = \sqrt{n!}\,[z^n]\, f$.

Thus $\mathfrak{B}|n\rangle = z^n/\sqrt{n!}$, which for finite ι is the orthonormal basis of Bargmann's space of entire functions [7, Sec. 1]. So $\mathfrak{B}$ maps $\Gamma_s(\mathfrak{h})$ onto the Taylor series of the Bargmann space, and the formal Fock space onto all of $R[[z]]$, including, for a single mode and $R = \mathbb{C}$, series such as $\sum_m m!\, z^m = \mathfrak{B}\psi$, $\psi(m) = m!\sqrt{m!}$, with radius of convergence 0. A vector ψ vanishes below k

exactly when $\mathfrak{B}\psi$ has order at least k (no monomial of total degree below k), so a map that raises the particle number raises the order of power series by at least one.

Lemma 6.10. Lean ✓ *If $n_j \geq 1$, then $n! = n_j\,(n - e_j)!$.*

Proof. n and $n - e_j$ differ only in the j-th entry, and $n_j! = n_j\,(n_j - 1)!$. □

Lemma 6.11. Lean ✓ *For every $f \in R[[z]]$: $[z^n]\,(z_j^m f) = [z^{n-me_j}]\,f$ if $m \leq n_j$, and 0 if $m > n_j$.*

Proof. This is the coefficient formula for multiplication by the monomial z_j^m; it holds for non-commutative R because z_j^m has a scalar coefficient. □

Theorem 6.12 (Intertwining)**.** Lean ✓ *For every $\psi \in \mathcal{F}(\iota, R)$ and $X \in R$:*

(a) $\mathfrak{B}(a_j^\dagger \psi) = z_j\,\mathfrak{B}\psi$;

(b) $\mathfrak{B}(a_j \psi) = \partial_j \mathfrak{B}\psi$;

(c) $\mathfrak{B}(X\psi) = X\,\mathfrak{B}\psi$;

(d) $\mathfrak{B}|0\rangle = 1$.

Proof. (a) At n with $n_j \geq 1$, Lemma 6.10 gives

$$[z^n]\,\mathfrak{B}(a_j^\dagger \psi) = \frac{\sqrt{n_j}\,\psi(n - e_j)}{\sqrt{n!}} = \frac{\psi(n - e_j)}{\sqrt{(n - e_j)!}},$$

which is $[z^n]\,(z_j \mathfrak{B}\psi)$ by Lemma 6.11. If $n_j = 0$, both sides vanish.

(b) By Lemma 6.10 at $n + e_j$,

$$[z^n]\,\mathfrak{B}(a_j \psi) = \frac{\sqrt{n_j + 1}\,\psi(n + e_j)}{\sqrt{n!}} = (n_j + 1)\,\frac{\psi(n + e_j)}{\sqrt{(n + e_j)!}},$$

which is $[z^n]\,(\partial_j \mathfrak{B}\psi)$.

(c) $1/\sqrt{n!}$ is a complex scalar and commutes with X.

(d) $[z^0]\,\mathfrak{B}|0\rangle = 1/\sqrt{0!} = 1$, and all other coefficients vanish. □

For $X \in R$, $j \in \iota$ and $p \geq 1$, the formal exponential $e^{Xz_j^p}$ is the series $\sum_{k \geq 0} X^k z_j^{pk}/k!$ in the single variable z_j.

Theorem 6.13 (Exponentials in the Bargmann picture)**.** Lean ✓ *For $X \in R$, $j \in \iota$, $p \geq 1$ and every $\psi \in \mathcal{F}(\iota, R)$: $\mathfrak{B}\big(e^{X \otimes (a_j^\dagger)^p} \psi\big) = e^{Xz_j^p}\,\mathfrak{B}\psi$.*

Proof. Write $T = X(a_j^\dagger)^p$ and $f = \mathfrak{B}\psi$. By Theorem 6.12, and since z_j commutes with every series, induction on k gives $\mathfrak{B}(T^k \psi) = X^k z_j^{pk} f$. Fix n. Then

$$[z^n]\,\mathfrak{B}(e^T \psi) = \sum_{k=0}^{|n|} \frac{1}{k!} [z^n]\,(X^k z_j^{pk} f).$$

The coefficient of a product Pf at n involves only the coefficients of P at multi-indices $n' \leq n$. For such n', the coefficient of $e^{Xz_j^p}$ at n' equals that of the polynomial $\sum_{k \leq |n|} X^k z_j^{pk}/k!$: a term with $k > |n|$ sits at a multi-index of total degree $pk \geq k > |n| \geq |n'|$. Hence $[z^n]\,(e^{Xz_j^p} f)$ is the same finite sum. □

The truncation in Definition 6.6 therefore cuts only terms whose coefficient at n vanishes.

Notes and references

The formal Fock space and the exponential summed coefficient by coefficient are the route of these notes, chosen so that no topology is needed. In algebraic terms, an operator that raises the particle number is topologically nilpotent for the filtration by particle number, under which $\mathcal{F}(\iota, R)$ is complete, and e^T is its exponential, as e^f is for a formal power series f without constant term. The construction is modelled on the completion $\prod_k \otimes_s^k \mathfrak{h}$ of the finite-particle space and on $R[[z]]$ as a module over the Weyl algebra, on which $a_j^\dagger$ and a_j act as z_j and ∂_j. This realization of the creation and annihilation operators goes back to Fock [25]; Bargmann [7] introduced the Hilbert space of entire functions on which multiplication by z_j and ∂_j are adjoint to each other. Normal and anti-normal ordering of products of a and $a^\dagger$ are standard; for ordered expansions of operators see Cahill and Glauber [33]. Proposition 6.5 is the identity used in Example 1 of Tjoa and Cirac [4], and Theorem 6.4 is the reduction behind their remark after Proposition 2 that the proposition extends to all polynomials in a and $a^\dagger$.

7 The bosonic Fock space and closed forms

The bosonic Fock space is the space of square-summable vectors of the formal Fock space of Section 6. A linear map whose coefficients are finite linear combinations of coefficients acts on every formal vector, and its restriction to the vectors of the Fock space that it maps back into the Fock space is a closed operator. Finite sums of squared norms of such maps are therefore closed quadratic forms in the sense of Kato [8, VI §1.3], the setting of the representation theorem for self-adjoint operators [8, VI §2]; the parent forms of the bosonic matrix product states of [4] are of this kind, and, where they are densely defined, their operators are the parent Hamiltonians.

Notation for this section.

$\mathbb{N}^{(\iota)}$	the multi-indices $n = (n_i)_{i\in\iota}$, finitely many nonzero (Section 6)
$\|\psi\|^2$	squared norm of a formal vector, in $[0,\infty]$ (Definition 7.1)
$\Gamma_s(\iota)$, $\langle\varphi,\psi\rangle$, $\jmath$	bosonic Fock space, its inner product, its embedding into $\mathcal{F}(\iota,\mathbb{C})$ (Definition 7.1)
$\mathcal{S}_n$	a finite set of multi-indices (Definition 7.3)
$\bigotimes_j \chi_j$, $\psi_{j,n}$, S_j	product vector, mode-j fiber, a single-mode map S acting on mode j (Definition 7.4)
$\bar{T}$, $\operatorname{dom}\bar{T}$	maximal realization of T and its domain (Definition 7.6)
$(T_k)_{k\in\mathcal{K}}$, $\operatorname{dom}\mathfrak{q}_\Gamma$, $\mathfrak{q}_\Gamma$	a finite family of maps, its form domain and its form (Definition 7.8)
$\lvert q,\kappa,\ell\rangle$, $\mathcal{P}_{\kappa,\ell}$	photon-added Gaussian state and photon-added class (Definition 7.12)

In this section ι is any set of modes unless it is said to be finite, T and T' are linear maps on $\mathcal{F}(\iota,\mathbb{C})$, S is a linear map on the single-mode space $\mathcal{F}(\{1\},\mathbb{C})$, whose multi-indices we identify with $\mathbb{N}$, and p is a polynomial, not the exponent $p \geq 1$ of Section 6.

7.1 The bosonic Fock space

Definition 7.1 (Squared norm; bosonic Fock space)**.** Let ι be a set of modes.

(a) For $\psi \in \mathcal{F}(\iota,\mathbb{C})$ the *squared norm* is $\|\psi\|^2 = \sum_n |\psi(n)|^2 \in [0,\infty]$, the sum of the non-negative terms over all multi-indices.

(b) The *bosonic Fock space* $\Gamma_s(\iota) = \ell^2(\mathbb{N}^{(\iota)})$ is the Hilbert space of the square-summable families $(\psi(n))_n$ over the set $\mathbb{N}^{(\iota)}$ of multi-indices n, with the inner product $\langle\varphi,\psi\rangle = \sum_n \overline{\varphi(n)}\,\psi(n)$, antilinear in φ. The *embedding* $\jmath : \Gamma_s(\iota) \to \mathcal{F}(\iota,\mathbb{C})$ is $\jmath(\psi)(n) = \psi(n)$.

The number states $|n\rangle$ form an orthonormal basis of $\Gamma_s(\iota)$. For finite ι this is the space $\Gamma_s(\mathfrak{h})$, $\mathfrak{h} = \mathbb{C}^\iota$, of Section 6 written in the number basis; for infinite ι it is in the same way the Fock space over $\ell^2(\iota)$ [5]. The formal vector with the coefficients $\sqrt{(2k)!}/k!$ at $|2k\rangle$ of Section 6 has infinite squared norm.

Lemma 7.2. Lean ✓ *Let ι be a set of modes.*

(a) *$\jmath$ is linear and injective, and $\|\jmath\psi\|^2 = \|\psi\|^2$ for $\psi \in \Gamma_s(\iota)$, the right side being the squared Hilbert norm.*

(b) *A formal vector $\varphi \in \mathcal{F}(\iota, \mathbb{C})$ lies in $\jmath(\Gamma_s(\iota))$ if and only if $\|\varphi\|^2 < \infty$.*

(c) *The finitely supported vectors, the finite linear combinations of number states, are dense in $\Gamma_s(\iota)$.*

Proof. (a) Linearity holds coefficientwise, and $\jmath\psi$ has the coefficients of ψ. The squared Hilbert norm $\sum_n |\psi(n)|^2$ is the real sum of a summable non-negative family, which equals its sum in $[0, \infty]$.

(b) A sum of non-negative terms in $[0, \infty]$ is finite exactly when the family of terms is summable, that is, when φ is square-summable.

(c) For $\psi \in \Gamma_s(\iota)$ and a finite set F of multi-indices, $\|\psi - \sum_{n \in F} \psi(n)|n\rangle\|^2 = \sum_{n \notin F} |\psi(n)|^2$, which is as small as we like for F large, by summability. □

7.2 Coefficient-finite maps and maximal realizations

Definition 7.3 (Coefficient-finite)**.** A linear map T on $\mathcal{F}(\iota, \mathbb{C})$ is *coefficient-finite* if for every multi-index n there is a finite set $\mathcal{S}_n$ of multi-indices such that $(T\psi)(n) = (T\varphi)(n)$ whenever $\psi(m) = \varphi(m)$ for every $m \in \mathcal{S}_n$.

Since ψ agrees on $\mathcal{S}_n$ with $\sum_{m \in \mathcal{S}_n} \psi(m)|m\rangle$, linearity gives $(T\psi)(n) = \sum_{m \in \mathcal{S}_n} T_{nm}\, \psi(m)$ with $T_{nm} = (T|m\rangle)(n)$. A coefficient-finite map is thus one whose matrix in the number basis is *row-finite*, and every row-finite matrix defines a coefficient-finite map by this formula. The ladder operators are of this kind, and so is every exponential of Definition 6.6 on finitely many modes.

Definition 7.4 (Product vectors; operators acting on one mode)**.** Let ι be a set of modes.

(a) For finite ι and $\chi_j \in \mathcal{F}(\{1\}, \mathbb{C})$, $j \in \iota$, the *product vector* $\bigotimes_j \chi_j \in \mathcal{F}(\iota, \mathbb{C})$ has the coefficients $(\bigotimes_j \chi_j)(n) = \prod_j \chi_j(n_j)$.

(b) For $\psi \in \mathcal{F}(\iota, \mathbb{C})$, $j \in \iota$ and a multi-index n, the *mode-j fiber* $\psi_{j,n} \in \mathcal{F}(\{1\}, \mathbb{C})$ is $\psi_{j,n}(m) = \psi(n + (m - n_j)e_j)$, the coefficient at n with n_j replaced by m.

(c) For a linear map S on $\mathcal{F}(\{1\}, \mathbb{C})$ and $j \in \iota$, the *map S acting on mode j* is the linear map S_j on $\mathcal{F}(\iota, \mathbb{C})$ with $(S_j\psi)(n) = (S\psi_{j,n})(n_j)$.

For example, a acting on mode j is a_j: $(a\psi_{j,n})(n_j) = \sqrt{n_j + 1}\, \psi(n + e_j)$. The fiber does not depend on n_j, and the fiber of a product vector along j is a multiple of its factor at j.

Lemma 7.5 (Coefficient-finite maps)**.** Lean ✓ *Let ι be a set of modes.*

(a) *The maps 0, 1, a_j and $a_j^\dagger$ are coefficient-finite. If T and T' are coefficient-finite, so are $T+T'$, TT', cT for $c \in \mathbb{C}$, every power T^k, every finite sum of coefficient-finite maps and $p(T)$ for every polynomial $p \in \mathbb{C}[X]$. If S is a coefficient-finite map on $\mathcal{F}(\{1\}, \mathbb{C})$, then S_j is coefficient-finite for every $j \in \iota$. If T is coefficient-finite and raises the particle number, then e^T is coefficient-finite.*

(b) *If ι is finite, every linear map on $\mathcal{F}(\iota, \mathbb{C})$ that raises the particle number is coefficient-finite.*

(c) *If T is coefficient-finite, then for every multi-index n the map $\psi \mapsto (T\jmath\psi)(n)$ is continuous on $\Gamma_s(\iota)$.*

Proof. (a) For 0 and 1 take $\mathcal{S}_n = \emptyset$ and $\{n\}$; for a_j take $\{n + e_j\}$, and for $a_j^\dagger$ take $\{n - e_j\}$ if $n_j \geq 1$ and $\emptyset$ otherwise (Definition 6.2). For $T + T'$ and cT the union of the two sets serves. For TT', $(TT'\psi)(n)$ depends on the coefficients $(T'\psi)(m)$, $m \in \mathcal{S}_n^T$, each of which depends on ψ on $\mathcal{S}_m^{T'}$; the union of these finitely many finite sets serves. Powers follow by induction, and finite sums and $p(T)$ by the first claims. For S_j, $(S_j\psi)(n)$ depends on $\psi_{j,n}$ at the finitely many $m \in \mathcal{S}_{n_j}^S$, that is, on ψ at the multi-indices $n + (m - n_j)e_j$. For e^T, $(e^T\psi)(n) = \sum_{r \leq |n|} (T^r\psi)(n)/r!$ (Definition 6.6) depends on ψ on the union of the sets of the maps T^r, $r \leq |n|$.

(b) By Lemma 6.7(b), $(T\psi)(n)$ depends only on the coefficients $\psi(n')$ with $|n'| < |n|$, and for finite ι there are finitely many such n'. In particular every polynomial in the creation operators with zero constant term is coefficient-finite (Lemma 6.7(c)).

(c) $(T\jmath\psi)(n) = \sum_{m \in \mathcal{S}_n} T_{nm}\, \psi(m)$ is a finite linear combination of coordinates, and $|\psi(m) - \psi'(m)| \leq \|\psi - \psi'\|$ in $\Gamma_s(\iota)$. □

An operator A on $\Gamma_s(\iota)$, defined on a subspace $\operatorname{dom} A$, is *closed* if its graph $\{(\psi, A\psi) : \psi \in \operatorname{dom} A\}$ is closed in $\Gamma_s(\iota) \times \Gamma_s(\iota)$, and *densely defined* if $\operatorname{dom} A$ is dense [8, III §5].

Definition 7.6 (Maximal realization)**.** For a linear map T on $\mathcal{F}(\iota, \mathbb{C})$, the *maximal realization* $\bar{T}$ is the operator on $\Gamma_s(\iota)$ with the domain $\operatorname{dom} \bar{T} = \{\psi \in \Gamma_s(\iota) : \|T\jmath\psi\|^2 < \infty\}$ and $\bar{T}\psi$ the vector with $\jmath(\bar{T}\psi) = T\jmath\psi$ (Lemma 7.2(b)).

The domain is a subspace, since a sum of square-summable vectors is square-summable. For a row-finite matrix, $\bar{T}$ is the maximal operator of the matrix on ℓ^2: the largest operator on $\Gamma_s(\iota)$ that acts on each vector of its domain by the matrix. No domain has to be chosen.

Theorem 7.7 (Closedness of the maximal realization)**.** Lean ✓ *Let T be coefficient-finite. Then $\bar{T}$ is closed: if $\psi^{(r)} \in \operatorname{dom} \bar{T}$, $\psi^{(r)} \to \psi$ and $\bar{T}\psi^{(r)} \to \chi$ in $\Gamma_s(\iota)$, then $\psi \in \operatorname{dom} \bar{T}$ and $\bar{T}\psi = \chi$.*

Proof. Fix n. By Lemma 7.5(c), $(T\jmath\psi^{(r)})(n) \to (T\jmath\psi)(n)$, and since coordinates are continuous on $\Gamma_s(\iota)$, $(T\jmath\psi^{(r)})(n) = (\bar{T}\psi^{(r)})(n) \to \chi(n)$. Hence $T\jmath\psi = \jmath\chi$, which has finite squared norm; so $\psi \in \operatorname{dom} \bar{T}$ and $\bar{T}\psi = \chi$. □

7.3 Closed forms

We use the forms of [8, VI §1] with the arguments in the order of the inner product. A *form* $\mathfrak{t}$ on $\Gamma_s(\iota)$ is a map $\mathfrak{t}(\varphi, \psi)$ on $\operatorname{dom}\mathfrak{t} \times \operatorname{dom}\mathfrak{t}$, for a subspace $\operatorname{dom}\mathfrak{t}$, the *form domain*, antilinear in φ and linear in ψ; it is *Hermitian* if $\overline{\mathfrak{t}(\varphi, \psi)} = \mathfrak{t}(\psi, \varphi)$ and *non-negative* if $\mathfrak{t}(\psi, \psi) \geq 0$. A non-negative Hermitian form is *closed* if $\psi^{(r)} \in \operatorname{dom}\mathfrak{t}$, $\psi^{(r)} \to \psi$ and $\mathfrak{t}(\psi^{(r)} - \psi^{(s)}, \psi^{(r)} - \psi^{(s)}) \to 0$ as $r, s \to \infty$ imply $\psi \in \operatorname{dom}\mathfrak{t}$ and $\mathfrak{t}(\psi^{(r)} - \psi, \psi^{(r)} - \psi) \to 0$ [8, VI §1.3]; equivalently, $\operatorname{dom}\mathfrak{t}$ is complete in the norm $(\|\psi\|^2 + \mathfrak{t}(\psi, \psi))^{1/2}$ [8, VI, Theorem 1.11]. It is *densely defined* if $\operatorname{dom}\mathfrak{t}$ is dense, and a *form core* is a subspace of $\operatorname{dom}\mathfrak{t}$ dense in that norm.

Definition 7.8 (The form of a finite family)**.** Let $(T_k)_{k\in\mathcal{K}}$ be a finite family of linear maps on $\mathcal{F}(\iota,\mathbb{C})$. Its *form domain* is $\operatorname{dom}\mathfrak{q}_\Gamma = \bigcap_k \operatorname{dom}\bar{T}_k$, and its *form* is

$$\mathfrak{q}_\Gamma(\varphi,\psi) = \sum_{k\in\mathcal{K}} \langle \bar{T}_k\varphi, \bar{T}_k\psi\rangle, \qquad \varphi,\psi\in\operatorname{dom}\mathfrak{q}_\Gamma.$$

On the formal Fock space the same family has the *formal form* $\sum_k\|T_k\varphi\|^2\in[0,\infty]$, defined for every $\varphi\in\mathcal{F}(\iota,\mathbb{C})$. The next lemma identifies $\mathfrak{q}_\Gamma$ with the formal form read on $\Gamma_s(\iota)$.

Lemma 7.9 (Form domain and form)**.** Lean ✓ *Let $(T_k)_{k\in\mathcal{K}}$ be a finite family of linear maps on $\mathcal{F}(\iota,\mathbb{C})$.*

(a) *$\operatorname{dom}\mathfrak{q}_\Gamma$ is a subspace of $\Gamma_s(\iota)$, and $\operatorname{dom}\mathfrak{q}_\Gamma = \{\psi\in\Gamma_s(\iota) : \sum_k\|T_k\jmath\psi\|^2<\infty\}$.*

(b) *$\mathfrak{q}_\Gamma$ is antilinear in φ, linear in ψ and Hermitian, and on the diagonal it is real and equals the formal form: $\mathfrak{q}_\Gamma(\psi,\psi)=\sum_k\|T_k\jmath\psi\|^2$ for $\psi\in\operatorname{dom}\mathfrak{q}_\Gamma$. In particular $\mathfrak{q}_\Gamma$ is non-negative.*

Proof. (a) An intersection of subspaces is a subspace, and a finite sum in $[0,\infty]$ is finite exactly when each term is.

(b) Each term is the inner product composed with the linear map $\bar{T}_k$ in both arguments, which gives the linearity and $\overline{\langle \bar{T}_k\varphi,\bar{T}_k\psi\rangle} = \langle \bar{T}_k\psi,\bar{T}_k\varphi\rangle$. On the diagonal, by Lemma 7.2(a),

$$\langle \bar{T}_k\psi,\bar{T}_k\psi\rangle = \|\bar{T}_k\psi\|^2 = \|T_k\jmath\psi\|^2. \qquad \square$$

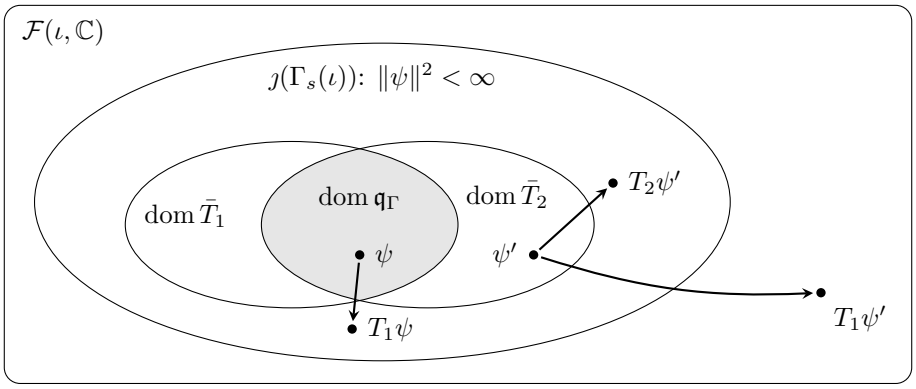


Figure 3: The form domain of a family T_1, T_2 (Definition 7.8). The maps act on the whole formal Fock space; the domain of the maximal realization $\bar{T}_k$ consists of the vectors of $\Gamma_s(\iota)$ that T_k maps back into $\Gamma_s(\iota)$, and the form domain is the intersection of these domains. Here ψ' lies in $\operatorname{dom}\bar{T}_2$ but not in $\operatorname{dom}\bar{T}_1$: $T_2\psi'$ is square-summable and $T_1\psi'$ is not.

Theorem 7.10 (Closed form)**.** Lean ✓ *Let $(T_k)_{k\in\mathcal{K}}$ be a finite family of coefficient-finite maps on $\mathcal{F}(\iota,\mathbb{C})$. If $\psi^{(r)}\in\operatorname{dom}\mathfrak{q}_\Gamma$, $\psi^{(r)}\to\psi$ in $\Gamma_s(\iota)$ and $\sum_k\|T_k\jmath(\psi^{(r)}-\psi^{(s)})\|^2\to 0$ as $r,s\to\infty$, then $\psi\in\operatorname{dom}\mathfrak{q}_\Gamma$ and $\sum_k\|T_k\jmath(\psi^{(r)}-\psi)\|^2\to 0$. So $\mathfrak{q}_\Gamma$ is closed.*

Proof. Each term of the sum tends to zero, and $\|T_k\jmath(\psi^{(r)}-\psi^{(s)})\| = \|\bar{T}_k\psi^{(r)}-\bar{T}_k\psi^{(s)}\|$ by linearity and Lemma 7.2(a), so each sequence $\bar{T}_k\psi^{(r)}$ is Cauchy in $\Gamma_s(\iota)$ and converges to some χ_k. By Theorem 7.7, $\psi\in\operatorname{dom}\bar{T}_k$ and $\bar{T}_k\psi=\chi_k$. Hence $\psi\in\operatorname{dom}\mathfrak{q}_\Gamma$, and $\|T_k\jmath(\psi^{(r)}-\psi)\|^2 = \|\bar{T}_k\psi^{(r)}-\chi_k\|^2\to 0$ for each of the finitely many k. By Lemma 7.9(b) this is closedness in the sense above. $\square$

The theorem applies two results of Kato to the closed operators $\bar{T}_k$ of Theorem 7.7: the form $\|T\psi\|^2$ of a closed operator T, on its domain, is closed [8], and a finite sum of closed non-negative forms, on the intersection of their domains, is closed [8]; the proof above checks this directly. Closedness needs no hypothesis beyond coefficient-finiteness; whether $\mathfrak{q}_\Gamma$ is densely defined depends on the family.

7.4 Product vectors and the photon-added class

Lemma 7.11 (Squared norm of a product vector). Lean ✓

(a) *For a finite set ι, a set A and functions $g_i : A \to [0,\infty]$, $i \in \iota$,*

$$\sum_{f:\iota\to A} \prod_{i\in\iota} g_i(f(i)) = \prod_{i\in\iota} \sum_{a\in A} g_i(a) \quad \text{in } [0,\infty],$$

with the convention $0 \cdot \infty = 0$.

(b) *For finite ι and $\chi_j \in \mathcal{F}(\{1\},\mathbb{C})$, $\|\bigotimes_j \chi_j\|^2 = \prod_j \|\chi_j\|^2$ in $[0,\infty]$.*

Proof. (a) Induction on the number of elements of ι. For $\iota = \emptyset$ both sides are 1. Adding an element i_0, the functions on $\iota \cup \{i_0\}$ are the pairs of a value a at i_0 and a function on ι; sums of non-negative terms in $[0,\infty]$ may be taken in any order and are multiplied termwise by a constant factor, so the sum is $\sum_a g_{i_0}(a)$ times the sum over ι.

(b) On finitely many modes the multi-indices are the functions $\iota \to \mathbb{N}$, and $|\prod_j \chi_j(n_j)|^2 = \prod_j |\chi_j(n_j)|^2$; apply (a) with $A = \mathbb{N}$. □

For a single mode and $\kappa, \ell \in \mathbb{C}$, the map $\kappa(a^\dagger)^2 + \ell a^\dagger$ raises the particle number (Lemma 6.7(c)); its exponential $e^{\kappa(a^\dagger)^2+\ell a^\dagger}$ is the *Gaussian operator*; the name is used here only for these exponentials of creation operators, not for Gaussian unitaries.

Definition 7.12 (Photon-added Gaussian states; the photon-added class). For $q \in \mathbb{N}$ and $\kappa, \ell \in \mathbb{C}$, the *photon-added Gaussian state* is $|q,\kappa,\ell\rangle = (a^\dagger)^q\, e^{\kappa(a^\dagger)^2+\ell a^\dagger}|0\rangle \in \mathcal{F}(\{1\},\mathbb{C})$, and the *photon-added class* is $\mathcal{P}_{\kappa,\ell} = \operatorname{span}\{|q,\kappa,\ell\rangle : q \in \mathbb{N}\}$.

The state $|0,\kappa,\ell\rangle$ is a formal vector, which for $|\kappa| < \frac{1}{2}$ is a multiple of a squeezed coherent state; the states $|q,\kappa,\ell\rangle$ are the photon-added Gaussian states of [4, Sec. II A 1]; for $\kappa = 0$ they are multiples of the photon-added coherent states of Agarwal and Tara [26]. The maps $\kappa(a^\dagger)^2$ and $\ell a^\dagger$ commute, so by Theorem 6.8 the Gaussian operator is $e^{\kappa(a^\dagger)^2} e^{\ell a^\dagger}$, and by Theorems 6.13 and 6.12 the Bargmann series of $|q,\kappa,\ell\rangle$ is $z^q E$ with $E = e^{\kappa z^2} e^{\ell z}$. So $\mathcal{P}_{\kappa,\ell}$ consists of the formal vectors whose Bargmann series are $p(z)E$, p a polynomial.

Lemma 7.13 (The photon-added class). Lean ✓ *Let $\kappa, \ell \in \mathbb{C}$.*

(a) *$|q,\kappa,\ell\rangle = (a^\dagger)^q|0,\kappa,\ell\rangle$, and $\mathcal{P}_{\kappa,\ell} = \{p(a^\dagger)|0,\kappa,\ell\rangle : p \in \mathbb{C}[X]\}$.*

(b) *$\mathcal{P}_{\kappa,\ell}$ is invariant under a and under $a^\dagger$, hence under $p(a)$ and under $p(a^\dagger)$ for every polynomial p.*

(c) *The Gaussian operator $e^{\kappa(a^\dagger)^2+\ell a^\dagger}$ maps $\mathcal{P}_{0,0}$ into $\mathcal{P}_{\kappa,\ell}$.*

(d) *$|q,0,0\rangle = \sqrt{q!}\,|q\rangle$ for every q, so every number state lies in $\mathcal{P}_{0,0}$.*

Proof. The Bargmann map is injective, so identities may be checked on Bargmann series, where $a^\dagger$ is multiplication by z, a is the derivative ∂ and $p(a^\dagger)$ is multiplication by p (Theorem 6.12).

(a) The first claim is the definition, and a linear combination of the $(a^\dagger)^q|0,\kappa,\ell\rangle$ is $p(a^\dagger)|0,\kappa,\ell\rangle$ for the polynomial with these coefficients.

(b) $a^\dagger p(a^\dagger) = (Xp)(a^\dagger)$. For a: from $e^{cz^m} = \sum_k c^k z^{mk}/k!$ the coefficientwise derivative is $\partial e^{cz^m} = mc\, z^{m-1} e^{cz^m}$ for $m \geq 1$, so the Leibniz rule gives $\partial E = (2\kappa z + \ell)E$ and $\partial(pE) = (p' + (2\kappa z + \ell)p)E$. Hence $a\, p(a^\dagger)|0,\kappa,\ell\rangle = (p' + (2\kappa X + \ell)p)(a^\dagger)|0,\kappa,\ell\rangle$. A subspace invariant under a map is invariant under its powers and their linear combinations.

(c) The Bargmann series of $e^{\kappa(a^\dagger)^2+\ell a^\dagger} p(a^\dagger)|0\rangle$ is $E\,p$, that of $p(a^\dagger)|0,\kappa,\ell\rangle$.

(d) The exponential of the zero map is the identity (Definition 6.6), so $|q,0,0\rangle = (a^\dagger)^q|0\rangle = \sqrt{q!}\,|q\rangle$ (after Theorem 6.3). □

By (b) the class is also invariant under every finite sum of products of a and $a^\dagger$. It contains no information about norms: $|0,\kappa,0\rangle$ is square-summable exactly when $|\kappa| < \frac{1}{2}$, its squared moduli being $\binom{2k}{k}|\kappa|^{2k}$ at $|2k\rangle$.

Notes and references

Closed forms, form cores and the representation theorems that turn a closed, densely defined, non-negative form into a self-adjoint operator are those of Kato [8, VI §§1–2]. The forms here are built from maximal realizations, which act on every vector they can, so no dense domain has to be chosen and closedness reduces to the continuity of the coefficients (Lemma 7.5(c)). The usual construction starts from the operators on a dense domain $\mathcal{D}$, such as the finitely supported vectors, and takes the closure of the form $\sum_k \|T_k\psi\|^2$ on $\mathcal{D}$. When $\mathcal{D} \subseteq \operatorname{dom}\mathfrak{q}_\Gamma$, that closure is the restriction of the closed form $\mathfrak{q}_\Gamma$ of Theorem 7.10 to the closure of $\mathcal{D}$ in the norm $(\|\psi\|^2 + \mathfrak{q}_\Gamma(\psi,\psi))^{1/2}$, and the two forms agree exactly when $\mathcal{D}$ is a form core.

8 Second quantization of a matrix

A complex matrix g on the one-particle space $\mathbb{C}^\iota$ acts on states of p particles as $g^{\otimes p}$. In the Bargmann picture of Section 6 this action is the linear change of variables $f(z) \mapsto f(g^T z)$, which is defined on the whole formal Fock space for every matrix g: substitution of linear forms preserves degrees, so each coefficient of the image is a finite sum. For unitary g the map restricts to a strongly continuous unitary representation on the bosonic Fock space of Section 7, and the second quantization of e^h is the exponential, summed coefficient by coefficient, of the quadratic operator $\sum_{j,k} h_{jk} a_j^\dagger a_k$. For $h = i\mathsf{M}$ with M Hermitian, on the bosonic Fock space this exponential is e^{iA_M} with A_M self-adjoint, so the passive linear-optical unitaries of Tjoa and Cirac [4] are the unitary operators $\mathsf{U}_\mathcal{U}$ obtained in this way.

Notation for this section.

ι, N	a finite set of $N \geq 1$ modes
$M_\iota(\mathbb{C})$; g, h, M	complex $\iota \times \iota$ matrices; elements of $M_\iota(\mathbb{C})$, M Hermitian where stated
g^T, g^*, e^h, I	transpose, conjugate transpose, matrix exponential $\sum_r h^r/r!$, identity matrix
$\mathrm{U}(\iota)$; $\mathcal{U}$, $\mathcal{V}$	the group of unitary matrices, with the topology of the entries; its elements
$\mathcal{F}$, $\mathcal{F}_p$, $[\psi]_p$	$\mathcal{F}(\iota,\mathbb{C})$, its p-particle subspace, the component of ψ in it (Definition 8.1)
n_{tot}, $\langle\varphi,\psi\rangle$	total number operator; pairing (Definition 8.1)
$\ell_j(g)$, $f(g^T z)$	linear forms and substitution (8.1)
$\Gamma(g)$, $\Gamma(g)_{mn}$	second quantization of g and its matrix elements (Definition 8.2)
$\mathsf{U}_\mathcal{U}$	the unitary operator on $\Gamma_s(\iota)$ given by $\Gamma(\mathcal{U})$, the passive unitary (Theorem 8.7)
$d\Gamma(h)$, e^T	second quantization of a generator, coefficientwise exponential (Definition 8.8)
$\|h\|_1$	$\sum_{j,k}\|h_{jk}\|$ (Lemma 8.9)
O, $O^\mathcal{U}$, $\bar{O}$	a linear map of $\mathcal{F}$, its conjugate $\Gamma(\mathcal{U}^*)\,O\,\Gamma(\mathcal{U})$, its maximal realization (Definition 7.6)
A_M, $A_{\mathsf{M},p}$	the closure of $d\Gamma(\mathsf{M})$, its restriction to $\mathcal{F}_p$ (Proposition 8.12)

In this section f, f_1, f_2 are formal power series in $(z_i)_{i\in\iota}$ over $\mathbb{C}$, p is a particle number, r, s, t are summation indices (t is a real parameter after Proposition 8.12), $[f]_p$ is the homogeneous component of degree p of f, and δ is a derivation. The letter Γ carries three standard meanings, kept apart by their arguments: $\Gamma(g)$, $d\Gamma(h)$ and the Hilbert space $\Gamma_s(\iota)$.

8.1 The definition and particle-number sectors

Throughout, ι is a finite set of $N \geq 1$ modes, and $\mathcal{F} = \mathcal{F}(\iota, \mathbb{C})$. The particle-number grading of Section 6 enters every argument below, so we name its pieces.

Definition 8.1 (Particle-number sectors; total number operator; pairing)**.** (a) For $p \in \mathbb{N}$ and $\psi \in \mathcal{F}$, the *component of ψ with p particles* is $[\psi]_p$, with $[\psi]_p(n) = \psi(n)$ if $|n| = p$ and $[\psi]_p(n) = 0$ otherwise. The *particle-number sector* $\mathcal{F}_p$, or *p-particle subspace*, consists of the ψ with $\psi = [\psi]_p$.

(b) The *total number operator* is $n_{\mathrm{tot}} = \sum_{j\in\iota} a_j^\dagger a_j$ on $\mathcal{F}$.

(c) For a finitely supported $\varphi \in \mathcal{F}$ and any $\psi \in \mathcal{F}$, the *pairing* is the finite sum $\langle \varphi, \psi\rangle = \sum_n \overline{\varphi(n)}\, \psi(n)$, antilinear in φ and linear in ψ.

By Definition 6.2, $(a_j^\dagger a_j \psi)(n) = n_j \psi(n)$, so $(n_{\mathrm{tot}}\psi)(n) = |n|\, \psi(n)$. The sector $\mathcal{F}_p$ is finite-dimensional, with basis the number states $|n\rangle$, $|n| = p$, and the Bargmann map sends $[\psi]_p$ to the homogeneous component of degree p of $\mathfrak{B}\psi$. If $\varphi = \jmath\varphi'$ and $\psi = \jmath\psi'$ with $\varphi', \psi' \in \Gamma_s(\iota)$ and φ finitely supported, the pairing is the inner product $\langle \varphi', \psi'\rangle$ of Definition 7.1; for φ not finitely supported we use only the inner product of $\Gamma_s(\iota)$.

For $g \in M_\iota(\mathbb{C})$ and $j \in \iota$ we write $\ell_j(g) = (g^T z)_j = \sum_k g_{kj} z_k$, a linear form, and for $f \in \mathbb{C}[[z]]$ we write $f(g^T z)$ for the series obtained by substituting $\ell_j(g)$ for z_j:

$$[z^m]\, f(g^T z) = \sum_{|d|=|m|} [z^d]\, f\, [z^m]\, \ell(g)^d, \qquad \ell(g)^d = \prod_j \ell_j(g)^{d_j}. \tag{8.1}$$

Since $\ell(g)^d$ is a homogeneous polynomial of degree $|d|$, only the finitely many d with $|d| = |m|$ contribute, and (8.1) is the substitution of formal power series without constant term. It is a unital algebra endomorphism of $\mathbb{C}[[z]]$ that sends z_j to $\ell_j(g)$.

Write $[f]_p$ for the homogeneous component of degree p of $f \in \mathbb{C}[[z]]$, a polynomial, so that $\mathfrak{B}[\psi]_p = [\mathfrak{B}\psi]_p$. A linear map T of $\mathbb{C}[[z]]$ *acts degree by degree* if $[Tf]_p = T[f]_p$ for every f and p; then the coefficient of Tf at m depends only on $[f]_{|m|}$. By (8.1) every substitution of linear forms acts degree by degree, and a composite of maps that act degree by degree does too. Two maps T, T' that act degree by degree and agree on every polynomial agree on $\mathbb{C}[[z]]$, since $[Tf]_p = T[f]_p = T'[f]_p = [T'f]_p$.

Definition 8.2 (Second quantization)**.** For $g \in M_\iota(\mathbb{C})$, the *second quantization* $\Gamma(g)$ is the linear map of $\mathcal{F}$ with

$$\mathfrak{B}\,\Gamma(g)\psi = (\mathfrak{B}\psi)(g^T z), \qquad \psi \in \mathcal{F}.$$

Its *matrix elements* are $\Gamma(g)_{mn} = (\Gamma(g)|n\rangle)(m)$.

The Bargmann map is a linear bijection of $\mathcal{F}$ onto $\mathbb{C}[[z]]$ (Definition 6.9), so the equation defines $\Gamma(g)$ on all of $\mathcal{F}$, for every complex matrix g.

Lemma 8.3 (Coefficients and sectors)**.** Lean ✓ *Let $g \in M_\iota(\mathbb{C})$.*

(a) *For every $\psi \in \mathcal{F}$ and every multi-index m,*

$$(\Gamma(g)\psi)(m) = \sum_{|n|=|m|} \Gamma(g)_{mn}\, \psi(n), \qquad \Gamma(g)_{mn} = \frac{\sqrt{m!}}{\sqrt{n!}}\, [z^m] \prod_j \ell_j(g)^{n_j},$$

a finite sum, and $\Gamma(g)_{mn} = 0$ *if* $|m| \neq |n|$. *For all* m, n *there is a polynomial in* N^2 *variables whose value at the entries of* g *is* $\Gamma(g)_{mn}$, *for every* g. *Moreover* $\Gamma(g)[\psi]_p = [\Gamma(g)\psi]_p$ *for every* p, $\Gamma(g)n_{\mathrm{tot}} = n_{\mathrm{tot}}\Gamma(g)$, *and* $\Gamma(g)$ *is coefficient-finite (Definition 7.3).*

(b) $\Gamma(g)(c|0\rangle) = c|0\rangle$ *for every* $c \in \mathbb{C}$, *and* $\Gamma(g)|e_j\rangle = \sum_k g_{kj}|e_k\rangle$ *for every* j.

(c) *If* φ *is finitely supported, so is* $\Gamma(g)\varphi$.

Proof. (a) Since $[z^n]\,\mathfrak{B}\psi = \psi(n)/\sqrt{n!}$, formula (8.1) gives

$$[z^m]\,\mathfrak{B}\,\Gamma(g)\psi = \sum_{|n|=|m|} \frac{\psi(n)}{\sqrt{n!}}\,[z^m]\,\ell(g)^n;$$

multiplying by $\sqrt{m!}$ gives the coefficient formula, and $\psi = |n\rangle$ gives the matrix elements, which vanish for $|n| \neq |m|$. Expanding the product of linear forms, $[z^m]\,\ell(g)^n$ is a polynomial with integer coefficients in the entries of g. The coefficient of $\Gamma(g)\psi$ at m involves only $[\psi]_{|m|}$, which gives $\Gamma(g)[\psi]_p = [\Gamma(g)\psi]_p$; since n_{tot} multiplies the coefficients at m and at every n with $|n| = |m|$ by the same number $|m|$, it commutes with $\Gamma(g)$. The finite set $\{n : |n| = |m|\}$ serves as $\mathcal{S}_m$ in Definition 7.3.

(b) $\mathfrak{B}(c|0\rangle) = c$ is a constant, which substitution fixes; and $\mathfrak{B}|e_j\rangle = z_j$ is sent to $\ell_j(g) = \sum_k g_{kj}z_k = \mathfrak{B}\sum_k g_{kj}|e_k\rangle$.

(c) A finitely supported vector has nonzero components in finitely many sectors; by (a), so does its image, and each sector is finite-dimensional. □

By (b), $\Gamma(g)$ acts on $\mathcal{F}_1 \cong \mathbb{C}^\iota$, $|e_j\rangle \leftrightarrow e_j$, as the matrix g itself. Standard texts define the second quantization of a contraction g as $\bigoplus_p g^{\otimes p}$ on the symmetric tensors [5]. The two definitions agree on every sector. Identify the symmetric tensors $\otimes^p_s\mathbb{C}^\iota$ with the homogeneous polynomials of degree p by sending the symmetrization of $e_{w_1} \otimes \cdots \otimes e_{w_p}$ to $z_{w_1} \cdots z_{w_p}$. Then $g^{\otimes p}$ sends this monomial to $\prod_q \ell_{w_q}(g)$, which is the substitution. The normalized symmetric tensor with occupation numbers n, $|n| = p$, corresponds to $(p!/n!)^{1/2}z^n = \sqrt{p!}\,\mathfrak{B}|n\rangle$, so the identification is $\sqrt{p!}\,\mathfrak{B}$ on $\mathcal{F}_p$, a multiple of $\mathfrak{B}$, under which both maps are the same substitution. The agreement is standard, and no later statement depends on it.

The standard $\Gamma(g)$ is bounded on $\Gamma_s(\iota)$ when g is a contraction; Definition 8.2 needs no such condition, but then $\Gamma(g)$ need not preserve $\Gamma_s(\iota)$. For one mode, $g = u \in \mathbb{C}$ and $\Gamma(u)|m\rangle = u^m|m\rangle$, that is, $\Gamma(u) = u^{n_{\mathrm{tot}}}$ with $0^0 = 1$; so $\Gamma(2)$ sends the square-summable vector with coefficients 2^{-m} to the vector with all coefficients 1. For $g = 0$, $\Gamma(0)\psi = \psi(0)|0\rangle$ is the projection onto the vacuum.

8.2 Representation, intertwining and adjoint

Theorem 8.4 (Representation). Lean ✓ $\Gamma(I) = 1$ *and* $\Gamma(gg') = \Gamma(g)\Gamma(g')$ *for all* $g, g' \in M_\iota(\mathbb{C})$, *so* $g \mapsto \Gamma(g)$ *is a monoid homomorphism from* $M_\iota(\mathbb{C})$ *to the linear maps of* $\mathcal{F}$. *For invertible* g, $\Gamma(g^{-1})\Gamma(g) = \Gamma(g)\Gamma(g^{-1}) = 1$.

Proof. For $g = I$ the linear forms are the variables, and substitution of the variables is the identity. Applying $\Gamma(g')$ and then $\Gamma(g)$ substitutes first $\ell_j(g')$ for z_j and then $\ell_k(g)$ for z_k. The composite is a unital algebra endomorphism of $\mathbb{C}[[z]]$ that acts degree by degree and sends z_j to

$$\ell_j(g')(g^T z) = \sum_k g'_{kj} \sum_l g_{lk}z_l = \sum_l (gg')_{lj}z_l = \ell_j(gg').$$

So it agrees with the substitution of $\ell(gg')$ on every polynomial, hence on $\mathbb{C}[[z]]$ by the remark after (8.1), and $\Gamma(g)\Gamma(g') = \Gamma(gg')$. The last statement follows from the first two. □

The ladder operators become multiplication and differentiation in the Bargmann picture (Theorem 6.12), so the substitution transforms them by the chain rule; this is how $\Gamma(g)$ intertwines them.

Theorem 8.5 (Intertwining). Lean ✓ *For every $g \in M_\iota(\mathbb{C})$ and every mode j, on all of $\mathcal{F}$,*

$$\Gamma(g)\, a_j^\dagger = \Big(\sum_k g_{kj} a_k^\dagger\Big)\Gamma(g), \qquad a_j\, \Gamma(g) = \Gamma(g)\Big(\sum_k g_{jk} a_k\Big).$$

The second identity rests on the chain rule $\partial_j[f(g^T z)] = \sum_k g_{jk}(\partial_k f)(g^T z)$, which holds for every $f \in \mathbb{C}[[z]]$.

Proof. Write $f = \mathfrak{B}\psi$. By Theorem 6.12(a), the first identity reads $(z_j f)(g^T z) = \ell_j(g)\, f(g^T z)$ under $\mathfrak{B}$, which holds because substitution is multiplicative, and $\ell_j(g) = \sum_k g_{kj} z_k$.

By Theorem 6.12(b), the second identity is the chain rule. Both of its sides are linear in f, and the coefficient of either side at m is a finite linear combination of the coefficients of the single homogeneous component $[f]_{|m|+1}$: the substitution acts degree by degree and each ∂_k lowers degrees by one. So it suffices to prove the chain rule for polynomials, and by linearity for $f = z^n$. With $\ell = \ell(g)$ and $\partial_j \ell_k = g_{jk}$, the product rule gives $\partial_j \ell^n = \sum_k n_k\, g_{jk}\, \ell^{n-e_k}$, while $(\partial_k z^n)(g^T z) = n_k \ell^{n-e_k}$; the terms with $n_k = 0$ vanish on both sides. □

Theorem 8.6 (Adjoint). Lean ✓ *Let $g \in M_\iota(\mathbb{C})$. For every finitely supported φ and every $\psi \in \mathcal{F}$, $\langle \varphi, \Gamma(g)\psi\rangle = \langle \Gamma(g^*)\varphi, \psi\rangle$, and $\Gamma(g^*)_{nm} = \overline{\Gamma(g)_{mn}}$ for all multi-indices m, n.*

Since $\Gamma(g^*)\varphi$ is finitely supported by Lemma 8.3(c), the right side of the first identity is defined.

Proof. We prove the identity of matrix elements by induction on $|m|$, for all n at once, and then sum. For $m = 0$, Lemma 8.3(b) gives $\Gamma(g^*)_{n0} = 1$ if $n = 0$ and 0 otherwise, and $\Gamma(g)_{0n}$ is the same by Lemma 8.3(a),(b).

If $m_i \geq 1$, apply the second identity of Theorem 8.5 to $|n\rangle$ and take the coefficient at $m - e_i$; apply the first identity, for g^*, whose entries are $(g^*)_{ki} = \overline{g_{ik}}$, to $|m - e_i\rangle$ and take the coefficient at n. With the formulas of Definition 6.2 this gives

$$\sqrt{m_i}\, \Gamma(g)_{mn} = \sum_{k:\, n_k \geq 1} g_{ik} \sqrt{n_k}\, \Gamma(g)_{m-e_i,\, n-e_k},$$
$$\sqrt{m_i}\, \Gamma(g^*)_{nm} = \sum_{k:\, n_k \geq 1} \overline{g_{ik}} \sqrt{n_k}\, \Gamma(g^*)_{n-e_k,\, m-e_i}.$$

By the induction hypothesis $\Gamma(g^*)_{n-e_k, m-e_i} = \overline{\Gamma(g)_{m-e_i, n-e_k}}$, so the second right side is the complex conjugate of the first, and dividing by $\sqrt{m_i} > 0$ gives $\Gamma(g^*)_{nm} = \overline{\Gamma(g)_{mn}}$.

For the pairing, let φ be supported in a finite set F. By Lemma 8.3(a) and the identity just proved, both $\langle \varphi, \Gamma(g)\psi\rangle$ and $\langle \Gamma(g^*)\varphi, \psi\rangle$ equal the finite double sum

$$\sum_{m \in F} \sum_{|n| = |m|} \overline{\varphi(m)}\, \Gamma(g)_{mn}\, \psi(n),$$

summed first over n for the left side and first over m for the right side. □

8.3 Unitary matrices

Theorem 8.7 (Unitarity and strong continuity). Lean ✓ *Let $\mathcal{U} \in \mathrm{U}(\iota)$.*

(a) $\|\Gamma(\mathcal{U})\psi\|^2 = \|\psi\|^2$ *in* $[0, \infty]$ *for every $\psi \in \mathcal{F}$; and $\|\psi\|^2 = \sum_p \|[\psi]_p\|^2$.*

(b) $\Gamma(\mathcal{U})$ *maps* $\jmath(\Gamma_s(\iota))$ *onto* $\jmath(\Gamma_s(\iota))$. *The operator* $\mathsf{U}_\mathcal{U}$ *on* $\Gamma_s(\iota)$ *with* $\jmath\,\mathsf{U}_\mathcal{U}\psi = \Gamma(\mathcal{U})\jmath\psi$ *is a surjective linear isometry, with* $\mathsf{U}_\mathcal{U}^{-1} = \mathsf{U}_{\mathcal{U}^*}$ *and* $\mathsf{U}_{\mathcal{U}\mathcal{V}} = \mathsf{U}_\mathcal{U}\mathsf{U}_\mathcal{V}$ *for* $\mathcal{V} \in \mathrm{U}(\iota)$. *The map* $\mathcal{U} \mapsto \mathsf{U}_\mathcal{U}$ *is a group homomorphism from* $\mathrm{U}(\iota)$ *to the unitary group of the bounded operators on* $\Gamma_s(\iota)$, *and* $\mathsf{U}_\mathcal{U}^* = \mathsf{U}_{\mathcal{U}^*}$.

(c) *On all of* $\mathcal{F}$,

$$\Gamma(\mathcal{U}^*)\,a_j\,\Gamma(\mathcal{U}) = \sum_k \mathcal{U}_{jk} a_k, \qquad \Gamma(\mathcal{U}^*)\,a_j^\dagger\,\Gamma(\mathcal{U}) = \sum_k \overline{\mathcal{U}_{jk}}\,a_k^\dagger,$$
$$\Gamma(\mathcal{U})\,a_j^\dagger\,\Gamma(\mathcal{U}^*) = \sum_k \mathcal{U}_{kj}\,a_k^\dagger, \qquad \Gamma(\mathcal{U})\,n_{\mathrm{tot}}\,\Gamma(\mathcal{U}^*) = n_{\mathrm{tot}}.$$

(d) *For every* $\psi \in \Gamma_s(\iota)$, *the map* $\mathcal{U} \mapsto \mathsf{U}_\mathcal{U}\psi$ *is continuous from* $\mathrm{U}(\iota)$ *to* $\Gamma_s(\iota)$.

Proof. (a) The squared norm is a sum of non-negative terms in $[0,\infty]$, which may be regrouped by particle number; this is the second identity. For a finitely supported χ, $\|\chi\|^2 = \langle\chi,\chi\rangle$. Take $\chi = [\psi]_p$, which is finitely supported; so is $\Gamma(\mathcal{U})\chi$ (Lemma 8.3(c)), and Theorem 8.6 with $\varphi = \Gamma(\mathcal{U})\chi$, followed by Theorem 8.4, gives

$$\|\Gamma(\mathcal{U})\chi\|^2 = \langle\Gamma(\mathcal{U}^*)\Gamma(\mathcal{U})\chi,\chi\rangle = \langle\Gamma(\mathcal{U}^*\mathcal{U})\chi,\chi\rangle = \|\chi\|^2.$$

Since $[\Gamma(\mathcal{U})\psi]_p = \Gamma(\mathcal{U})[\psi]_p$, summing over p gives $\|\Gamma(\mathcal{U})\psi\|^2 = \|\psi\|^2$.

(b) By (a) and Lemma 7.2(b), $\Gamma(\mathcal{U})$ and $\Gamma(\mathcal{U}^*)$ map $\jmath(\Gamma_s(\iota))$ into itself, and they are mutually inverse by Theorem 8.4, since $\mathcal{U}^*\mathcal{U} = \mathcal{U}\mathcal{U}^* = I$; so the map is onto, and $\mathsf{U}_\mathcal{U}^{-1} = \mathsf{U}_{\mathcal{U}^*}$. By Lemma 7.2(a), $\mathsf{U}_\mathcal{U}$ is an isometry, and $\mathsf{U}_{\mathcal{U}\mathcal{V}} = \mathsf{U}_\mathcal{U}\mathsf{U}_\mathcal{V}$ is Theorem 8.4. A surjective linear isometry of a Hilbert space is unitary, so $\mathsf{U}_\mathcal{U}^* = \mathsf{U}_\mathcal{U}^{-1} = \mathsf{U}_{\mathcal{U}^*}$.

(c) Multiply $a_j\Gamma(\mathcal{U}) = \Gamma(\mathcal{U})\sum_k \mathcal{U}_{jk}a_k$ (Theorem 8.5) on the left by $\Gamma(\mathcal{U}^*)$. Multiply the first identity of Theorem 8.5 for $\mathcal{U}^*$, whose entries are $(\mathcal{U}^*)_{kj} = \overline{\mathcal{U}_{jk}}$, on the right by $\Gamma(\mathcal{U})$, and the same identity for $\mathcal{U}$ on the right by $\Gamma(\mathcal{U}^*)$. Finally $\Gamma(\mathcal{U})n_{\mathrm{tot}}\Gamma(\mathcal{U}^*) = n_{\mathrm{tot}}\Gamma(\mathcal{U})\Gamma(\mathcal{U}^*) = n_{\mathrm{tot}}$ by Lemma 8.3(a). Each step uses $\Gamma(\mathcal{U}^*)\Gamma(\mathcal{U}) = \Gamma(\mathcal{U})\Gamma(\mathcal{U}^*) = 1$.

(d) For a number state, $\mathsf{U}_\mathcal{U}|m\rangle = \sum_{|n|=|m|}\Gamma(\mathcal{U})_{nm}|n\rangle$ is a finite sum whose coefficients are polynomials in the entries of $\mathcal{U}$ (Lemma 8.3(a)), so $\mathcal{U} \mapsto \mathsf{U}_\mathcal{U}\chi$ is continuous for every finitely supported χ. Let $\psi \in \Gamma_s(\iota)$, $\mathcal{U} \in \mathrm{U}(\iota)$ and $\varepsilon > 0$, and choose a finitely supported χ with $\|\psi - \chi\| < \varepsilon/3$ (Lemma 7.2(c)). Since every $\mathsf{U}_\mathcal{V}$ is an isometry,

$$\|\mathsf{U}_\mathcal{U}\psi - \mathsf{U}_\mathcal{V}\psi\| \le \|\mathsf{U}_\mathcal{U}(\psi-\chi)\| + \|\mathsf{U}_\mathcal{U}\chi - \mathsf{U}_\mathcal{V}\chi\| + \|\mathsf{U}_\mathcal{V}(\chi-\psi)\| < \tfrac{2}{3}\varepsilon + \|\mathsf{U}_\mathcal{U}\chi - \mathsf{U}_\mathcal{V}\chi\|,$$

which is below ε for $\mathcal{V}$ near $\mathcal{U}$. □

So $\mathcal{U} \mapsto \mathsf{U}_\mathcal{U}$ is a strongly continuous unitary representation of $\mathrm{U}(\iota)$ on $\Gamma_s(\iota)$, and $\Gamma(\mathcal{U})|0\rangle = |0\rangle$ by Lemma 8.3(b). In quantum optics the unitary operator $\mathsf{U}_\mathcal{U}$ on $\Gamma_s(\iota)$ is the *passive* (linear-optical) unitary of the interferometer $\mathcal{U}$: by (c) it maps annihilation operators to linear combinations of annihilation operators and preserves the total particle number. The map $\Gamma(\mathcal{U})$ itself acts on the formal Fock space, which carries no inner product, and $\mathsf{U}_\mathcal{U}$ is its restriction to $\Gamma_s(\iota)$. The first identity of (c) is the convention $\mathsf{U}^\dagger a_j \mathsf{U} = \sum_k \mathcal{U}_{jk}a_k$ of [4].

8.4 The generator and the exponential

The passive unitary of [4, (12)] is written as the exponential of a quadratic operator. We take that exponential coefficient by coefficient, as in Definition 6.6, but now as a series in r, and show that the series always converges.

Definition 8.8 (Second quantization of a generator; coefficientwise exponential)**.** For $h \in M_\iota(\mathbb{C})$, $d\Gamma(h) = \sum_{j,k} h_{jk}\, a_j^\dagger a_k$ on $\mathcal{F}$. For a linear map T of $\mathcal{F}$ and $\psi \in \mathcal{F}$ such that the series below converges absolutely for every multi-index n, $e^T\psi$ is the vector with the coefficients

$$(e^T\psi)(n) = \sum_{r\geq 0} \frac{(T^r\psi)(n)}{r!}.$$

The map $h \mapsto d\Gamma(h)$ is linear, so $d\Gamma(ch) = c\, d\Gamma(h)$ for $c \in \mathbb{C}$, and $d\Gamma(h)$ is the standard second quantization of the one-particle operator h, written in the number basis [5]. If T raises the particle number, the series terminates at $r = |n|$ (Lemma 6.7(a)) and e^T is the exponential of Definition 6.6. Here $T = d\Gamma(h)$, which preserves every sector, and the next lemma shows that its series converges absolutely for every ψ.

Lemma 8.9 (The generator)**.** Lean ✓ *Let $h \in M_\iota(\mathbb{C})$ and $\|h\|_1 = \sum_{j,k}|h_{jk}|$.*

(a) *For every $\psi \in \mathcal{F}$, $\mathfrak{B}\, d\Gamma(h)\psi = \sum_{j,k} h_{jk}\, z_j\partial_k \mathfrak{B}\psi$, a sum of derivations $z_j\partial_k$ of $\mathbb{C}[[z]]$. For every mode l and every $r \in \mathbb{N}$, $\mathfrak{B}\, d\Gamma(h)|e_l\rangle = \ell_l(h)$ and $\mathfrak{B}\, d\Gamma(h)^r|e_l\rangle = \ell_l(h^r)$. Moreover $d\Gamma(h)[\psi]_p = [d\Gamma(h)\psi]_p$, $d\Gamma(h)|e_j\rangle = \sum_k h_{kj}|e_k\rangle$, and $d\Gamma(ch) = c\, d\Gamma(h)$ for every $c \in \mathbb{C}$.*

(b) *For every derivation δ of a commutative $\mathbb{C}$-algebra A, every $r \in \mathbb{N}$ and all $x, y \in A$, $\delta^r(xy) = \sum_{s+t=r} \binom{r}{s} \delta^s x\, \delta^t y$.*

(c) *Let $\psi \in \mathcal{F}$, $p \in \mathbb{N}$ and $C \geq 0$ with $|[z^m]\,\mathfrak{B}\psi| \leq C$ for every m with $|m| = p$. Then $|[z^n]\,\mathfrak{B}\, d\Gamma(h)^r\psi| \leq (p\|h\|_1)^r C$ for every r and every n with $|n| = p$; in particular this holds with $C = \max_{|m|=|n|}|[z^m]\,\mathfrak{B}\psi|$. For every ψ and n the series*

$$\sum_{r\geq 0} \frac{|[z^n]\,\mathfrak{B}\, d\Gamma(h)^r\psi|}{r!} \quad \text{and} \quad \sum_{r\geq 0} \frac{|(d\Gamma(h)^r\psi)(n)|}{r!}$$

converge. Hence $e^{d\Gamma(h)}$ is defined on all of $\mathcal{F}$; it is linear, and $e^{d\Gamma(h)}[\psi]_p = [e^{d\Gamma(h)}\psi]_p$ (also a consequence of Theorem 8.10).

Proof. (a) By Theorem 6.12, $\mathfrak{B}a_j^\dagger a_k\psi = z_j\partial_k\mathfrak{B}\psi$, and $z_j\partial_k$ is a derivation because ∂_k is. Write $\delta_h = \sum_{j,k} h_{jk} z_j\partial_k$, so that $\mathfrak{B}\, d\Gamma(h)^r = \delta_h^r\mathfrak{B}$. Then $\delta_h z_l = \sum_j h_{jl} z_j = \ell_l(h)$, and by induction $\delta_h\ell_l(h^r) = \sum_k (h^r)_{kl} \sum_j h_{jk} z_j = \ell_l(h^{r+1})$. Each $z_j\partial_k$ maps a monomial of degree p to a multiple of a monomial of degree p, so $d\Gamma(h)$ preserves sectors; $d\Gamma(h)|e_j\rangle$ is $\mathfrak{B}^{-1}\ell_j(h)$, and $d\Gamma(ch) = c\, d\Gamma(h)$ is linearity in h.

(b) Induction on r: apply δ to the sum for r, use $\delta(xy) = \delta x\, y + x\, \delta y$, and collect terms with Pascal's rule $\binom{r}{s-1} + \binom{r}{s} = \binom{r+1}{s}$, as for the binomial theorem.

(c) The coefficient of $z_j\partial_k f$ at n is $((n - e_j)_k + 1)[z^{n-e_j+e_k}]\, f$ if $n_j \geq 1$ and 0 otherwise, and the factor $(n - e_j)_k + 1$ is at most $|n| = p$. So $|[z^n]\,\delta_h f| \leq p\|h\|_1 \max_{|m|=p}|[z^m]\, f|$, and induction on r gives the bound. The first series is then dominated by $Ce^{p\|h\|_1}$, and the second is the first times $\sqrt{n!}$, since $(d\Gamma(h)^r\psi)(n) = \sqrt{n!}\,[z^n]\,\mathfrak{B}\, d\Gamma(h)^r\psi$. Each term of the series of $(e^{d\Gamma(h)}\psi)(n)$ is linear in ψ, and so is the sum of an absolutely convergent series; and $d\Gamma(h)^r[\psi]_p = [d\Gamma(h)^r\psi]_p$ by (a), which gives the last identity termwise. □

Theorem 8.10 (Exponential form; cf. [4, (12), (A20)–(A26)])**.** Lean ✓ *Let $h \in M_\iota(\mathbb{C})$. For every $\psi \in \mathcal{F}$ and every multi-index n, the series $\sum_r [z^n]\,\mathfrak{B}\, d\Gamma(h)^r\psi/r!$ converges to $[z^n]\,(\mathfrak{B}\psi)((e^h)^T z)$, and the series $\sum_r (d\Gamma(h)^r\psi)(n)/r!$ converges to $(\Gamma(e^h)\psi)(n)$. Hence $\Gamma(e^h) = e^{d\Gamma(h)}$ on $\mathcal{F}$, and $\Gamma(e^{i\mathsf{M}}) = e^{i\, d\Gamma(\mathsf{M})}$ for every $\mathsf{M} \in M_\iota(\mathbb{C})$.*

Proof. With δ_h as in the proof of Lemma 8.9, every $f \in \mathbb{C}[[z]]$ is $\mathfrak{B}\psi$ for some ψ, so by Lemma 8.9(c) the series $[z^n]\,E(f) = \sum_r [z^n]\,\delta_h^r f/r!$ converges absolutely and defines a linear map E of $\mathbb{C}[[z]]$. We show $E(f) = f((e^h)^T z)$: first E is multiplicative, then it agrees with the substitution on the variables, and we conclude degree by degree.

Multiplicativity. By Lemma 8.9(b) and the product of power series,

$$\frac{[z^n]\,\delta_h^r(f_1 f_2)}{r!} = \sum_{s+t=r} \sum_{n'+n''=n} \frac{[z^{n'}]\,\delta_h^s f_1}{s!}\,\frac{[z^{n''}]\,\delta_h^t f_2}{t!}.$$

For each of the finitely many pairs $n'+n''=n$, the series $\sum_s [z^{n'}]\,\delta_h^s f_1/s!$ and $\sum_t [z^{n''}]\,\delta_h^t f_2/t!$ converge absolutely, by Lemma 8.9(c) with $p=|n'|$ and $p=|n''|$, so their Cauchy product converges to the product of their sums. Summing over r gives $[z^n]\,E(f_1 f_2) = \sum_{n'+n''=n} [z^{n'}]\,E(f_1)\,[z^{n''}]\,E(f_2)$, that is, $E(f_1 f_2) = E(f_1)E(f_2)$; and $E(1)=1$ because $\delta_h 1 = 0$.

The variables. By Lemma 8.9(a), $\delta_h^r z_l = \ell_l(h^r) = \sum_k (h^r)_{kl} z_k$, so $E(z_l) = \sum_k (e^h)_{kl} z_k = \ell_l(e^h)$, the entries of e^h being the sums of the entries of its series.

Conclusion. E and the substitution of $\ell(e^h)$ are linear, multiplicative and unital and agree on the variables, so they agree on every polynomial. Both act degree by degree: the substitution by the remark after (8.1), and E because δ_h does, so that $[E(f)]_p = E([f]_p)$ coefficient by coefficient. By the same remark they agree on $\mathbb{C}[[z]]$. With $f = \mathfrak{B}\psi$ this is the first series; multiplying by $\sqrt{n!}$ gives the second, whose sum is the coefficient of $e^{d\Gamma(h)}\psi$ at n. For the last statement take $h = i\mathsf{M}$, with $d\Gamma(i\mathsf{M}) = i\,d\Gamma(\mathsf{M})$. □

The paper relates its two forms of the passive unitary only for $h = i\mathsf{M}$ with M Hermitian, through [4, (A23)–(A26)]. For one mode, $d\Gamma(h) = h\,n_{\mathrm{tot}}$ and Theorem 8.10 reads $e^{hm} = (e^h)^m$ on $|m\rangle$. To write a given passive unitary in the form of [4, (12)] we need a Hermitian logarithm of $\mathcal{U}$.

Lemma 8.11 (Unitary matrices are exponentials)**.** Lean ✓ *Every $\mathcal{U} \in \mathrm{U}(\iota)$ is $e^{i\mathsf{M}}$ for a Hermitian $\mathsf{M} \in M_\iota(\mathbb{C})$, and $e^{i\mathsf{M}} \in \mathrm{U}(\iota)$ for every Hermitian M.*

Proof. A unitary matrix is normal, so by the spectral theorem $\mathcal{U} = W\,\mathrm{diag}(\lambda_k)W^*$ with W unitary and $|\lambda_k| = 1$. Put $\theta_k = \arg\lambda_k$ and $\mathsf{M} = W\,\mathrm{diag}(\theta_k)W^*$, which is Hermitian; then $e^{i\mathsf{M}} = W\,\mathrm{diag}(e^{i\theta_k})W^* = \mathcal{U}$. Conversely $(e^{i\mathsf{M}})^* = e^{-i\mathsf{M}^*} = e^{-i\mathsf{M}} = (e^{i\mathsf{M}})^{-1}$. □

Read as an operator on $\Gamma_s(\iota)$, the exponential in [4, (12)] is e^{iA_M}, where A_M is the closure of $d\Gamma(\mathsf{M})$ on the finitely supported vectors. The next proposition identifies it with $\mathsf{U}_{e^{i\mathsf{M}}}$, using only the functional calculus of orthogonal direct sums of self-adjoint operators [9, Ch. VIII].

Proposition 8.12 (The passive unitary as an operator exponential)**.** *Let $\mathsf{M} \in M_\iota(\mathbb{C})$ be Hermitian, let $A_{\mathsf{M},p}$ be the restriction of $d\Gamma(\mathsf{M})$ to the sector $\mathcal{F}_p$, and let A_M be the closure of $d\Gamma(\mathsf{M})$ on the finitely supported vectors of $\Gamma_s(\iota)$. Then each $A_{\mathsf{M},p}$ is a Hermitian matrix, $A_\mathsf{M} = \bigoplus_p A_{\mathsf{M},p}$ is self-adjoint and is the maximal realization of $d\Gamma(\mathsf{M})$ (Definition 7.6), and $e^{iA_\mathsf{M}} = \mathsf{U}_{e^{i\mathsf{M}}}$.*

Proof. Each sector $\mathcal{F}_p$ consists of finitely supported vectors, so it lies in $\jmath(\Gamma_s(\iota))$; we identify it with its preimage, a finite-dimensional subspace of $\Gamma_s(\iota)$. The number states form an orthonormal basis, so $\Gamma_s(\iota)$ is the Hilbert direct sum $\bigoplus_p \mathcal{F}_p$, and every ψ is $\sum_p \psi_p$ with $\psi_p = [\psi]_p$. By Lemma 8.9(a), $d\Gamma(\mathsf{M})$ maps $\mathcal{F}_p$ into itself. By Definition 6.2 the matrices of $a_j^\dagger$ and a_j in the number basis are real and transposes of each other, so on $\mathcal{F}_p$ the adjoint of $a_j^\dagger a_k$ is $a_k^\dagger a_j$, and $A_{\mathsf{M},p}^* = \sum_{j,k} \overline{\mathsf{M}_{jk}}\, a_k^\dagger a_j = A_{\mathsf{M},p}$: each $A_{\mathsf{M},p}$ is a Hermitian matrix.

The direct sum $\bigoplus_p A_{\mathsf{M},p}$, with the domain $\{\psi : \sum_p \|A_{\mathsf{M},p}\psi_p\|^2 < \infty\}$ and $(\bigoplus_p A_{\mathsf{M},p})\psi = \sum_p A_{\mathsf{M},p}\psi_p$, is self-adjoint, as a direct sum of self-adjoint operators. On the finitely supported vectors it is $d\Gamma(\mathsf{M})$, and they form a core for it: for ψ in the domain the partial sums $\sum_{p\le P}\psi_p$ converge to ψ, and their images $\sum_{p\le P} A_{\mathsf{M},p}\psi_p$ converge to $(\bigoplus_p A_{\mathsf{M},p})\psi$. Hence $A_\mathsf{M} = \bigoplus_p A_{\mathsf{M},p}$ is

self-adjoint. Since $d\Gamma(\mathsf{M})\jmath\psi$ has the components $A_{\mathsf{M},p}\psi_p$, its squared norm is $\sum_p \|A_{\mathsf{M},p}\psi_p\|^2$, so the domain and the action of A_{M} are those of the maximal realization. The functional calculus of direct sums gives $e^{iA_{\mathsf{M}}} = \bigoplus_p e^{iA_{\mathsf{M},p}}$, with $e^{iA_{\mathsf{M},p}}$ the matrix exponential on $\mathcal{F}_p$. For $\psi \in \mathcal{F}_p$, $d\Gamma(i\mathsf{M})^r\psi = (iA_{\mathsf{M},p})^r\psi$, so Theorem 8.10 with $h = i\mathsf{M}$ gives $e^{iA_{\mathsf{M},p}}\psi = \Gamma(e^{i\mathsf{M}})\psi$. The bounded operators $e^{iA_{\mathsf{M}}}$ and $\mathsf{U}_{e^{i\mathsf{M}}}$ therefore agree on every sector, hence on the dense finitely supported vectors, hence on $\Gamma_s(\iota)$. □

With Lemma 8.11, the passive unitaries of [4, (12)] are exactly the operators $\mathsf{U}_{\mathcal{U}}$, $\mathcal{U} \in \mathrm{U}(\iota)$. For real t, replacing M by $t\mathsf{M}$ gives $A_{t\mathsf{M}} = tA_{\mathsf{M}}$ and $e^{itA_{\mathsf{M}}} = \mathsf{U}_{e^{it\mathsf{M}}}$, so A_{M} is also the self-adjoint generator that Stone's theorem [9, Sec. VIII.4] gives to the strongly continuous unitary group $t \mapsto \mathsf{U}_{e^{it\mathsf{M}}}$ (Theorem 8.7(b),(d)), in the sign convention of Reed and Simon; in that of Definition 13.1, A_{M} generates $t \mapsto \mathsf{U}_{e^{-it\mathsf{M}}}$.

8.5 Conjugation of operators

Expectation values in a state $\Gamma(\mathcal{U})\Psi$ reduce to expectation values in Ψ of the conjugated operator $\Gamma(\mathcal{U}^*)\,O\,\Gamma(\mathcal{U})$. The observables of interest are polynomials in the ladder operators, which are unbounded on $\Gamma_s(\iota)$, so we state the reduction for an arbitrary linear map O of $\mathcal{F}$ through its maximal realization $\bar{O}$ (Definition 7.6), which needs no closure, core or self-adjointness.

Theorem 8.13 (Conjugation of operators)**.** Lean ✓ *Let $\mathcal{U} \in \mathrm{U}(\iota)$, let O be a linear map of $\mathcal{F}$, and write $O^{\mathcal{U}} = \Gamma(\mathcal{U}^*)\,O\,\Gamma(\mathcal{U})$.*

(a) *$(OO')^{\mathcal{U}} = O^{\mathcal{U}}O'^{\mathcal{U}}$, $(O+O')^{\mathcal{U}} = O^{\mathcal{U}} + O'^{\mathcal{U}}$ and $(cO)^{\mathcal{U}} = c\,O^{\mathcal{U}}$ for all linear maps O, O' of $\mathcal{F}$ and all $c \in \mathbb{C}$. For every noncommutative polynomial P with complex coefficients in $2N$ variables x_j, y_j, $j \in \iota$,*

$$P(a^\dagger, a)^{\mathcal{U}} = P\Big(\sum_k \overline{\mathcal{U}_{jk}}\, a_k^\dagger,\ \sum_k \mathcal{U}_{jk} a_k\Big),$$

where $P(a^\dagger, a)$ substitutes $a_j^\dagger$ for x_j and a_j for y_j, and the right side substitutes the displayed sums.

(b) *For $\psi \in \Gamma_s(\iota)$, $\mathsf{U}_{\mathcal{U}}\psi \in \operatorname{dom}\bar{O}$ if and only if $\psi \in \operatorname{dom}\overline{O^{\mathcal{U}}}$. Thus $\operatorname{dom}\overline{O^{\mathcal{U}}} = \mathsf{U}_{\mathcal{U}}^{-1}(\operatorname{dom}\bar{O})$, and $\overline{O^{\mathcal{U}}}\psi = \mathsf{U}_{\mathcal{U}}^{-1}\bar{O}\,\mathsf{U}_{\mathcal{U}}\psi$ for ψ in it. Moreover $\langle \mathsf{U}_{\mathcal{U}}\psi, \bar{O}\,\mathsf{U}_{\mathcal{U}}\psi\rangle = \langle \psi, \overline{O^{\mathcal{U}}}\psi\rangle$ whenever $\mathsf{U}_{\mathcal{U}}\psi \in \operatorname{dom}\bar{O}$.*

(c) *If A is a bounded operator on $\Gamma_s(\iota)$ with $\jmath(A\psi) = O(\jmath\psi)$ for every $\psi \in \Gamma_s(\iota)$, then $\jmath(\mathsf{U}_{\mathcal{U}}^* A\,\mathsf{U}_{\mathcal{U}}\psi) = O^{\mathcal{U}}(\jmath\psi)$ for every ψ, and $\operatorname{dom}\overline{O^{\mathcal{U}}} = \Gamma_s(\iota)$. For every bounded operator A and every $\psi \in \Gamma_s(\iota)$, $\langle \mathsf{U}_{\mathcal{U}}\psi, A\,\mathsf{U}_{\mathcal{U}}\psi\rangle = \langle \psi, \mathsf{U}_{\mathcal{U}}^* A\,\mathsf{U}_{\mathcal{U}}\psi\rangle$.*

Proof. (a) Inserting $\Gamma(\mathcal{U})\Gamma(\mathcal{U}^*) = 1$ between O and O' gives multiplicativity; additivity and homogeneity hold because composition is bilinear; and $1^{\mathcal{U}} = \Gamma(\mathcal{U}^*)\Gamma(\mathcal{U}) = 1$. So $O \mapsto O^{\mathcal{U}}$ is a unital algebra endomorphism of the linear maps of $\mathcal{F}$, and $P \mapsto P(a^\dagger, a)^{\mathcal{U}}$ and the right side are two unital algebra homomorphisms from the noncommutative polynomials that agree on the variables by Theorem 8.7(c). They therefore agree on every P.

(b) $O^{\mathcal{U}}\jmath\psi = \Gamma(\mathcal{U}^*)\,O\,\jmath\,\mathsf{U}_{\mathcal{U}}\psi$, and $\Gamma(\mathcal{U}^*)$ preserves the squared norm of every formal vector (Theorem 8.7(a)), so $\|O^{\mathcal{U}}\jmath\psi\|^2 < \infty$ exactly when $\|O\jmath\,\mathsf{U}_{\mathcal{U}}\psi\|^2 < \infty$; this is the equivalence of domains. On the domain, $\jmath\,\overline{O^{\mathcal{U}}}\psi = \Gamma(\mathcal{U}^*)\jmath\,\bar{O}\,\mathsf{U}_{\mathcal{U}}\psi = \jmath\,\mathsf{U}_{\mathcal{U}^*}\bar{O}\,\mathsf{U}_{\mathcal{U}}\psi$, and $\mathsf{U}_{\mathcal{U}^*} = \mathsf{U}_{\mathcal{U}}^{-1}$. Since $\mathsf{U}_{\mathcal{U}}$ is unitary, $\langle \mathsf{U}_{\mathcal{U}}\psi, \bar{O}\,\mathsf{U}_{\mathcal{U}}\psi\rangle = \langle \mathsf{U}_{\mathcal{U}}\psi, \mathsf{U}_{\mathcal{U}}\overline{O^{\mathcal{U}}}\psi\rangle = \langle \psi, \overline{O^{\mathcal{U}}}\psi\rangle$.

(c) $\jmath(\mathsf{U}_{\mathcal{U}}^* A\,\mathsf{U}_{\mathcal{U}}\psi) = \Gamma(\mathcal{U}^*)\jmath(A\,\mathsf{U}_{\mathcal{U}}\psi) = \Gamma(\mathcal{U}^*)\,O\,\Gamma(\mathcal{U})\jmath\psi$ by Theorem 8.7(b), and this vector is square-summable, so every ψ lies in $\operatorname{dom}\overline{O^{\mathcal{U}}}$. The last identity is the definition of the adjoint $\mathsf{U}_{\mathcal{U}}^*$. □

Notes and references

Tjoa and Cirac [4] define the passive unitary by their (12), $\mathsf{U} = e^{i\sum_{j,k}\mathsf{M}_{jk}a_j^\dagger a_k}$ with M Hermitian, and relate it to the single-particle unitary $\mathcal{U} \in \mathrm{U}(N)$ by their (29), $\mathsf{U}^\dagger a_j \mathsf{U} = \sum_k \mathcal{U}_{jk} a_k$. Their Appendix A derives the same relation as (A26), with $\mathcal{U} = e^{i\mathsf{M}}$ (A20), from the commutator $[G_\mathsf{H}, a_l] = -\sum_k \mathsf{H}_{lk} a_k$ (A24), where $G_\mathsf{H} = \sum_{j,k} \mathsf{H}_{jk} a_j^\dagger a_k$ (A23) is $d\Gamma(\mathsf{H})$ in our notation, and the Baker–Campbell–Hausdorff formula (A25); (29) and (A26) agree, and the first identity of Theorem 8.7(c) proves them with U and $\mathsf{U}^\dagger$ read as $\Gamma(\mathcal{U})$ and $\Gamma(\mathcal{U}^*)$ on the formal Fock space. The inverse conjugation $\Gamma(\mathcal{U}) a_j^\dagger \Gamma(\mathcal{U})^{-1} = \sum_k \mathcal{U}_{kj} a_k^\dagger$ is a consequence and does not appear in the paper. We instead define $\Gamma(g)$ by substitution in the Bargmann picture [7] for every matrix, prove the representation, intertwining, adjoint and continuity properties without exponentials of unbounded operators, and obtain the exponential form for every matrix h, not only for $h = i\mathsf{M}$; the smallest case is one mode, $\Gamma(e^{i\theta}) = e^{i\theta a^\dagger a}$. Proposition 8.12 identifies the coefficientwise $e^{i\,d\Gamma(\mathsf{M})}$ with the exponential of i times the self-adjoint closure of $d\Gamma(\mathsf{M})$, by the functional calculus of orthogonal direct sums. The second quantization of operators on Fock space is treated in [5], and the functional calculus and Stone's theorem in [9, Secs. VIII.3 and VIII.4]; for the identity $\Gamma(e^{itA}) = e^{it\,d\Gamma(A)}$, A self-adjoint, see [9]. For the action of the unitary group on the Bargmann space by a linear substitution of the variables, in the Fock model of the metaplectic representation, see [16, Ch. 4]; Definition 8.2 extends this substitution to every matrix on the formal Fock space.

9 The Bloch–Messiah decomposition

Section 8 lifts unitary $N \times N$ matrices to Fock space; on phase space these unitaries are the orthogonal symplectic matrices, and every real symplectic matrix S factors through them as $S = O\,\Sigma(r)\,O'$, with O and O' orthogonal and symplectic and $\Sigma(r) = \mathrm{diag}(e^{-r}, e^{r})$ a diagonal squeezing matrix with $r \geq 0$. In optics this says that every linear canonical transformation of N modes is an interferometer, followed by N single-mode squeezers, followed by a second interferometer. The singular values of S are the $e^{\pm r_k}$, so the squeezing parameters are fixed up to order, while the orthogonal factors are not, and the polar decomposition of S follows from the factorization. The lift of the factorization to Fock space is planned for a later section of these notes.

Notation for this section.

N, Ω, I	number of modes, $N \geq 1$; $\Omega = \Omega_N$ of Section 1; identity matrix
$v^T w$	Euclidean inner product of column vectors in $\mathbb{R}^{2N}$
$\mathrm{Sp}(2N,\mathbb{R})$, $\mathrm{O}(2N)$, $\mathrm{U}(N)$	symplectic, orthogonal and unitary groups (Lemma 9.1)
O, O', $\tilde{O}$, $\tilde{O}'$	orthogonal matrices, the factors of a decomposition
$\mathcal{U} = X + iY$, $O_\mathcal{U}$	a unitary matrix with real and imaginary parts X, Y; its orthogonal symplectic matrix (Definition 9.2)
r, s, $\Sigma(r)$	squeezing parameters in $\mathbb{R}^N$, $r \geq 0$ meaning $r_k \geq 0$ for every k; squeezing matrix (Definition 9.5)
S, M	a symplectic matrix; a positive definite symplectic matrix
$\ker(M - \mu)$	the eigenspace of M for the eigenvalue μ
f_k, V_k	vectors of an eigenbasis; the span of $f_1, \ldots, f_k, -\Omega f_1, \ldots, -\Omega f_k$
P, Q	the factors of the polar decomposition (Corollary 9.14)

The roman O denotes the orthogonal group, an italic O a matrix. The letters O, M, P, r and s do not carry their meanings from Section 8.

9.1 Orthogonal symplectic matrices

Throughout, $N \geq 1$, and phase-space vectors are the columns $(q_1, \ldots, q_N, p_1, \ldots, p_N)^T$ of $\mathbb{R}^{2N}$, with the positions first. The symplectic form is that of Section 1,

$$\Omega = \Omega_N = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \qquad \Omega^T = \Omega^{-1} = -\Omega.$$

The *symplectic group* $\mathrm{Sp}(2N, \mathbb{R})$ is the group of the symplectic matrices of Section 1, $S\Omega S^T = \Omega$; $\mathrm{O}(2N)$ is the orthogonal group and $\mathrm{U}(N)$ the group of unitary $N \times N$ matrices. A real matrix A is *positive definite* if it is symmetric and $v^T A v > 0$ for every $v \neq 0$. Both signs of the standard symplectic matrix occur in the literature, and the first statement of the next lemma shows that they define the same group.

Lemma 9.1 (The symplectic group)**.** Lean ✓ *Let S be a real $2N \times 2N$ matrix and $J = -\Omega = \left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$. Then $S\Omega S^T = \Omega$ if and only if $SJS^T = J$, if and only if $S^T \Omega S = \Omega$. Every $S \in \mathrm{Sp}(2N, \mathbb{R})$ is invertible.*

Proof. The first equivalence is multiplication by -1. If $S\Omega S^T = \Omega$, then $\det(S)^2 \det \Omega = \det \Omega$ and $\det \Omega = 1$, so S is invertible; inverting both sides and using $\Omega^{-1} = -\Omega$ gives $(S^T)^{-1} \Omega S^{-1} = \Omega$, that is, $S^T \Omega S = \Omega$. The converse is the same argument for S^T. □

We identify $\mathbb{C}^N$ with $\mathbb{R}^{2N}$ by $q + ip \leftrightarrow (q, p)$, the identification behind $a = (q + ip)/\sqrt{2}$ in [4, (A18)]. Multiplication by i sends (q, p) to $(-p, q)$, so it is the matrix $-\Omega$. For $u \leftrightarrow f$ and $u' \leftrightarrow f'$, the Hermitian product of $\mathbb{C}^N$, antilinear in its first argument, is

$$u^* u' = f^T f' + i\, f^T \Omega f'. \tag{9.1}$$

A complex matrix acts on $\mathbb{C}^N$ as a real-linear map, and its real $2N \times 2N$ matrix is the following.

Definition 9.2 (Orthogonal symplectic matrix of a unitary)**.** For $\mathcal{U} = X + iY \in \mathrm{U}(N)$, with X and Y real $N \times N$ matrices, the *orthogonal symplectic matrix of* $\mathcal{U}$ is

$$O_{\mathcal{U}} = \begin{pmatrix} X & -Y \\ Y & X \end{pmatrix}, \qquad O_{\mathcal{U}} \begin{pmatrix} q \\ p \end{pmatrix} = \begin{pmatrix} Xq - Yp \\ Yq + Xp \end{pmatrix} \leftrightarrow \mathcal{U}(q + ip).$$

This is the block form of [4, (A14)], with $\mathcal{U} = X + iY$ as in [4, (A16)]. That $O_{\mathcal{U}}$ is orthogonal and symplectic is part of Theorem 9.4, which rests on a block computation.

Lemma 9.3 (Commutant of Ω)**.** Lean ✓ *An orthogonal matrix O lies in $\mathrm{Sp}(2N, \mathbb{R})$ if and only if $O\Omega = \Omega O$. A real $2N \times 2N$ matrix O satisfies $O\Omega = \Omega O$ if and only if $O = \left(\begin{smallmatrix} X & -Y \\ Y & X \end{smallmatrix}\right)$ for some real $N \times N$ matrices X and Y, that is, if and only if O is the real matrix of $X + iY$ acting on $\mathbb{C}^N \cong \mathbb{R}^{2N}$.*

Proof. For orthogonal O, multiplying $O\Omega O^T = \Omega$ on the right by O gives $O\Omega = \Omega O$, and conversely. For $O = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$,

$$O\Omega = \begin{pmatrix} -B & A \\ -D & C \end{pmatrix}, \qquad \Omega O = \begin{pmatrix} C & D \\ -A & -B \end{pmatrix},$$

which agree exactly when $C = -B$ and $D = A$, that is, when O has the stated form with $X = A$ and $Y = C$. □

Theorem 9.4 (Orthogonal symplectic and unitary matrices; [4, (A14)–(A16)])**.** Lean ✓

(a) *For all $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$: $O_{\mathcal{U}\mathcal{V}} = O_{\mathcal{U}} O_{\mathcal{V}}$, $O_I = I$, $O_{\mathcal{U}^*} = O_{\mathcal{U}}^T$, and $\mathcal{U} \mapsto O_{\mathcal{U}}$ is injective. Moreover $O_{-\mathcal{U}} = -O_{\mathcal{U}}$, and $O_{iI} = -\Omega$.*

(b) $O_{\mathcal{U}} \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ *for every* $\mathcal{U} \in \mathrm{U}(N)$*, and a matrix* O *lies in* $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ *if and only if* $O = O_{\mathcal{U}}$ *for some* $\mathcal{U} \in \mathrm{U}(N)$*. In blocks,* $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ *if and only if* $O = \left(\begin{smallmatrix} X & -Y \\ Y & X \end{smallmatrix}\right)$ *with* X*,* Y *real and* $X + iY \in \mathrm{U}(N)$*.*

(c) *The map that sends* $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$*, written in* $N \times N$ *blocks as* $O = \left(\begin{smallmatrix} X & B \\ Y & D \end{smallmatrix}\right)$*, to* $X + iY$ *is a group isomorphism* $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N) \to \mathrm{U}(N)$*, with inverse* $\mathcal{U} \mapsto O_{\mathcal{U}}$*.*

Proof. The block formula of Definition 9.2 assigns a real matrix to every complex matrix $X + iY$, and the computations in (a) hold for all complex matrices.

(a) $(X_1 + iY_1)(X_2 + iY_2) = (X_1X_2 - Y_1Y_2) + i(X_1Y_2 + Y_1X_2)$, and the block product of the two real matrices has the same blocks. For I the blocks are I and 0. The adjoint $X^T - iY^T$ has the real matrix $\left(\begin{smallmatrix} X^T & Y^T \\ -Y^T & X^T \end{smallmatrix}\right) = O_{\mathcal{U}}^T$. Injectivity holds because X and Y are blocks of $O_{\mathcal{U}}$. The real matrix is linear in $X + iY$, which gives $O_{-\mathcal{U}} = -O_{\mathcal{U}}$, and iI has $X = 0$, $Y = I$.

(b) By (a), $O_{\mathcal{U}}^T O_{\mathcal{U}} = O_{\mathcal{U}^*\mathcal{U}} = O_I = I$, so $O_{\mathcal{U}}$ is orthogonal; it commutes with Ω by Lemma 9.3, so it is symplectic by the same lemma. Conversely, an $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ commutes with Ω, so $O = \left(\begin{smallmatrix} X & -Y \\ Y & X \end{smallmatrix}\right)$, and $O^TO = I$ is the real form of

$$(X + iY)^*(X + iY) = (X^TX + Y^TY) + i(X^TY - Y^TX) = I,$$

the identity [4, (A15), (A16)]; by injectivity, $X + iY$ is unitary and $O = O_{X+iY}$.

(c) By (b) the map is the inverse of the bijection $\mathcal{U} \mapsto O_{\mathcal{U}}$ from $\mathrm{U}(N)$ onto $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, which is a homomorphism by (a). □

The orthogonal symplectic matrices are the *passive* transformations: on quadrature operators $\hat{q}_j = (a_j + a_j^\dagger)/\sqrt{2}$ and $\hat{p}_j = (a_j - a_j^\dagger)/(i\sqrt{2})$ they preserve $\sum_j(\hat{q}_j^2 + \hat{p}_j^2) = 2n_{\mathrm{tot}} + N$, hence the total particle number.

If a unitary V on Fock space implements $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ in the sense that $\mathsf{V}^\dagger \hat{\mathbf{R}} \mathsf{V} = O\hat{\mathbf{R}}$ for the column $\hat{\mathbf{R}} = (\hat{q}_1, \ldots, \hat{q}_N, \hat{p}_1, \ldots, \hat{p}_N)^T$, then on $a = (\hat{q} + i\hat{p})/\sqrt{2}$ it acts by $\mathsf{V}^\dagger a \mathsf{V} = (X + iY)a$ [4, (A17)–(A19)]. This is the convention $\mathsf{U}^\dagger a_j \mathsf{U} = \sum_k \mathcal{U}_{jk} a_k$ of [4, (29), (A26)] with $\mathcal{U} = X + iY$, which Theorem 8.7(c) proves for $\Gamma(\mathcal{U})$; both $\mathcal{U} \mapsto O_{\mathcal{U}}$ and $\mathcal{U} \mapsto \Gamma(\mathcal{U})$ are multiplicative. The lift to Fock space, the statement that $\Gamma(\mathcal{U})$ implements $O_{\mathcal{U}}$, is planned for a later section of these notes. The other parametrization $\left(\begin{smallmatrix} X & Y \\ -Y & X \end{smallmatrix}\right)$ of the same group is $O_{\bar{\mathcal{U}}}$, $\bar{\mathcal{U}} = X - iY$, and gives $(X - iY)a$ in this reading.

9.2 The squeezing matrix

Definition 9.5 (Squeezing matrix)**.** For $r = (r_1, \ldots, r_N) \in \mathbb{R}^N$, the *squeezing matrix* is $\Sigma(r) = \mathrm{diag}(e^{-r_1}, \ldots, e^{-r_N}, e^{r_1}, \ldots, e^{r_N})$.

It scales the position q_k by e^{-r_k} and the momentum p_k by e^{r_k}, as the single-mode squeezers of [4, (A10), (A28)–(A29)]. In coordinates with the momenta first, the same transformation has the matrix $\Sigma(-r)$.

Lemma 9.6 (Squeezing matrices)**.** Lean ✓ *For all* $r, s \in \mathbb{R}^N$*:* $\Sigma(r) \in \mathrm{Sp}(2N,\mathbb{R})$*,* $\Sigma(r + s) = \Sigma(r)\Sigma(s)$*,* $\Sigma(0) = I$*,* $\Sigma(r)^T = \Sigma(r)$*,* $\Sigma(r)$ *is positive definite, and* $\Sigma(-r)\Sigma(r) = \Sigma(r)\Sigma(-r) = I$*. In particular* $\Sigma(r)^2 = \Sigma(2r)$*.*

Proof. All matrices involved are diagonal with positive entries, and $e^{-r_k}e^{-s_k} = e^{-(r_k+s_k)}$. With $D = \mathrm{diag}(e^{r_1}, \ldots, e^{r_N})$,

$$\Sigma(r)\,\Omega\,\Sigma(r)^T = \begin{pmatrix} D^{-1} & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} \begin{pmatrix} D^{-1} & 0 \\ 0 & D \end{pmatrix} = \begin{pmatrix} 0 & D^{-1}D \\ -DD^{-1} & 0 \end{pmatrix} = \Omega. \qquad \square$$

9.3 Positive definite symplectic matrices

The squeezing parameters of S are read off from the positive definite matrix $M = SS^T$, which forgets the orthogonal factor on the right. The next two lemmas collect what is used about M.

Lemma 9.7 (The matrix SS^T). Lean ✓ *For $S \in \mathrm{Sp}(2N,\mathbb{R})$, SS^T is positive definite and lies in $\mathrm{Sp}(2N,\mathbb{R})$. Every symmetric $M \in \mathrm{Sp}(2N,\mathbb{R})$ satisfies $M\Omega = \Omega M^{-1}$.*

Proof. $S^T \in \mathrm{Sp}(2N,\mathbb{R})$ by Lemma 9.1, so $SS^T \in \mathrm{Sp}(2N,\mathbb{R})$; it is symmetric, and $v^T SS^T v = |S^T v|^2 > 0$ for $v \neq 0$ because S is invertible. If $M = M^T \in \mathrm{Sp}(2N,\mathbb{R})$, then $M\Omega M = \Omega$, which is $M\Omega = \Omega M^{-1}$. □

A subspace $V \subseteq \mathbb{R}^{2N}$ is *isotropic* (with respect to Ω) if $v^T\Omega w = 0$ for all $v, w \in V$.

Lemma 9.8 (Eigenspaces; [4, (A6)–(A7)]). Lean ✓ *Every eigenvalue of a matrix $M \in \mathrm{Sp}(2N,\mathbb{R})$ is nonzero. Let $M \in \mathrm{Sp}(2N,\mathbb{R})$ be symmetric.*

(i) *Let $\mu \neq 0$. If $Mv = \mu v$, then $M\Omega v = \mu^{-1}\Omega v$, and Ω maps $\ker(M-\mu)$ onto $\ker(M-\mu^{-1})$.*

(ii) *If $Mv = \mu v$, $Mw = \nu w$ and $\mu\nu \neq 1$, then $v^T\Omega w = 0$. If M is positive definite and $\mu \neq 1$, then $\ker(M-\mu)$ is isotropic.*

(iii) *Ω maps $\ker(M-1)$ into itself.*

Proof. M is invertible (Lemma 9.1), so its eigenvalues are nonzero.

(i) $M^{-1}v = \mu^{-1}v$, so $M\Omega v = \Omega M^{-1}v = \mu^{-1}\Omega v$ by Lemma 9.7. Applied to μ^{-1}, this shows that Ω maps $\ker(M-\mu^{-1})$ into $\ker(M-\mu)$; so every $w \in \ker(M-\mu^{-1})$ is $\Omega(-\Omega w)$ with $-\Omega w \in \ker(M-\mu)$, since $\Omega^2 = -I$.

(ii) If $w = 0$ there is nothing to prove; otherwise $\nu \neq 0$, and by Lemma 9.7

$$\mu\, v^T\Omega w = (Mv)^T\Omega w = v^T M\Omega w = v^T\Omega M^{-1}w = \nu^{-1}v^T\Omega w,$$

with $\mu \neq \nu^{-1}$. For the second statement, if $\ker(M-\mu) \neq 0$, then μ is an eigenvalue of the positive definite M, so $\mu > 0$ and $\mu^2 \neq 1$ when $\mu \neq 1$; otherwise there is nothing to prove.

(iii) This is (i) with $\mu = 1$. □

So the eigenvalues of M come in pairs μ, μ^{-1}, and we look for an orthonormal basis of eigenvectors of the form $f_k, -\Omega f_k$. For $\mu < 1$ any orthonormal basis of $\ker(M-\mu)$ could serve as the f_k, by (ii). On $\ker(M-1)$, which (iii) makes a complex subspace of $\mathbb{C}^N$, the f_k must be orthonormal for the Hermitian product (9.1). The construction below chooses one vector at a time and treats every eigenvalue alike; it uses only (i), an eigenvector of a symmetric matrix on an invariant subspace, and orthogonal complements.

Lemma 9.9 (Orthonormal families $f_k, -\Omega f_k$, and invariant subspaces). Lean ✓ *For vectors $f_1, \dots, f_k \in \mathbb{R}^{2N}$ let*

$$V_k = \mathrm{span}\{f_1, \dots, f_k, -\Omega f_1, \dots, -\Omega f_k\}.$$

(a) *If $f_1, \dots, f_k$ are orthonormal and $f_i^T\Omega f_j = 0$ for all i, j, then the $2k$ vectors f_j and $-\Omega f_j$, $j \leq k$, are orthonormal, and $\dim V_k^\perp + 2k = 2N$.*

(b) *If $M \in \mathrm{Sp}(2N,\mathbb{R})$ is symmetric and $Mf_j = \mu_j f_j$ for $j \leq k$, then $v \in V_k^\perp$ implies $Mv \in V_k^\perp$ and $\Omega v \in V_k^\perp$.*

(c) *If $M \in \mathrm{Sp}(2N,\mathbb{R})$ is symmetric, $f_1, \dots, f_N$ are orthonormal, $f_i^T\Omega f_j = 0$ for all i, j, and $Mf_j = \mu_j f_j$ for every j, then the matrix O with the columns $f_1, \dots, f_N, -\Omega f_1, \dots, -\Omega f_N$ lies in $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, and $MO = O\,\mathrm{diag}(\mu_1, \dots, \mu_N, \mu_1^{-1}, \dots, \mu_N^{-1})$.*

(d) *If M is positive definite and $W \neq 0$ is a subspace of $\mathbb{R}^{2N}$ with $MW \subseteq W$, there are $v \in W$ with $v^T v = 1$ and $\mu > 0$ with $Mv = \mu v$.*

Proof. (a) Ω is orthogonal, so $(\Omega f_i)^T(\Omega f_j) = f_i^T f_j = \delta_{ij}$, and $f_i^T(-\Omega f_j) = 0$ by hypothesis. Orthonormal vectors are linearly independent, so $\dim V_k = 2k$.

(b) V_k is invariant under M, since $M(-\Omega f_j) = \mu_j^{-1}(-\Omega f_j)$ by Lemma 9.8(i) when $f_j \neq 0$, and then $\mu_j \neq 0$, and under Ω, since $\Omega f_j = -(-\Omega f_j)$ and $\Omega(-\Omega f_j) = f_j$. For $v \in V_k^\perp$ and $w \in V_k$, $w^T M v = (Mw)^T v = 0$ because M is symmetric, and $w^T \Omega v = -(\Omega w)^T v = 0$ because $\Omega^T = -\Omega$.

(c) The columns are orthonormal by (a), so $O^T O = I$. The blocks of $O^T \Omega O$ are $f_i^T \Omega f_j = 0$, $f_i^T \Omega(-\Omega f_j) = f_i^T f_j = \delta_{ij}$, $(-\Omega f_i)^T \Omega f_j = f_i^T \Omega^2 f_j = -\delta_{ij}$ and $(-\Omega f_i)^T \Omega(-\Omega f_j) = -f_i^T \Omega^3 f_j = f_i^T \Omega f_j = 0$, so $O^T \Omega O = \Omega$ and $O \in \mathrm{Sp}(2N, \mathbb{R})$ by Lemma 9.1. Each μ_j is nonzero, since $f_j \neq 0$ and M is invertible (Lemma 9.8), and the eigenvalue equation for MO holds column by column, by Lemma 9.8(i) for the second half.

(d) The restriction of M to W is a symmetric linear map of $W \neq 0$, so by the spectral theorem it has a unit eigenvector $v \in W$, with eigenvalue $\mu = v^T M v > 0$. □

The sign of the second block of columns matters: with the columns f_k and $+\Omega f_k$ the same computation gives $O^T \Omega O = -\Omega$.

Theorem 9.10 (Orthogonal symplectic diagonalization of positive definite symplectic matrices; [4, (A8)–(A10)]). Lean ✓ *Let $M \in \mathrm{Sp}(2N, \mathbb{R})$ be positive definite.*

(a) *For every $k \leq N$ there are orthonormal vectors $f_1, \dots, f_k$ and numbers $\mu_1, \dots, \mu_k \in (0, 1]$ with $Mf_j = \mu_j f_j$ for every j and $f_i^T \Omega f_j = 0$ for all i, j.*

(b) *There are orthonormal $f_1, \dots, f_N$ with $f_j^T \Omega f_k = 0$ for all j, k and $r \in \mathbb{R}^N$, $r \geq 0$, with $Mf_k = e^{-2r_k} f_k$, such that the matrix O with the columns $f_1, \dots, f_N, -\Omega f_1, \dots, -\Omega f_N$ lies in $\mathrm{Sp}(2N, \mathbb{R}) \cap \mathrm{O}(2N)$ and $M = O\,\Sigma(r)^2\,O^T$. In particular $M = O\,\Sigma(r)^2\,O^T$ for some $O \in \mathrm{Sp}(2N, \mathbb{R}) \cap \mathrm{O}(2N)$ and some $r \geq 0$.*

Proof. (a) Induction on k; for $k = 0$ the family is empty. Given a family for some $k < N$, the subspace $W = V_k^\perp$ has dimension $2N - 2k > 0$ and is invariant under M and Ω (Lemma 9.9(a),(b)). By Lemma 9.9(d) it contains a unit eigenvector v of M with an eigenvalue $\mu > 0$. If $\mu \leq 1$, put $f_{k+1} = v$ and $\mu_{k+1} = \mu$. Otherwise put $f_{k+1} = \Omega v$, a unit vector in W with the eigenvalue $\mu_{k+1} = \mu^{-1} < 1$ (Lemma 9.8(i)). Since f_{k+1} is orthogonal to f_j and to $-\Omega f_j$ for $j \leq k$, we get $f_{k+1}^T f_j = f_{k+1}^T \Omega f_j = 0$ and $f_j^T \Omega f_{k+1} = -(\Omega f_j)^T f_{k+1} = 0$; and $f_{k+1}^T \Omega f_{k+1} = 0$ because Ω is antisymmetric.

(b) Take $k = N$ in (a) and $r_k = -\frac{1}{2} \log \mu_k$, which is ≥ 0 because $\mu_k \in (0, 1]$. Then, by Lemma 9.6,

$$\mathrm{diag}(\mu_1, \dots, \mu_N, \mu_1^{-1}, \dots, \mu_N^{-1}) = \Sigma(2r) = \Sigma(r)^2,$$

so Lemma 9.9(c) gives $O \in \mathrm{Sp}(2N, \mathbb{R}) \cap \mathrm{O}(2N)$ and $MO = O\Sigma(r)^2$, that is, $M = O\Sigma(r)^2 O^T$. □

Read through (9.1), the conditions on the f_k say that the complex vectors $u_k \leftrightarrow f_k$ form an orthonormal basis of $\mathbb{C}^N$, and $-\Omega f_k \leftrightarrow i u_k$; so $O = O_{\mathcal{U}}$ for the unitary matrix $\mathcal{U}$ with the columns u_k, as Theorem 9.4 requires.

9.4 The decomposition

Theorem 9.11 (Bloch–Messiah decomposition; [4, (A11)])**.** Lean ✓ *Every $S \in \mathrm{Sp}(2N,\mathbb{R})$ can be written as $S = O\,\Sigma(r)\,O'$ with $O, O' \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ and $r \in \mathbb{R}^N$, $r \geq 0$. Moreover, if $S \in \mathrm{Sp}(2N,\mathbb{R})$, $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, $r \in \mathbb{R}^N$ and $SS^T = O\Sigma(r)^2 O^T$, then $\Sigma(-r)O^T S \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$.*

Proof. We prove the second statement first. Put $O' = \Sigma(-r)O^T S$. It is a product of symplectic matrices, $O^T = O^{-1}$ being symplectic, and it is orthogonal: by Lemma 9.6,

$$O'O'^T = \Sigma(-r)O^T SS^T O\,\Sigma(-r) = \Sigma(-r)\Sigma(r)^2\Sigma(-r) = I.$$

For the first statement, Lemma 9.7 and Theorem 9.10 give $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ and $r \geq 0$ with $SS^T = O\Sigma(r)^2 O^T$; with this O', $O\Sigma(r)O' = O\Sigma(r)\Sigma(-r)O^T S = S$. □

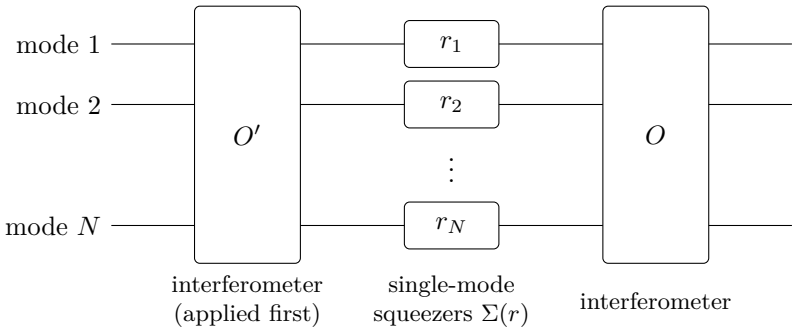


Figure 4: The standard optical reading of Theorem 9.11. A phase-space vector v is mapped to $Sv = O(\Sigma(r)(O'v))$: first the passive transformation O', then on each mode k the squeezer that scales q_k by e^{-r_k} and p_k by e^{r_k}, then the passive transformation O.

The squeezing parameters are determined by S up to order, through the singular values of S, the square roots of the eigenvalues of $S^T S$ counted with multiplicity.

Proposition 9.12 (Singular values)**.** Lean ✓

(a) *Let $O, O' \in \mathrm{O}(2N)$, $r \in \mathbb{R}^N$ and $S = O\Sigma(r)O'$. Then SS^T and $S^T S$ have the characteristic polynomial $\prod_{k=1}^{N}(t - e^{-2r_k})(t - e^{2r_k})$, and the singular values of S, with multiplicity, are $e^{-r_1}, \ldots, e^{-r_N}, e^{r_1}, \ldots, e^{r_N}$. If $r \geq 0$, the map $t \mapsto \frac{1}{2}|\log t|$ sends the multiset of roots of this polynomial to two copies of the multiset $\{r_1, \ldots, r_N\}$.*

(b) *Let $O, O', \tilde{O}, \tilde{O}' \in \mathrm{O}(2N)$ and $r, \tilde{r} \in \mathbb{R}^N$ with $r, \tilde{r} \geq 0$, and let $S = O\Sigma(r)O' = \tilde{O}\Sigma(\tilde{r})\tilde{O}'$. Then $\{r_k\} = \{\tilde{r}_k\}$ as multisets, and $\tilde{r}_k = r_{\pi(k)}$ for a permutation π of $\{1, \ldots, N\}$.*

Proof. (a) $SS^T = O\Sigma(2r)O^T$ and $S^T S = O'^T\Sigma(2r)O'$ are similar to the diagonal matrix $\Sigma(2r)$, whose characteristic polynomial is the product. The singular values are the square roots of its roots, and $\frac{1}{2}|\log e^{\mp 2r_k}| = r_k$ for $r_k \geq 0$.

(b) By (a), the two decompositions give the same polynomial, so two copies of $\{r_k\}$ equal two copies of $\{\tilde{r}_k\}$. Each value occurs in two copies of a multiset exactly twice as often as in the multiset, so $\{r_k\} = \{\tilde{r}_k\}$; two families of N numbers with the same multiset of values differ by a permutation of the indices. □

Proposition 9.13 (Non-uniqueness of the orthogonal factors)**.** Lean ✓ *$-I \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, and $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ implies $-O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$. Every $O \in \mathrm{O}(2N)$ satisfies $-O \neq O$. If $S = O\Sigma(r)O'$, then $S = (-O)\Sigma(r)(-O')$. Hence the orthogonal factors of Theorem 9.11 are never unique.*

Proof. $(-O)\Omega(-O)^T = O\Omega O^T$, $(-O)(-O)^T = OO^T$, $(-O)\Sigma(r)(-O') = O\Sigma(r)O'$, and $I \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$. If $-O = O$, then $O = 0$, which is not orthogonal since $N \geq 1$. □

The non-uniqueness is larger when squeezing parameters coincide. Let $r_j = r_k = r_0$ for two modes $j \neq k$, and let $\mathcal{U} = X + iY$ be a unitary matrix acting on these two modes only. On the coordinates (q_j, q_k, p_j, p_k), $\Sigma(r) = \mathrm{diag}(e^{-r_0} I_2, e^{r_0} I_2)$ and

$$\Sigma(r)\, O_{\mathcal{U}} - O_{\mathcal{U}}\, \Sigma(r) = (e^{r_0} - e^{-r_0}) \begin{pmatrix} 0 & Y \\ Y & 0 \end{pmatrix},$$

so for $r_0 > 0$ the unitaries of the two modes that commute with $\Sigma(r)$ are exactly the real ones, $Y = 0$, and for $r_0 = 0$ all of them commute. Each such $\mathcal{U}$ gives the decomposition $S = (OO_{\mathcal{U}})\Sigma(r)(O_{\mathcal{U}}^T O')$. The exchange of two modes, $(q_j, p_j) \leftrightarrow (q_k, p_k)$, is $O_{\mathcal{U}}$ for a permutation matrix $\mathcal{U}$ and exchanges r_j and r_k. For $r = 0$, that is, for $S \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, every $O \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ occurs, with $O' = O^T S$.

Corollary 9.14 (Polar decomposition; [4, (A4)–(A5)])**.** Lean ✓ *Let $O, O' \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, $r \in \mathbb{R}^N$, $S = O\Sigma(r)O'$ and $P = O\Sigma(r)O^T$. Then P is positive definite, $P \in \mathrm{Sp}(2N,\mathbb{R})$, $P^2 = SS^T$, $P^{-1}S = OO'$, $OO' \in \mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$ and $S = P\,(OO')$. If only $O, O' \in \mathrm{O}(2N)$, still $(SS^T)^{1/2} = P$, where $(\cdot)^{1/2}$ is the positive semidefinite square root. For every $S \in \mathrm{Sp}(2N,\mathbb{R})$, $P = (SS^T)^{1/2}$ is positive definite and lies in $\mathrm{Sp}(2N,\mathbb{R})$, $Q = P^{-1}S$ lies in $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$, and $S = PQ$. If a real $2N \times 2N$ matrix S equals $P'Q'$ with P' positive definite and $Q' \in \mathrm{O}(2N)$, then $P' = (SS^T)^{1/2}$ and $Q' = ((SS^T)^{1/2})^{-1}S$.*

Proof. P is conjugate to $\Sigma(r)$ by the orthogonal matrix O, so it is symmetric with the positive eigenvalues $e^{\pm r_k}$, and it is a product of symplectic matrices. Since $O'O'^T = I$, $P^2 = O\Sigma(r)^2 O^T = O\Sigma(r)O'O'^T\Sigma(r)O^T = SS^T$, which uses only the orthogonality of O and O'; and $P^{-1}S = O\Sigma(-r)O^T O\Sigma(r)O' = OO'$, a product in the group $\mathrm{Sp}(2N,\mathbb{R}) \cap \mathrm{O}(2N)$. A positive semidefinite matrix has exactly one positive semidefinite square root, so $P = (SS^T)^{1/2}$. For $S \in \mathrm{Sp}(2N,\mathbb{R})$, Theorem 9.11 provides the factors. If $S = P'Q'$ as stated, then $SS^T = P'Q'Q'^T P' = P'^2$, so $P' = (SS^T)^{1/2}$, and $Q' = P'^{-1}S$ because P' is invertible. □

Notes and references

Tjoa and Cirac [4, App. A] define $P = (SS^T)^{1/2}$ and $O = P^{-1}S$ (A4), our Q, with P positive and symplectic and O orthogonal symplectic (A5). As Ω maps an eigenvector of P for λ to one for λ^{-1} (A7), they arrange the reciprocal pairs into an orthonormal symplectic eigenbasis, in their notation $P = O_{\mathsf{U}}\Sigma O_{\mathsf{U}}^T$ with $r_j \geq 0$ (A8)–(A10), and with $O_{\mathsf{W}} = O_{\mathsf{U}}^T O$ conclude $S = O_{\mathsf{U}}\Sigma O_{\mathsf{W}}$ (A11). Their O_{U} and O_{W} are our O and O', and their Σ is our $\Sigma(r)$; they call the $e^{\pm r_j}$ the symplectic singular values of S.

We apply the same pairing to $M = SS^T$ instead of P, so the existence proof takes no square root; the arrangement into an orthonormal symplectic eigenbasis is the induction of Theorem 9.10, which includes the eigenvalue 1, where the basis must be orthonormal for (9.1). Corollary 9.14 recovers (A4)–(A5), with $O' = O^T Q$, and Proposition 9.12 makes the uniqueness of r precise.

For one mode, $S\Omega S^T = (\det S)\,\Omega$, so $\mathrm{Sp}(2,\mathbb{R}) = \mathrm{SL}(2,\mathbb{R})$ and $\mathrm{Sp}(2,\mathbb{R}) \cap \mathrm{O}(2) = \mathrm{SO}(2)$, with $O_{e^{i\theta}}$ the rotation by θ. The square root of a positive definite 2×2 matrix M of determinant 1 is $(M + I)/\sqrt{\mathrm{tr}\, M + 2}$, since $(M + I)^2 = (\mathrm{tr}\, M + 2)M$ by the Cayley–Hamilton theorem.

The decomposition carries the names of Bloch and Messiah [10], whose paper concerns the canonical form of an antisymmetric tensor and its application to the theory of superconductivity; it proves the canonical form of the fermionic Bogoliubov transformation, the fermionic analogue of the decomposition that carries their name. Braunstein [11] decomposes a linear optical circuit into an interferometer, single-mode squeezers and a second interferometer through the Bloch–Messiah reduction, and is the reference that [4] gives for the decomposition. In their account of the real symplectic groups in quantum mechanics and optics, Arvind, Dutta, Mukunda and

Simon [12, Sec. IV] write every real symplectic matrix as $S(X_1, Y_1)\, D(\kappa)\, S(X_2, Y_2)$, with unitary outer factors and a diagonal positive middle factor, without the name of Bloch and Messiah. The same section identifies the orthogonal symplectic matrices with the unitary group, as in Theorem 9.4 [12, Sec. IV, Eq. (50)], with the unitary $X - iY$ of the other parametrization noted after that theorem, and gives the polar decomposition of Corollary 9.14 [12, Sec. IV, Eq. (60)]; see also Simon, Mukunda and Dutta [18].

10 Exponential vectors, displacements and squeezing

On the bosonic Fock space the displacement operator $D(\alpha) = e^{\alpha a^\dagger - \bar{\alpha} a}$ and the squeezing operator $\mathsf{S}(\zeta) = e^{(\bar{\zeta} a^2 - \zeta a^{\dagger 2})/2}$ are exponentials of unbounded operators. Their normal-ordered forms, a lowering exponential followed by a raising one, act coefficient by coefficient on the formal Fock space of Section 6, and on the exponential vectors $\varepsilon(z)$ they reduce to identities between exponentials of numbers. Since the exponential vectors span a dense subspace of $\Gamma_s(\iota)$, these identities determine unitary operators, and the same density proves that the displacements act irreducibly.

Notation for this section.

ι, N	a finite set of $N \geq 1$ modes; $\mathcal{F} = \mathcal{F}(\iota, \mathbb{C})$, $\jmath : \Gamma_s(\iota) \to \mathcal{F}$ (Definition 7.1)
$\langle w, z\rangle$, $\lvert z\rvert$	$\sum_j \bar{w}_j z_j$ and $\langle z, z\rangle^{1/2}$ for $w, z \in \mathbb{C}^\iota$
z^n, $n!$, $\lvert n\rvert$	$\prod_j z_j^{n_j}$, $\prod_j n_j!$, $\sum_j n_j$ for a multi-index n
$\varepsilon(z)$	exponential vector (Definition 10.1)
$\Lambda(\kappa, \ell)$, $\lvert \kappa, \ell\rangle$	$\sum_j (\kappa_j a_j^{\dagger 2} + \ell_j a_j^\dagger)$; the one-mode vector $e^{\kappa a^{\dagger 2} + \ell a^\dagger}\lvert 0\rangle = \lvert 0, \kappa, \ell\rangle$ (Definition 10.6)
$e^T \psi$	the exponential, coefficientwise (Definitions 6.6 and 8.8)
$D^{\mathrm{ord}}(\alpha)$, $D(\alpha)$	normal-ordered and unitary displacement (Definition 10.11, Theorem 10.12)
$\zeta = re^{i\phi}$; c, s, τ	squeezing parameter; $\cosh r$, $\sinh r$, $e^{i\phi}\tanh r$ (Definition 10.20)
$\mathsf{S}^{\mathrm{ord}}(\zeta)$, $\mathsf{S}(\zeta)$, $R(\theta)$	squeezing and rotation operators (Definition 10.20, Theorem 10.21)
L_ζ	$(L_\zeta \beta)_j = c_j \beta_j - e^{i\phi_j} s_j \bar{\beta}_j$ (Theorem 10.21)
$\mathcal{S}(y)$	$\sum_k \binom{2k}{k} y^k$, the binomial series of $(1 - 4y)^{-1/2}$ (Lemma 10.18)
$\mathsf{U}_\mathcal{U}$, $\bar{T}$, $\psi_{j,n}$	passive unitary (Theorem 8.7); maximal realization (Definition 7.6); mode-j fiber (Definition 7.4)

In this section $z, w \in \mathbb{C}^\iota$ are the arguments of exponential vectors, not the variables of the Bargmann picture, which appears only in the proof of Lemma 10.19, in the variable x. The letter φ denotes a finitely supported vector and ϕ a squeezing phase; k is a summation index.

10.1 Exponential vectors and totality

Definition 10.1 (Exponential vector)**.** For $z \in \mathbb{C}^\iota$, the *exponential vector* $\varepsilon(z) \in \mathcal{F}$ has the coefficients $\varepsilon(z)(n) = z^n/\sqrt{n!}$. When it is square-summable (Lemma 10.3), $\varepsilon(z)$ also denotes the vector of $\Gamma_s(\iota)$ with these coefficients, and $\lvert 0\rangle \in \Gamma_s(\iota)$ is the vacuum.

The vector $e^{-|z|^2/2}\varepsilon(z)$ is the coherent state with amplitudes z_j [27]. Inner products of exponential vectors are sums over multi-indices of products over the modes, and we use the following form of the product of absolutely convergent series.

Lemma 10.2 (Products of absolutely convergent series)**.** Lean ✓ *Let A be a complete normed commutative ring, ι a finite set, and $g_i : \mathbb{N} \to A$, $i \in \iota$, with $\sum_k \lVert g_i(k)\rVert < \infty$ for every i. Then the family $f \mapsto \prod_i g_i(f(i))$, indexed by the maps $f : \iota \to \mathbb{N}$, is summable in norm, and its sum is $\prod_i \sum_k g_i(k)$.*

Proof. By Lemma 7.11(a) and $\|xy\| \le \|x\|\|y\|$, $\sum_f \|\prod_i g_i(f(i))\| \le \prod_i \sum_k \|g_i(k)\| < \infty$. For the sum we argue by induction on the number of elements of ι; for $\iota = \emptyset$ both sides are 1. Adding an element i_0, the maps on $\iota \cup \{i_0\}$ are the pairs (k, f) of a value at i_0 and a map on ι. An absolutely summable family over $\mathbb{N} \times \mathbb{N}^\iota$ may be summed first over f and then over k, and its sum is $\sum_k g_{i_0}(k) \sum_f \prod_{i \in \iota} g_i(f(i))$, the product of the two sums. □

Lemma 10.3 (Exponential vectors)**.** Lean ✓ *Let $w, z \in \mathbb{C}^\iota$.*

(a) *$\varepsilon(z) = e^{z \cdot a^\dagger}|0\rangle$, the exponential of $z \cdot a^\dagger = \sum_j z_j a_j^\dagger$ of Definition 6.6. The series $\sum_n \overline{\varepsilon(w)(n)}\, \varepsilon(z)(n)$ converges absolutely to $e^{\langle w,z\rangle}$. Hence $\varepsilon(z)$ is square-summable, $\langle \varepsilon(w), \varepsilon(z)\rangle = e^{\langle w,z\rangle}$, $\|\varepsilon(z)\|^2 = e^{|z|^2}$, and $z \mapsto \varepsilon(z)$ is continuous from $\mathbb{C}^\iota$ to $\Gamma_s(\iota)$.*

(b) *On $\mathcal{F}$, $a_j \varepsilon(z) = z_j \varepsilon(z)$ for every mode j, and $\Gamma(g)\varepsilon(z) = \varepsilon(gz)$ for every $g \in M_\iota(\mathbb{C})$.*

(c) *$\varepsilon(z) = \bigotimes_j \varepsilon(z_j)$ with one-mode exponential vectors $\varepsilon(z_j)$, and $\varepsilon(0) = |0\rangle$.*

Proof. (a) The operators $z_j a_j^\dagger$ commute, and $(a^\dagger)^n|0\rangle = \sqrt{n!}\,|n\rangle$ by Definition 6.2, so the multinomial theorem gives $(z \cdot a^\dagger)^k|0\rangle = \sum_{|n|=k} (k!/n!)\, z^n \sqrt{n!}\,|n\rangle$. At the coefficient n only the term $k = |n|$ contributes, and it is $z^n/\sqrt{n!}$. The terms of the series are $\prod_j (\bar{w}_j z_j)^{n_j}/n_j!$, so Lemma 10.2 with $g_j(k) = (\bar{w}_j z_j)^k/k!$ gives absolute convergence to $\prod_j e^{\bar{w}_j z_j} = e^{\langle w,z\rangle}$. With $w = z$ the squared norm is $e^{|z|^2} < \infty$, so $\varepsilon(z) \in \jmath(\Gamma_s(\iota))$ (Lemma 7.2(b)), and the inner product is the sum of the series. Finally $\|\varepsilon(z) - \varepsilon(w)\|^2 = e^{|z|^2} + e^{|w|^2} - 2\,\mathrm{Re}\, e^{\langle w,z\rangle}$ tends to 0 as $w \to z$.

(b) $(a_j \varepsilon(z))(n) = \sqrt{n_j + 1}\, z^{n+e_j}/\sqrt{(n+e_j)!} = z_j \varepsilon(z)(n)$. By Theorem 8.5 and Lemma 8.3(b), $\Gamma(g)(z \cdot a^\dagger)^k|0\rangle = ((gz) \cdot a^\dagger)^k|0\rangle$, since $\sum_j z_j \sum_k g_{kj} a_k^\dagger = (gz) \cdot a^\dagger$. Both sides lie in the sector of k particles, and $\Gamma(g)$ acts sector by sector (Lemma 8.3(a)), so summing the sector components gives $\Gamma(g)\varepsilon(z) = \varepsilon(gz)$.

(c) $z^n = \prod_j z_j^{n_j}$ and $\sqrt{n!} = \prod_j \sqrt{n_j!}$, since $n! = \prod_j n_j!$ and square roots of non-negative numbers are multiplicative; Definition 7.4(a) then gives the product. At $z = 0$ only $n = 0$ contributes. □

The density of the exponential vectors rests on an identity theorem for their inner products, which are power series in $\bar{z}$; the device of (iii) below, rotating the argument by a phase, separates the two variables z and $\bar{z}$.

Lemma 10.4 (Identity theorem for exponential vectors)**.** Lean ✓ *Let $\psi, \psi' \in \Gamma_s(\iota)$.*

(i) *If $x_k \in \mathbb{C}$ and $\sum_k x_k t^k$ converges to 0 for every real t, then $x_k = 0$ for every k.*

(ii) *For every $w \in \mathbb{C}^\iota$ and real t, $\sum_k t^k \sum_{|n|=k} \bar{w}^n \psi(n)/\sqrt{n!}$ converges to $\langle \varepsilon(tw), \psi\rangle$.*

(iii) *If $\langle \varepsilon(z), \psi\rangle = \langle \psi', \varepsilon(z)\rangle$ for every $z \in \mathbb{C}^\iota$, then $\psi(n) = \psi'(n) = 0$ for every $n \neq 0$, and $\psi(0) = \overline{\psi'(0)}$.*

Proof. (i) A power series that converges at every real t has infinite radius of convergence, so its sum is a real-analytic function of t whose kth derivative at 0 is $k!\, x_k$; the sum vanishes identically, hence so do its derivatives.

(ii) $\langle \varepsilon(tw), \psi\rangle = \sum_n t^{|n|} \bar{w}^n \psi(n)/\sqrt{n!}$ converges absolutely, by the Cauchy–Schwarz inequality for the square-summable families $\varepsilon(tw)$ and ψ, so its terms may be grouped into the finite sums over $|n| = k$.

(iii) Fix w and write $P_k(w) = \sum_{|n|=k} \bar{w}^n \psi(n)/\sqrt{n!}$ and $Q_k(w) = \sum_{|n|=k} w^n \overline{\psi'(n)}/\sqrt{n!}$. By (ii) and its complex conjugate, $\sum_k t^k (P_k(w) - Q_k(w))$ converges to 0 for every real t, so $P_k(w) = Q_k(w)$ by (i), for every w. Replacing w by $e^{i\theta}w$ multiplies $P_k(w)$ by $e^{-ik\theta}$ and $Q_k(w)$ by $e^{ik\theta}$; for $k \geq 1$ and $\theta = \pi/(2k)$ this gives $-iP_k(w) = iQ_k(w)$. Together with $P_k(w) = Q_k(w)$, both vanish for every w. A polynomial in $\bar{w}$, or in w, that vanishes on $\mathbb{C}^\iota$ has zero coefficients, so $\psi(n) = \psi'(n) = 0$ for $|n| = k \geq 1$. At $k = 0$, $P_0 = Q_0$ reads $\psi(0) = \overline{\psi'(0)}$. □

Theorem 10.5 (Totality). Lean ✓ *The exponential vectors $\varepsilon(z)$, $z \in \mathbb{C}^\iota$, span a dense subspace of $\Gamma_s(\iota)$. Hence two continuous linear maps from $\Gamma_s(\iota)$ to a Hausdorff topological vector space over $\mathbb{C}$ that agree on every $\varepsilon(z)$ are equal, and $\psi = \psi'$ in $\Gamma_s(\iota)$ if $\langle \varepsilon(w), \psi\rangle = \langle \varepsilon(w), \psi'\rangle$ for every $w \in \mathbb{C}^\iota$.*

Proof. If ψ is orthogonal to every $\varepsilon(z)$, Lemma 10.4(iii) with $\psi' = 0$ gives $\psi = 0$; so the orthogonal complement of the span is 0, and the span is dense. Two continuous linear maps that agree on the span agree on its closure, since the set where two continuous maps into a Hausdorff space agree is closed. The last statement is the case of the maps $\psi \mapsto \langle \varepsilon(w), \psi\rangle$, applied to $\psi - \psi'$, which is orthogonal to every $\varepsilon(w)$. □

10.2 Lowering exponentials and extension

The normal-ordered forms of $D(\alpha)$ and $\mathsf{S}(\zeta)$ end with an exponential of creation operators, which raises the particle number. We name the exponents that occur.

Definition 10.6 (Quadratic raising operators; vectors with at most ν particles). (a) For $\kappa, \ell \in \mathbb{C}^\iota$, $\Lambda(\kappa, \ell) = \sum_j (\kappa_j a_j^{\dagger 2} + \ell_j a_j^\dagger)$ on $\mathcal{F}$. For one mode and $\kappa, \ell \in \mathbb{C}$, $|\kappa, \ell\rangle = e^{\kappa a^{\dagger 2} + \ell a^\dagger}|0\rangle$, the vector $|0, \kappa, \ell\rangle$ of Definition 7.12.

(b) For $\nu \in \mathbb{N}$, a vector $\psi \in \mathcal{F}$ *vanishes above* ν if $\psi(n) = 0$ for every n with $|n| > \nu$.

By Lemma 6.7(c), $\Lambda(\kappa, \ell)$ raises the particle number, so $e^{\Lambda(\kappa,\ell)}$ is a finite sum at each coefficient; $\Lambda(0, 0) = 0$, with exponential the identity. The vector $|\kappa, \ell\rangle$ is the paper's $|\kappa, \ell\rangle$ of [4, (8)], and $\Lambda(0, z) = z \cdot a^\dagger$, so $\varepsilon(z) = e^{\Lambda(0,z)}|0\rangle$. A vector vanishes above ν when it lies in the sum of the sectors of $0, \ldots, \nu$ particles; $|m\rangle$ vanishes above $|m|$, and for $\nu = 0$ the vector is a multiple of $|0\rangle$.

Lemma 10.7 (Quadratic raising operators). Lean ✓ *Let $\kappa, \ell \in \mathbb{C}^\iota$ and $\nu, \nu' \in \mathbb{N}$.*

(a) *If ψ vanishes above ν, it vanishes above every $\nu' \geq \nu$. Sums, finite sums and scalar multiples of vectors that vanish above ν vanish above ν, and so does $\Gamma(g)\psi$ for every $g \in M_\iota(\mathbb{C})$; $a_j^\dagger \psi$ vanishes above $\nu + 1$ and $\Lambda(\kappa, \ell)\psi$ above $\nu + 2$.*

(b) *A vector that vanishes above some ν is finitely supported and lies in $\jmath(\Gamma_s(\iota))$; for every finitely supported $\psi \in \Gamma_s(\iota)$, $\jmath\psi$ vanishes above some ν.*

(c) *For one-mode vectors $\chi_j \in \mathcal{F}(\{1\}, \mathbb{C})$, $e^{\Lambda(\kappa,\ell)} \bigotimes_j \chi_j = \bigotimes_j e^{\kappa_j a^{\dagger 2} + \ell_j a^\dagger} \chi_j$; in particular $e^{\Lambda(\kappa,\ell)}|0\rangle = \bigotimes_j |\kappa_j, \ell_j\rangle$.*

Proof. (a) The first three claims hold coefficientwise; $\Gamma(g)$ preserves each sector (Lemma 8.3(a)), and the coefficient $(a_j^\dagger \psi)(n)$ involves only $\psi(n - e_j)$.

(b) On finitely many modes there are finitely many multi-indices with $|n| \leq \nu$, and a finitely supported vector is square-summable. Conversely take ν the largest $|n|$ over the support.

(c) The summands $\kappa_j a_j^{\dagger 2} + \ell_j a_j^\dagger$ commute and raise the particle number, so by Theorem 6.8 $e^{\Lambda(\kappa,\ell)}$ is the product of their exponentials. The exponential of the summand of mode j is the one-mode Gaussian operator acting on mode j (Definition 7.4(c)), which on a product vector changes only the factor at j. The vacuum is $\bigotimes_j |0\rangle$. □

The lowering exponentials are series at every coefficient. On $\Gamma_s(\iota)$ each coefficient is an inner product with a vector of $\Gamma_s(\iota)$, the exponential of the adjoint raising operator applied to a number state; this is where the condition $|\tau_j| < 1$ on the squeezing enters.

Lemma 10.8 (Lowering exponentials). Lean ✓ *Let n be a multi-index.*

(a) *Let $\alpha \in \mathbb{C}^\iota$. For every $\psi \in \Gamma_s(\iota)$, $\sum_k |((-\bar\alpha \cdot a)^k \jmath\psi)(n)|/k! < \infty$, where $\bar\alpha \cdot a = \sum_j \bar\alpha_j a_j$; and there is $f_n \in \Gamma_s(\iota)$ with $\jmath f_n = e^{\Lambda(0,-\alpha)}|n\rangle$ and $(e^{-\bar\alpha\cdot a}\jmath\psi)(n) = \langle f_n, \psi\rangle$ for every $\psi \in \Gamma_s(\iota)$.*

(b) *Let $\tau \in \mathbb{C}^\iota$ with $|\tau_j| < 1$ for every j. The same holds for the lowering operator $\frac12 \sum_j \bar\tau_j a_j^2$, with $\jmath f_n = e^{\Lambda(\tau/2,0)}|n\rangle$. If $|\tau_j| \geq 1$ for some j, then $e^{\Lambda(\tau/2,0)}|0\rangle$ is not square-summable; for one mode, $\||\kappa, 0\rangle\|^2 = \infty$ when $|\kappa| \geq \frac12$.*

(c) *$\psi \mapsto (e^{-\bar\alpha\cdot a}\jmath\psi)(n)$ is continuous on $\Gamma_s(\iota)$, and so is $\psi \mapsto (e^{\frac12\sum_j \bar\tau_j a_j^2}\jmath\psi)(n)$ when $|\tau_j| < 1$ for every j.*

Proof. Write B for the lowering operator, $-\bar\alpha \cdot a$ or $\frac12 \sum_j \bar\tau_j a_j^2$, and $B^\dagger$ for $\Lambda(0,-\alpha)$ or $\Lambda(\tau/2, 0)$. For a finitely supported φ, $\langle a_j^\dagger \varphi, \psi\rangle = \langle \varphi, a_j\psi\rangle$ by Definition 6.2, so $(B^k\jmath\psi)(n) = \langle (B^\dagger)^k|n\rangle, \psi\rangle$. The vector $x_k = (B^\dagger)^k|n\rangle/k!$ lies in the sector of $|n| + k$, respectively $|n| + 2k$, particles; these sectors are distinct for distinct k. Let $P_k\psi$ be the component of ψ in that sector. By the Cauchy–Schwarz inequality in each sector and then over k,

$$\sum_k \frac{|(B^k\jmath\psi)(n)|}{k!} \leq \sum_k \|x_k\|\,\|P_k\psi\| \leq \Big(\sum_k \|x_k\|^2\Big)^{1/2}\|\psi\|.$$

The orthogonal sum $\sum_k x_k = e^{B^\dagger}|n\rangle$ is a product vector by Lemma 10.7(c), since $|n\rangle = \bigotimes_j |n_j\rangle$, and by Definition 6.2 its one-mode factors are

$$e^{\beta a^\dagger}|m\rangle = \sum_k \frac{\beta^k}{k!}\sqrt{\tfrac{(m+k)!}{m!}}\,|m+k\rangle, \qquad e^{\kappa a^{\dagger 2}}|m\rangle = \sum_k \frac{\kappa^k}{k!}\sqrt{\tfrac{(m+2k)!}{m!}}\,|m+2k\rangle,$$

with $\beta = -\alpha_j$, respectively $\kappa = \tau_j/2$. Their squared norms $\sum_k |\beta|^{2k}(m+k)!/(k!^2 m!)$ and $\sum_k |\kappa|^{2k}(m+2k)!/(k!^2 m!)$ have term ratios tending to 0 and to $4|\kappa|^2$, so they are finite when $|\kappa| < \frac12$. By Lemma 7.11(b), $\sum_k \|x_k\|^2 = \|e^{B^\dagger}|n\rangle\|^2 < \infty$, so $e^{B^\dagger}|n\rangle = \jmath f_n$ for some $f_n \in \Gamma_s(\iota)$, the series converges absolutely, and its sum is $\sum_k \langle x_k, \psi\rangle = \langle f_n, \psi\rangle$. This proves (a) and the first part of (b), and (c) follows, since $\psi \mapsto \langle f_n, \psi\rangle$ is a continuous linear functional.

For $m = 0$ and $|\kappa| \geq \frac12$ the terms of the second squared norm are $|2\kappa|^{2k}\binom{2k}{k}4^{-k} \geq \binom{2k}{k}4^{-k} \geq 1/(2\sqrt{k})$ for $k \geq 1$, so the series diverges. If $|\tau_j| \geq 1$, the factor of $e^{\Lambda(\tau/2,0)}|0\rangle = \bigotimes_j |\tau_j/2, 0\rangle$ at j has infinite squared norm and every factor has coefficient 1 at 0, so the product has infinite squared norm by Lemma 7.11(b). □

An operator on $\Gamma_s(\iota)$ is fixed by its values on the exponential vectors, and an isometry can be constructed from them. The next lemma is the general form of this construction, for any family with a dense span.

Lemma 10.9 (Extension of a map that preserves inner products). Lean ✓ *Let $\mathcal{H}_1$, $\mathcal{H}_2$ be complex inner product spaces, $\mathcal{H}_2$ complete, and $(x_a)_{a\in A}$, $(x'_a)_{a\in A}$ families in $\mathcal{H}_1$ and $\mathcal{H}_2$ with $\langle x_a, x_b\rangle = \langle x'_a, x'_b\rangle$ for all $a, b \in A$, such that the span of the x_a is dense. Then there is exactly one linear isometry $T : \mathcal{H}_1 \to \mathcal{H}_2$ with $Tx_a = x'_a$ for every a.*

Proof. For a finitely supported family of coefficients c, $\|\sum_a c_a x'_a\|^2 = \sum_{a,b} \bar c_a c_b \langle x'_a, x'_b\rangle = \|\sum_a c_a x_a\|^2$. Applied to the difference of two families, this shows that $\sum_a c_a x'_a$ depends only on $\sum_a c_a x_a$, so $T_0(\sum_a c_a x_a) = \sum_a c_a x'_a$ is a well-defined linear isometry on the span. A uniformly continuous map from a dense subspace into a complete space has exactly one continuous extension to the whole space, and the extension of T_0 is linear and isometric by continuity. Two continuous maps that agree on the dense span agree everywhere, which gives uniqueness. □

Lemma 10.10 (Extension from exponential vectors)**.** Lean ✓ *Let $T : \Gamma_s(\iota) \to \mathcal{F}$ be linear, with every coefficient $\psi \mapsto (T\psi)(n)$ continuous.*

(a) *If $V : \mathbb{C}^\iota \to \Gamma_s(\iota)$ satisfies $T\varepsilon(z) = \jmath V(z)$ and $\langle V(w), V(z)\rangle = \langle \varepsilon(w), \varepsilon(z)\rangle$ for all w, z, there is a linear isometry L of $\Gamma_s(\iota)$ with $\jmath L\psi = T\psi$ for every ψ.*

(b) *If T' is another such map and $T\varepsilon(z) = T'\varepsilon(z)$ for every z, then $T = T'$.*

Proof. (a) Lemma 10.9, with $x_z = \varepsilon(z)$, $x'_z = V(z)$ and the dense span of Theorem 10.5, gives a linear isometry L with $L\varepsilon(z) = V(z)$. The maps $\jmath \circ L$ and T are linear with continuous coefficients, since each coefficient of $\jmath$ is continuous on $\Gamma_s(\iota)$, and they agree on every $\varepsilon(z)$; by (b) they are equal.

(b) For each n, the continuous linear functionals $\psi \mapsto (T\psi)(n)$ and $\psi \mapsto (T'\psi)(n)$ agree on every $\varepsilon(z)$, hence everywhere by Theorem 10.5. □

10.3 Displacement operators

Definition 10.11 (Normal-ordered displacement)**.** For $\alpha \in \mathbb{C}^\iota$ and $\psi \in \Gamma_s(\iota)$,

$$D^{\mathrm{ord}}(\alpha)\jmath\psi = e^{-|\alpha|^2/2}\, e^{\Lambda(0,\alpha)}\big(e^{-\bar\alpha\cdot a}\jmath\psi\big) :$$

the lowering exponential, summed coefficientwise (Lemma 10.8(a)), then the raising exponential, a finite sum at each coefficient, then the factor $e^{-|\alpha|^2/2}$.

For commuting $A = \alpha \cdot a^\dagger$ and $B = -\bar\alpha \cdot a$ with central commutator $[A, B] = |\alpha|^2$, the Baker–Campbell–Hausdorff formula gives the formal identity $e^{A+B} = e^{-|\alpha|^2/2}e^A e^B$; the definition takes its right side, which is the standard normal-ordered displacement.

Theorem 10.12 (Displacement operators)**.** Lean ✓ *Let $\alpha, \beta, z \in \mathbb{C}^\iota$.*

(a) *$\psi \mapsto D^{\mathrm{ord}}(\alpha)\jmath\psi$ is a linear map $\Gamma_s(\iota) \to \mathcal{F}$ with continuous coefficients, and $D^{\mathrm{ord}}(\alpha)\varepsilon(z) = e^{-|\alpha|^2/2-\langle\alpha,z\rangle}\varepsilon(z+\alpha)$.*

(b) *There is a unitary operator $D(\alpha)$ on $\Gamma_s(\iota)$, the* displacement operator*, with $\jmath D(\alpha)\psi = D^{\mathrm{ord}}(\alpha)\jmath\psi$ for every $\psi \in \Gamma_s(\iota)$. It satisfies $D(\alpha)\varepsilon(z) = e^{-|\alpha|^2/2-\langle\alpha,z\rangle}\varepsilon(z+\alpha)$, $D(\alpha)D(-\alpha) = 1$, $D(\alpha)^* = D(-\alpha)$ and $D(0) = 1$.*

(c) *(Weyl relation) $D(\alpha)D(\beta) = \exp\big(\frac{1}{2}\sum_j(\alpha_j\bar\beta_j - \bar\alpha_j\beta_j)\big)D(\alpha+\beta) = e^{(\langle\beta,\alpha\rangle-\langle\alpha,\beta\rangle)/2}D(\alpha+\beta)$; if $\sum_j(\alpha_j\bar\beta_j - \bar\alpha_j\beta_j) = 0$, then $D(\alpha)D(\beta) = D(\alpha+\beta)$.*

(d) *$D(\alpha)|0\rangle = e^{-|\alpha|^2/2}\varepsilon(\alpha)$, the coherent state of [4, (5)]. For an enumeration $j_1, \dots, j_N$ of the modes, $D(\alpha) = D(\alpha_{j_1}e_{j_1})\cdots D(\alpha_{j_N}e_{j_N})$. For $\eta \in \mathbb{C}$, $\psi \in \Gamma_s(\iota)$ and a multi-index n, $(D(\eta e_j)\psi)(n) = (D(\eta)\psi_{j,n})(n_j)$, with $D(\eta)$ the one-mode displacement and $\psi_{j,n}$ the mode-j fiber.*

Proof. (a) Each coefficient of $D^{\mathrm{ord}}(\alpha)\jmath\psi$ is a finite linear combination of coefficients of $e^{-\bar\alpha\cdot a}\jmath\psi$, and these are the continuous linear functionals $\langle f_n, \psi\rangle$ of Lemma 10.8(a). By Lemma 10.3(b), $(-\bar\alpha \cdot a)^k\varepsilon(z) = (-\langle\alpha, z\rangle)^k\varepsilon(z)$, so the lowering exponential multiplies $\varepsilon(z)$ by $e^{-\langle\alpha,z\rangle}$. The raising operators $\Lambda(0,\alpha)$ and $\Lambda(0,z)$ commute, so by Theorem 6.8 $e^{\Lambda(0,\alpha)}\varepsilon(z) = e^{\Lambda(0,\alpha)}e^{\Lambda(0,z)}|0\rangle = \varepsilon(z+\alpha)$.

(b) With $V(z) = e^{-|\alpha|^2/2-\langle\alpha,z\rangle}\varepsilon(z+\alpha)$,

$$\langle V(w), V(z)\rangle = e^{-|\alpha|^2-\langle w,\alpha\rangle-\langle\alpha,z\rangle+\langle w+\alpha,z+\alpha\rangle} = e^{\langle w,z\rangle},$$

so Lemma 10.10(a) gives a linear isometry $D(\alpha)$ with $\jmath D(\alpha)\psi = D^{\mathrm{ord}}(\alpha)\jmath\psi$; this relation determines it, because $\jmath$ is injective. Applying the formula twice, $D(\alpha)D(-\alpha)\varepsilon(z) = \varepsilon(z)$, so $D(\alpha)D(-\alpha) = 1$ by Theorem 10.5. Hence $D(\alpha)$ is onto, a surjective isometry is unitary, and $D(\alpha)^* = D(\alpha)^{-1} = D(-\alpha)$. $D^{\mathrm{ord}}(0)$ is the identity.

(c) On $\varepsilon(z)$ the left side gives $e^{-|\beta|^2/2-\langle\beta,z\rangle-|\alpha|^2/2-\langle\alpha,z+\beta\rangle}\varepsilon(z+\alpha+\beta)$ and $D(\alpha+\beta)$ gives $e^{-|\alpha+\beta|^2/2-\langle\alpha+\beta,z\rangle}\varepsilon(z+\alpha+\beta)$. Since $|\alpha+\beta|^2=|\alpha|^2+|\beta|^2+\langle\alpha,\beta\rangle+\langle\beta,\alpha\rangle$, the ratio is $e^{(\langle\beta,\alpha\rangle-\langle\alpha,\beta\rangle)/2}$, and $\langle\beta,\alpha\rangle-\langle\alpha,\beta\rangle=\sum_j(\alpha_j\bar\beta_j-\bar\alpha_j\beta_j)$. The two bounded operators agree on every $\varepsilon(z)$, hence are equal by Theorem 10.5.

(d) The first statement is (b) at $z=0$. For vectors with disjoint supports the phase in (c) vanishes, so $D(\alpha)$ is the product of the $D(\alpha_j e_j)$ by induction along the enumeration. For the fiber, both sides are continuous linear functionals of ψ: the map $\psi\mapsto\psi_{j,n}$ from $\Gamma_s(\iota)$ to the one-mode Fock space has norm at most 1. On $\varepsilon(z)=\bigotimes_k\varepsilon(z_k)$, $D(\eta e_j)$ multiplies by $e^{-|\eta|^2/2-\bar\eta z_j}$ and replaces the factor $\varepsilon(z_j)$ by $\varepsilon(z_j+\eta)$, while the fiber of $\varepsilon(z)$ along j is $\prod_{k\neq j}\varepsilon(z_k)(n_k)$ times $\varepsilon(z_j)$, on which the one-mode $D(\eta)$ acts in the same way. Theorem 10.5 concludes. □

On the operators a_j, $a_j^\dagger$, which are unbounded on $\Gamma_s(\iota)$, the displacement acts through their maximal realizations of Definition 7.6, which need no choice of domain.

Theorem 10.13 (Displacement of the ladder operators). Lean ✓ *Let $\alpha\in\mathbb{C}^\iota$, j a mode and $\psi,\xi\in\Gamma_s(\iota)$. If $a_j\jmath\psi=\jmath\xi$, then $a_j\jmath D(\alpha)\psi=\jmath D(\alpha)\xi+\alpha_j\jmath D(\alpha)\psi$; if $a_j^\dagger\jmath\psi=\jmath\xi$, then $a_j^\dagger\jmath D(\alpha)\psi=\jmath D(\alpha)\xi+\bar\alpha_j\jmath D(\alpha)\psi$. Moreover $D(\alpha)\psi\in\operatorname{dom}\bar a_j$ if and only if $\psi\in\operatorname{dom}\bar a_j$. Thus $D(\alpha)$ maps $\operatorname{dom}\bar a_j$ onto itself, and on it $D(\alpha)^*\bar a_jD(\alpha)=\bar a_j+\alpha_j$ and $D(\alpha)^*\overline{a_j^\dagger}D(\alpha)=\overline{a_j^\dagger}+\bar\alpha_j$: the relation $D^\dagger aD=a+\alpha$ and [4, (A31)].*

Proof. Write $B=-\bar\alpha\cdot a$ and $A=\Lambda(0,\alpha)$. Since a_j commutes with B, and the series of $e^B\jmath\psi$ and of $e^Ba_j\jmath\psi$ converge absolutely at every coefficient (Lemma 10.8(a)), $a_je^B\jmath\psi=e^Ba_j\jmath\psi$ coefficient-wise: $(a_jx)(n)$ is a multiple of the single coefficient $x(n+e_j)$. Next, $[a_j,A]=\alpha_j$ commutes with A, so $a_jA^k=A^ka_j+k\alpha_jA^{k-1}$, and summing the finitely many terms at each coefficient gives $a_je^Ax=e^A(a_j+\alpha_j)x$ for every $x\in\mathcal{F}$. Together, $a_jD^{\mathrm{ord}}(\alpha)\jmath\psi=D^{\mathrm{ord}}(\alpha)a_j\jmath\psi+\alpha_jD^{\mathrm{ord}}(\alpha)\jmath\psi$, which is the first identity. For $a_j^\dagger$: it commutes with A, and $[B,a_j^\dagger]=-\bar\alpha_j$ gives $a_j^\dagger B^k=B^ka_j^\dagger+k\bar\alpha_jB^{k-1}$; when $a_j^\dagger\jmath\psi\in\jmath(\Gamma_s(\iota))$ both series converge absolutely and $a_j^\dagger e^B\jmath\psi=e^Ba_j^\dagger\jmath\psi+\bar\alpha_je^B\jmath\psi$.

If $\psi\in\operatorname{dom}\bar a_j$, the first identity shows $a_j\jmath D(\alpha)\psi\in\jmath(\Gamma_s(\iota))$, so $D(\alpha)\psi\in\operatorname{dom}\bar a_j$; the converse is the same statement for $D(-\alpha)=D(\alpha)^{-1}$. Applying $D(\alpha)^*=D(-\alpha)$ to the two identities gives the operator forms. □

The derivative of the one-parameter group $t\mapsto D(t\alpha)$ on finitely supported vectors needs a norm estimate of the raising exponential, uniform for small parameters. The same estimate serves the squeezing operators below.

Lemma 10.14 (Derivative of a quadratic raising exponential). Lean ✓ *Let $\kappa,\ell:\mathbb{R}\to\mathbb{C}^\iota$ vanish at 0 and be differentiable at 0, with derivatives κ',ℓ'. Let m be a multi-index, $W:\mathbb{R}\to\Gamma_s(\iota)$ with $\jmath W(t)=e^{\Lambda(\kappa(t),\ell(t))}|m\rangle$ for every t, and $G\in\Gamma_s(\iota)$ with $\jmath G=\Lambda(\kappa',\ell')|m\rangle$. Then W is differentiable at 0 with $W'(0)=G$.*

Proof. Write $T_t=\Lambda(\kappa(t),\ell(t))$ and $\|x\|_1=\sum_n|x(n)|$; for finitely supported x, $\|x\|\le\|x\|_1$. If x vanishes above ν, the factor $\sqrt{n_j}$ in $(a_j^\dagger x)(n)$ is at most $\sqrt{\nu+1}$ on the support, so $\|a_j^\dagger x\|_1\le\sqrt{\nu+1}\|x\|_1$, and $\|T_tx\|_1\le\mu(t)(\nu+2)\|x\|_1$ with $\mu(t)=\sum_j(|\kappa_j(t)|+|\ell_j(t)|)$. Since $T_t^k|m\rangle$ vanishes above $|m|+2k$ (Lemma 10.7(a)), induction gives

$$\frac{\|T_t^k|m\rangle\|_1}{k!}\le\frac{\mu(t)^k}{k!}\prod_{i=1}^k(|m|+2i)\le(2\mu(t))^k\binom{M+k}{k},\qquad M=|m|.$$

For $2\mu(t)\le\frac12$, $(2\mu(t))^k\le(2\mu(t))^2\,2^{2-k}$ for $k\ge2$, and the series $\sum_k\binom{M+k}{k}x^k=(1-x)^{-M-1}$ at $x=\frac12$ gives $\sum_{k\ge2}\|T_t^k|m\rangle\|/k!\le(2\mu(t))^2\,2^{M+3}$. At each coefficient $e^{T_t}|m\rangle$ is the finite sum of the terms $T_t^k|m\rangle/k!$, and these converge absolutely in $\Gamma_s(\iota)$, so $\|W(t)-|m\rangle-T_t|m\rangle\|\le(2\mu(t))^2 2^{M+3}$,

which is $o(t)$ because $\mu(t) = O(t)$. The vector $T_t|m\rangle$ is linear in $(\kappa(t), \ell(t))$ with values in the finite-dimensional span of finitely many number states, so $(T_t|m\rangle - t\,\Lambda(\kappa', \ell')|m\rangle)/t \to 0$. Since $W(0) = |m\rangle$, $(W(t) - W(0) - tG)/t \to 0$. □

Theorem 10.15 (One-parameter groups of displacements). Lean ✓ *Fix $\alpha \in \mathbb{C}^\iota$.*

(a) *$D(t\alpha)D(t'\alpha) = D((t+t')\alpha)$ for all real t, t'.*

(b) *For every $\psi \in \Gamma_s(\iota)$, $\alpha \mapsto D(\alpha)\psi$ is continuous from $\mathbb{C}^\iota$ to $\Gamma_s(\iota)$.*

(c) *Let $\varphi \in \Gamma_s(\iota)$ be finitely supported, and $G \in \Gamma_s(\iota)$ with $\jmath G = \sum_j (\alpha_j a_j^\dagger - \bar\alpha_j a_j)\jmath\varphi$. Then $t \mapsto D(t\alpha)\varphi$ is differentiable at $t = 0$ with derivative G.*

The vector G exists and is unique: the generator maps the finitely supported $\jmath\varphi$ to a finitely supported vector (Lemma 10.7), and $\jmath$ is injective.

Proof. (a) The phase of the Weyl relation is $\exp(tt' \sum_j (\alpha_j\bar\alpha_j - \bar\alpha_j\alpha_j)/2) = 1$.

(b) For $\psi = \varepsilon(z)$ the formula of Theorem 10.12(b) is continuous in α, by Lemma 10.3(a), and so is $D(\alpha)f$ for f in the span of the exponential vectors. For general ψ and $\epsilon > 0$ choose such f with $\|\psi - f\| < \epsilon/3$ (Theorem 10.5); since every $D(\beta)$ is an isometry, $\|D(\alpha)\psi - D(\beta)\psi\| < \frac{2}{3}\epsilon + \|D(\alpha)f - D(\beta)f\|$, which is below ϵ for β near α.

(c) By linearity it suffices to take $\varphi = |m\rangle$. The lowering exponential is the finite sum $e^{-t\bar\alpha\cdot a}|m\rangle = \sum_{k\le|m|}(-t)^k(\bar\alpha\cdot a)^k|m\rangle/k!$, so

$$\jmath D(t\alpha)|m\rangle = e^{-t^2|\alpha|^2/2}\Big(e^{\Lambda(0,t\alpha)}|m\rangle - t\,e^{\Lambda(0,t\alpha)}(\bar\alpha\cdot a)|m\rangle + \sum_{k\ge2}\frac{(-t)^k}{k!}e^{\Lambda(0,t\alpha)}(\bar\alpha\cdot a)^k|m\rangle\Big),$$

each $(\bar\alpha\cdot a)^k|m\rangle$ being a finite combination of number states. Each $e^{\Lambda(0,t\alpha)}|m'\rangle$ is square-summable, its one-mode factors $e^{t\alpha_j a^\dagger}|m'_j\rangle$ having the finite squared norms computed in the proof of Lemma 10.8. By Lemma 10.14 with $\kappa = 0$ and $\ell(t) = t\alpha$, $t \mapsto e^{\Lambda(0,t\alpha)}|m'\rangle$ is differentiable at 0 with derivative $\Lambda(0,\alpha)|m'\rangle = (\alpha\cdot a^\dagger)|m'\rangle$, and in particular continuous there. The scalar factor has derivative 0 at 0. By the product rule the derivative at 0 is $(\alpha\cdot a^\dagger)|m\rangle - (\bar\alpha\cdot a)|m\rangle$. □

10.4 Irreducibility

Theorem 10.16 (Irreducibility). Lean ✓ *Let T be a bounded operator on $\Gamma_s(\iota)$ with $TD(\alpha) = D(\alpha)T$ for every $\alpha \in \mathbb{C}^\iota$. Then $\langle\varepsilon(\alpha), T|0\rangle\rangle = \langle T^*|0\rangle, \varepsilon(-\alpha)\rangle$ for every α, $T|0\rangle = \langle 0|T|0\rangle\,|0\rangle$, and $T = \langle 0|T|0\rangle\,1$.*

Proof. T commutes with $D(-\alpha) = D(\alpha)^*$ as well. By Theorem 10.12(b),(d),

$$e^{-|\alpha|^2/2}\langle\varepsilon(\alpha), T|0\rangle\rangle = \langle D(\alpha)|0\rangle, T|0\rangle\rangle = \langle|0\rangle, TD(-\alpha)|0\rangle\rangle = e^{-|\alpha|^2/2}\langle T^*|0\rangle, \varepsilon(-\alpha)\rangle.$$

The passive unitary U_{-I} is self-adjoint (Theorem 8.7(b)) and maps $\varepsilon(\alpha)$ to $\varepsilon(-\alpha)$ (Lemma 10.3(b)), so $\langle\varepsilon(\alpha), T|0\rangle\rangle = \langle\mathsf{U}_{-I}T^*|0\rangle, \varepsilon(\alpha)\rangle$ for every α. Lemma 10.4(iii) with $\psi = T|0\rangle$ and $\psi' = \mathsf{U}_{-I}T^*|0\rangle$ gives $(T|0\rangle)(n) = 0$ for $n \ne 0$, that is, $T|0\rangle = \lambda|0\rangle$ with $\lambda = \langle 0|T|0\rangle$. Then $T\varepsilon(\alpha) = e^{|\alpha|^2/2}TD(\alpha)|0\rangle = e^{|\alpha|^2/2}D(\alpha)T|0\rangle = \lambda\varepsilon(\alpha)$, and $T = \lambda 1$ by Theorem 10.5. □

Corollary 10.17 (Consequences of irreducibility). Lean ✓

(a) *If V, V' are unitary operators on $\Gamma_s(\iota)$ with $V^*D(\alpha)V = V'^*D(\alpha)V'$ for every α, then $V = \lambda V'$ with $|\lambda| = 1$; and $V = V'$ if moreover $\langle 0|V|0\rangle = \langle 0|V'|0\rangle \ne 0$.*

(b) *A closed subspace K of $\Gamma_s(\iota)$ with $D(\alpha)K \subseteq K$ for every α is 0 or $\Gamma_s(\iota)$.*

(c) *The coherent states $D(\alpha)|0\rangle$ span a dense subspace of $\Gamma_s(\iota)$.*

Proof. (a) $T = VV'^*$ is unitary and $T^*D(\alpha)T = V'V^*D(\alpha)VV'^* = D(\alpha)$, so T commutes with every $D(\alpha)$, and $T = \lambda 1$ by Theorem 10.16; $|\lambda| = 1$ since T is unitary. Then $V = \lambda V'$, and $\langle 0|V|0\rangle = \lambda\langle 0|V'|0\rangle$ forces $\lambda = 1$ under the extra hypothesis.

(b) The family of the $D(\alpha)$ is closed under adjoints, so the orthogonal projection P onto K commutes with every $D(\alpha)$: $D(\alpha)P = PD(\alpha)P$ and its adjoint $PD(-\alpha) = PD(-\alpha)P$ for all α give $D(\alpha)P = PD(\alpha)$. By Theorem 10.16, $P = \lambda 1$, and $P^2 = P$ gives $\lambda \in \{0, 1\}$.

(c) $D(\alpha)|0\rangle$ is a nonzero multiple of $\varepsilon(\alpha)$ (Theorem 10.12(d)); apply Theorem 10.5. □

10.5 Squeezing and rotation operators

The inner products of squeezed vectors are the sums of the binomial series of $(1-4y)^{-1/2}$, for complex y in Lemma 10.19 and for real y of either sign in Theorem 10.24(a).

Lemma 10.18 (The binomial series). Lean ✓ *Let $\mathcal{S}(y) = \sum_k \binom{2k}{k} y^k$. For every real y with $4|y| < 1$, $\mathcal{S}(y) = (1-4y)^{-1/2}$.*

Proof. $\binom{2k}{k} \le 4^k$, so the series converges absolutely for $4|y| < 1$. Newton's binomial series $(1+x)^{-1/2} = \sum_k \binom{-1/2}{k} x^k$ holds for real $|x| < 1$, and $\binom{-1/2}{k} = (-1)^k \frac{1\cdot 3\cdots(2k-1)}{2^k k!} = (-\frac{1}{4})^k \binom{2k}{k}$; take $x = -4y$. □

Newton's series holds also for complex $|x| < 1$ with the principal branch, so $\mathcal{S}(y)$ is the principal value of $(1-4y)^{-1/2}$ for complex y with $4|y| < 1$; we use this in the next proof.

The overlaps of one-mode Gaussian vectors are given by Mehler's formula [34], which we prove in the scalar form needed here.

Lemma 10.19 (Overlaps of one-mode Gaussian vectors). Lean ✓ *Let $\kappa, \kappa', \ell, \ell' \in \mathbb{C}$ with $|\kappa| + |\kappa'| < 1$. Then $1 - 4\bar\kappa\kappa' \neq 0$, and the series $\sum_m \overline{|\kappa,\ell\rangle(m)}\,|\kappa',\ell'\rangle(m)$ converges absolutely to*

$$\mathcal{S}(\bar\kappa\kappa')\exp\frac{\kappa'\bar\ell^2 + \ell'\bar\ell + \bar\kappa\ell'^2}{1-4\bar\kappa\kappa'}.$$

Proof. By Definition 6.9 and the Bargmann series $e^{\kappa x^2+\ell x}$ of $|\kappa,\ell\rangle$ (the remark before Lemma 7.13, written in the variable x), $|\kappa,\ell\rangle(m) = \sqrt{m!}\sum_p \kappa^p \ell^{m-2p}/(p!\,(m-2p)!)$. Hence $\lambda^m|\kappa,\ell\rangle(m) = |\lambda^2\kappa, \lambda\ell\rangle(m)$ for real $\lambda > 0$, and the terms of the series are unchanged when $(\kappa,\ell,\kappa',\ell')$ is replaced by $(\lambda^2\kappa, \lambda\ell, \kappa'/\lambda^2, \ell'/\lambda)$, as is the claimed sum. Since $4|\kappa\kappa'| \le (|\kappa| + |\kappa'|)^2 < 1$, there is λ with $2|\kappa'| < \lambda^2$ and $2\lambda^2|\kappa| < 1$; so we may assume $|\kappa|, |\kappa'| < \frac{1}{2}$. Then both vectors are square-summable (proof of Lemma 10.8, with the factor $e^{\ell a^\dagger}$ handled in the same way), the series converges absolutely by the Cauchy–Schwarz inequality, and its sum is their inner product.

Their Bargmann functions $f(x) = e^{\kappa x^2+\ell x}$ and $f'(x) = e^{\kappa' x^2+\ell' x}$ are entire and square-integrable for the measure $e^{-|x|^2}d^2x$, and by Bargmann's theorem [7] the inner product equals

$$\frac{1}{\pi}\int_{\mathbb{C}} \overline{f(x)}\, f'(x)\, e^{-|x|^2} d^2x.$$

With $x = x_1 + ix_2$ the exponent of the integrand is $-X^TQX + J^TX$, $X = (x_1, x_2)$, where

$$Q = \begin{pmatrix} 1-A-B & -i(A-B) \\ -i(A-B) & 1+A+B \end{pmatrix}, \quad J = \begin{pmatrix} C+D \\ i(C-D) \end{pmatrix}, \quad (A,B,C,D) = (\kappa', \bar\kappa, \ell', \bar\ell).$$

Q is complex symmetric with positive definite real part, since $\mathrm{Re}(Ax^2 + B\bar x^2) \le (|A|+|B|)|x|^2$. For such Q the Gaussian integral is $\int e^{-X^TQX+J^TX}d^2X = \pi(\det Q)^{-1/2}e^{J^TQ^{-1}J/4}$, with the branch of the square root that is continuous on these matrices and positive on real ones. Here $\det Q =$

$1-4AB=1-4\bar\kappa\kappa'$, and multiplying out, $J^TQ^{-1}J/4=(CD+AD^2+BC^2)/(1-4AB)$, which is the claimed exponent. Along $t\mapsto(t\kappa,t\kappa')$, $0\le t\le 1$, the determinant $1-4t^2\bar\kappa\kappa'$ stays in the disc of radius 1 about 1, where the principal branch is continuous and positive at $t=0$; so the root is the principal one, and $(1-4\bar\kappa\kappa')^{-1/2}=\mathcal{S}(\bar\kappa\kappa')$ by the remark after Lemma 10.18. □

At $\kappa=0$ the lemma reads $\langle\varepsilon(w),|\kappa',\ell'\rangle\rangle=e^{\kappa'\bar w^2+\ell'\bar w}$ for one mode, the value of the Bargmann function at $\bar w$; by Lemma 10.2 the inner product of $\varepsilon(w)$ with a product vector $\bigotimes_j|\kappa_j,\ell_j\rangle$ is $\prod_j e^{\kappa_j\bar w_j^2+\ell_j\bar w_j}$.

Definition 10.20 (Squeezing and rotation operators)**.** For $\zeta\in\mathbb{C}$ write $\zeta=re^{i\phi}$ with $r=|\zeta|$, and put $c=\cosh r$, $s=\sinh r$ and $\tau=\tau(\zeta)=\zeta\tanh|\zeta|/|\zeta|$ for $\zeta\neq0$, $\tau(0)=0$; so $\tau=e^{i\phi}\tanh r$ (the paper's (C16)) and $|\tau|=\tanh r<1$. For $\zeta\in\mathbb{C}^\iota$ these are taken mode by mode, and for $\psi\in\Gamma_s(\iota)$

$$\mathsf{S}^{\mathrm{ord}}(\zeta)\jmath\psi=\Big(\prod_j c_j\Big)^{-1/2}e^{\Lambda(-\tau/2,0)}\,\Gamma\big(\mathrm{diag}(c_j^{-1})\big)\,e^{\frac12\sum_j\bar\tau_j a_j^2}\jmath\psi,$$

with the lowering exponential of Lemma 10.8(b), coefficientwise. For $\theta\in\mathbb{R}^\iota$ the *rotation operator* is $R(\theta)=\mathsf{U}_{\mathrm{diag}(e^{-i\theta_j})}$ (Theorem 8.7).

$\Gamma(\mathrm{diag}(c_j^{-1}))$ multiplies the coefficient at n by $\prod_j c_j^{-n_j}$, and $e^{\Lambda(-\tau/2,0)}=e^{-\frac12\sum_j\tau_j a_j^{\dagger2}}$. This is the standard normal-ordered form of $e^{\frac12\sum_j(\bar\zeta_j a_j^2-\zeta_j a_j^{\dagger2})}$ [15], with $(\cosh r)^{-(n+1/2)}=c^{-1/2}c^{-n}$ on $|n\rangle$. Its lowering exponential converges because $|\tau_j|<1$ for every ζ. By Theorem 8.10, $R(\theta)$ is $e^{-i\sum_j\theta_j\hat n_j}$ summed coefficientwise, with $\hat n_j=a_j^\dagger a_j$.

Theorem 10.21 (Squeezing operators)**.** Lean ✓ *Let $\zeta,\zeta'\in\mathbb{C}^\iota$ and $w,z\in\mathbb{C}^\iota$.*

(a) *In $\mathcal{F}$,*

$$\mathsf{S}^{\mathrm{ord}}(\zeta)\varepsilon(z)=\Big(\prod_j c_j\Big)^{-1/2}e^{\frac12\sum_j\bar\tau_j z_j^2}\bigotimes_j|-\tau_j/2,\,z_j/c_j\rangle,$$

$$\mathsf{S}^{\mathrm{ord}}(\zeta)|0\rangle=\Big(\prod_j c_j\Big)^{-1/2}\bigotimes_j|-\tau_j/2,0\rangle,$$

$$\langle\varepsilon(w),\mathsf{S}^{\mathrm{ord}}(\zeta)\varepsilon(z)\rangle=\Big(\prod_j c_j\Big)^{-1/2}e^{\frac12\sum_j\bar\tau_j z_j^2}\exp\sum_j\Big(-\tfrac12\tau_j\bar w_j^2+\frac{z_j\bar w_j}{c_j}\Big);$$

the vector in the second line is the squeezed vacuum of [4, (5)].

(b) *$\psi\mapsto\mathsf{S}^{\mathrm{ord}}(\zeta)\jmath\psi$ is linear with continuous coefficients, and there is a unitary operator $\mathsf{S}(\zeta)$ on $\Gamma_s(\iota)$, the* squeezing operator*, with $\jmath\mathsf{S}(\zeta)\psi=\mathsf{S}^{\mathrm{ord}}(\zeta)\jmath\psi$ for every $\psi\in\Gamma_s(\iota)$.*

(c) *(Covariance) $\mathsf{S}(\zeta)D(\beta)=D(L_\zeta\beta)\mathsf{S}(\zeta)$ for every $\beta\in\mathbb{C}^\iota$, where $(L_\zeta\beta)_j=c_j\beta_j-e^{i\phi_j}s_j\bar\beta_j$; $\mathsf{S}(\zeta)^*D(\alpha)\mathsf{S}(\zeta)=D(L_{-\zeta}\alpha)$, and $L_\zeta L_{-\zeta}=1$.*

(d) *$\mathsf{S}(\zeta)\mathsf{S}(-\zeta)=1$, $\mathsf{S}(\zeta)^*=\mathsf{S}(-\zeta)$, $\mathsf{S}(0)=1$.*

(e) *For an enumeration $j_1,\dots,j_N$ of the modes, $\mathsf{S}(\zeta)=\mathsf{S}(\zeta_{j_1}e_{j_1})\cdots\mathsf{S}(\zeta_{j_N}e_{j_N})$; $\mathsf{S}(\zeta+\zeta')=\mathsf{S}(\zeta)\mathsf{S}(\zeta')$ if $\zeta_j=0$ or $\zeta'_j=0$ for every j, so these factors commute; and $(\mathsf{S}(\eta e_j)\psi)(n)=(\mathsf{S}(\eta)\psi_{j,n})(n_j)$ for $\eta\in\mathbb{C}$, with $\mathsf{S}(\eta)$ the one-mode squeezing operator.*

Moreover, $\langle\mathsf{S}(\zeta)|0\rangle,\mathsf{S}(\zeta')|0\rangle\rangle=(\prod_j c_j)^{-1/2}(\prod_j c'_j)^{-1/2}\prod_j\mathcal{S}(\bar\tau_j\tau'_j/4)$, with c', τ' the parameters of ζ'.

Proof. (a) By Lemma 10.3(b) the lowering exponential multiplies $\varepsilon(z)$ by $e^{\frac12\sum_j\bar\tau_j z_j^2}$, and $\Gamma(\mathrm{diag}(c_j^{-1}))\varepsilon(z)=\varepsilon(z/c)$. By Theorem 6.8 and Lemma 10.7(c), $e^{\Lambda(-\tau/2,0)}e^{\Lambda(0,z/c)}|0\rangle=e^{\Lambda(-\tau/2,z/c)}|0\rangle=\bigotimes_j|-\tau_j/2,z_j/c_j\rangle$. At $z=0$ this is the second formula, and the third follows from the remark after Lemma 10.19.

(b) Linearity and continuity of the coefficients follow from Lemma 10.8(b),(c) as in Theorem 10.12(a). Inner products of product vectors are products of one-mode inner products (Lemma 10.2). For one mode, Lemma 10.19 with $\kappa = \kappa' = -\tau/2$, $\ell = w/c$, $\ell' = z/c$, where $|\kappa| + |\kappa'| = |\tau| < 1$ and $1 - |\tau|^2 = c^{-2}$, gives

$$c^{-1}e^{(\tau\bar{w}^2+\bar{\tau}z^2)/2}\,\langle|-\tfrac{\tau}{2},\tfrac{w}{c}\rangle, |-\tfrac{\tau}{2},\tfrac{z}{c}\rangle\rangle = c^{-1}e^{(\tau\bar{w}^2+\bar{\tau}z^2)/2}\cdot c\,e^{-\tau\bar{w}^2/2+\bar{w}z-\bar{\tau}z^2/2} = e^{\bar{w}z}.$$

So the vectors of (a) have the inner products of the exponential vectors, and Lemma 10.10(a) gives a linear isometry $\mathsf{S}(\zeta)$ with $\jmath\mathsf{S}(\zeta)\psi = \mathsf{S}^{\mathrm{ord}}(\zeta)\jmath\psi$, unique since $\jmath$ is injective. It is unitary by (d).

(c) Both sides are bounded. We compare $\langle\varepsilon(w), \cdot\,\varepsilon(z)\rangle$, writing $\gamma = L_\zeta\beta$ and computing one mode at a time. By Theorem 10.12(b) and (a), the left side gives the exponent

$$-\tfrac{1}{2}|\beta|^2 - \bar{\beta}z + \tfrac{1}{2}\bar{\tau}(z+\beta)^2 - \tfrac{1}{2}\tau\bar{w}^2 + (z+\beta)\bar{w}/c,$$

and the right side, as $\langle D(-\gamma)\varepsilon(w), \mathsf{S}(\zeta)\varepsilon(z)\rangle$, gives

$$-\tfrac{1}{2}|\gamma|^2 + \gamma\bar{w} + \tfrac{1}{2}\bar{\tau}z^2 - \tfrac{1}{2}\tau(\bar{w}-\bar{\gamma})^2 + z(\bar{w}-\bar{\gamma})/c,$$

with the same factor $(\prod_j c_j)^{-1/2}$. The terms in z^2, $\bar{w}^2$ and $z\bar{w}$ agree. The terms linear in z agree because $\bar{\gamma} = c(\bar{\beta} - \bar{\tau}\beta)$, that is $\gamma = c\beta - c\tau\bar{\beta} = c\beta - e^{i\phi}s\bar{\beta}$; the terms linear in $\bar{w}$ because $\gamma + \tau\bar{\gamma} = c(1-|\tau|^2)\beta = \beta/c$; and the constant terms because $-\frac{1}{2}|\gamma|^2 - \frac{1}{2}\tau\bar{\gamma}^2 = -\frac{1}{2}c^2(1-|\tau|^2)(|\beta|^2 - \bar{\tau}\beta^2) = -\frac{1}{2}|\beta|^2 + \frac{1}{2}\bar{\tau}\beta^2$. By Theorem 10.5 the vectors $\mathsf{S}(\zeta)D(\beta)\varepsilon(z)$ and $D(\gamma)\mathsf{S}(\zeta)\varepsilon(z)$ agree, and then so do the operators. Since $\tau(-\zeta) = -\tau$, $(L_{-\zeta}\beta)_j = c_j\beta_j + e^{i\phi_j}s_j\bar{\beta}_j$, and $L_\zeta L_{-\zeta}\beta = (c^2 - s^2)\beta = \beta$ mode by mode. With $\beta = L_{-\zeta}\alpha$ and the isometry property $\mathsf{S}(\zeta)^*\mathsf{S}(\zeta) = 1$, the covariance gives $\mathsf{S}(\zeta)^*D(\alpha)\mathsf{S}(\zeta) = D(L_{-\zeta}\alpha)$.

(d) By (c), $\mathsf{S}(\zeta)\mathsf{S}(-\zeta)D(\beta) = D(L_\zeta L_{-\zeta}\beta)\mathsf{S}(\zeta)\mathsf{S}(-\zeta) = D(\beta)\mathsf{S}(\zeta)\mathsf{S}(-\zeta)$, so by Theorem 10.16 $\mathsf{S}(\zeta)\mathsf{S}(-\zeta) = \lambda 1$ with λ its vacuum coefficient. The raising exponential and $\Gamma(\mathrm{diag}(c_j^{-1}))$ do not change the coefficient at 0, so by Lemma 10.8(b) and (a),

$$\lambda = \Big(\prod_j c_j\Big)^{-1/2}\langle e^{\Lambda(\tau/2,0)}|0\rangle, \mathsf{S}(-\zeta)|0\rangle\rangle = \Big(\prod_j c_j\Big)^{-1}\prod_j \||\tau_j/2,0\rangle\|^2 = \prod_j c_j^{-1}\mathcal{S}(|\tau_j|^2/4),$$

using $\tau(-\zeta) = -\tau$ and the coefficients $|\kappa,0\rangle(2k) = \kappa^k\sqrt{(2k)!}/k!$. By Lemma 10.18, $\mathcal{S}(|\tau|^2/4) = (1-|\tau|^2)^{-1/2} = c$, so $\lambda = 1$. Exchanging ζ and $-\zeta$, $\mathsf{S}(-\zeta)\mathsf{S}(\zeta) = 1$; so $\mathsf{S}(\zeta)$ is onto, hence unitary, with $\mathsf{S}(\zeta)^* = \mathsf{S}(\zeta)^{-1} = \mathsf{S}(-\zeta)$. $\mathsf{S}^{\mathrm{ord}}(0)$ is the identity.

(e) The inner product of squeezed vacua is $\prod_j (c_j c_j')^{-1/2}\langle|-\tau_j/2,0\rangle, |-\tau_j'/2,0\rangle\rangle$ by (a) and Lemma 10.2, and the one-mode inner product is $\sum_k (\bar{\tau}_j\tau_j'/4)^k(2k)!/k!^2 = \mathcal{S}(\bar{\tau}_j\tau_j'/4)$. If ζ and ζ' have disjoint supports, then in each mode one of L_ζ, $L_{\zeta'}$ is the identity, so $L_{-\zeta'}L_{-\zeta} = L_{-(\zeta+\zeta')}$ and, by (c), $\mathsf{S}(\zeta)\mathsf{S}(\zeta')$ and $\mathsf{S}(\zeta+\zeta')$ have the same conjugation action on every $D(\alpha)$. Their vacuum expectations agree: $\langle 0|\mathsf{S}(\zeta)\mathsf{S}(\zeta')|0\rangle = \langle \mathsf{S}(-\zeta)|0\rangle, \mathsf{S}(\zeta')|0\rangle\rangle = \prod_j (c_j c_j')^{-1/2}$, since every $\bar{\tau}(-\zeta_j)\tau_j'$ vanishes, and this is $\prod_j c(\zeta_j+\zeta_j')^{-1/2} = \langle 0|\mathsf{S}(\zeta+\zeta')|0\rangle \neq 0$. Corollary 10.17(a) gives $\mathsf{S}(\zeta+\zeta') = \mathsf{S}(\zeta)\mathsf{S}(\zeta')$, and the product over an enumeration follows by induction. The fiber formula is proved as in Theorem 10.12(d): on $\varepsilon(z)$ both sides are $\prod_{k\neq j}\varepsilon(z_k)(n_k)$ times the coefficient at n_j of the one-mode vector $\mathsf{S}(\eta)\varepsilon(z_j)$, by (a), since the modes $k \neq j$ have $\tau_k = 0$ and $c_k = 1$. □

By (c) and (d), $\mathsf{S}(\zeta)a\mathsf{S}(\zeta)^* = c\,a + e^{i\phi}s\,a^\dagger$ is the form of the covariance with ζ replaced by $-\zeta$. The matrix elements in the number basis are finite sums, because on a number state every lowering exponential terminates.

Lemma 10.22 (Matrix elements). Lean ✓ *For one mode, with* $\langle k|X|n\rangle = (X|n\rangle)(k)$, $\alpha, \zeta \in \mathbb{C}$ *and* $\theta \in \mathbb{R}$*:*

$$\langle m|D(\alpha)|n\rangle = e^{-|\alpha|^2/2} \sum_{i=0}^{\min(m,n)} \frac{\alpha^{m-i}(-\bar\alpha)^{n-i}\sqrt{m!\,n!}}{(m-i)!\,(n-i)!\,i!},$$

$$\langle k|\mathsf{S}(\zeta)|n\rangle = c^{-1/2} \sum_{\substack{0\le i\le n/2,\ \mu=n-2i\le k,\\ k-\mu=2h\ \text{even}}} \frac{(-\tau/2)^h}{h!}\sqrt{\tfrac{k!}{\mu!}}\; c^{-\mu}\; \frac{(\bar\tau/2)^i}{i!}\sqrt{\tfrac{n!}{\mu!}},$$

and $\langle k|R(\theta)|n\rangle = \delta_{kn}e^{-i\theta n}$*. In particular* $R(0) = 1$*, and* $R(\theta) = 1$ *when* $\theta_j = 2\pi$ *for every* j*, on any number of modes.*

Proof. $e^{-\bar\alpha a}|n\rangle = \sum_{i'}(-\bar\alpha)^{i'}\sqrt{n!/(n-i')!}\,|n-i'\rangle/i'!$, and the coefficient of $e^{\alpha a^\dagger}|n-i'\rangle$ at m is $\alpha^{m-n+i'}\sqrt{m!/(n-i')!}/(m-n+i')!$; with $i = n-i'$ this is the first formula. For S: the lowering exponential gives $\sum_i(\bar\tau/2)^i\sqrt{n!/\mu!}\,|\mu\rangle/i!$ with $\mu = n-2i$, then $\Gamma(c^{-1})$ multiplies by $c^{-\mu}$, and the coefficient of $e^{-\tau a^{\dagger 2}/2}|\mu\rangle$ at $k = \mu+2h$ is $(-\tau/2)^h\sqrt{k!/\mu!}/h!$. For $R(\theta)$, Lemma 8.3(b) gives $\Gamma(e^{-i\theta})|n\rangle = e^{-i\theta n}|n\rangle$; on several modes the diagonal matrix multiplies $|n\rangle$ by $\prod_j e^{-i\theta_j n_j}$, which is 1 for $\theta = 0$ and for $\theta_j = 2\pi$. □

On several modes the matrix elements of $D(\alpha)$ and $\mathsf{S}(\zeta)$ are the products of the one-mode ones, by the product and fiber formulas of Theorems 10.12(d) and 10.21(e). The matrix elements of $D(\alpha)$ are those of Cahill and Glauber [33], who write them with associated Laguerre polynomials; we read off those of $\mathsf{S}(\zeta)$ from the normal-ordered form of Definition 10.20.

Theorem 10.23 (Squeezing of the ladder operators). Lean ✓ *Let* $\zeta \in \mathbb{C}^\iota$ *and* j *a mode. Then* $\operatorname{dom}\overline{a_j^\dagger} = \operatorname{dom}\bar a_j$*. Let* $\psi, \xi, \xi' \in \Gamma_s(\iota)$ *with* $a_j\jmath\psi = \jmath\xi$ *and* $a_j^\dagger\jmath\psi = \jmath\xi'$*. Then*

$$a_j\jmath\mathsf{S}(\zeta)\psi = \jmath\mathsf{S}(\zeta)(c_j\xi - e^{i\phi_j}s_j\xi'), \qquad a_j^\dagger\jmath\mathsf{S}(\zeta)\psi = \jmath\mathsf{S}(\zeta)(c_j\xi' - e^{-i\phi_j}s_j\xi),$$

and $\mathsf{S}(\zeta)\psi \in \operatorname{dom}\bar a_j$ *if and only if* $\psi \in \operatorname{dom}\bar a_j$*. That is,* $\mathsf{S}(\zeta)$ *maps* $\operatorname{dom}\bar a_j$ *onto itself, and on it* $\mathsf{S}(\zeta)^*\bar a_j\mathsf{S}(\zeta) = c_j\bar a_j - e^{i\phi_j}s_j\overline{a_j^\dagger}$*.*

Proof. For every $\psi \in \mathcal{F}$, $\|a_j^\dagger\psi\|^2 = \sum_n(n_j+1)|\psi(n)|^2 = \|a_j\psi\|^2 + \|\psi\|^2$ in $[0,\infty]$, which gives the equality of the domains.

Write $E_- = e^{\frac12\sum_k\bar\tau_k a_k^2}$, $C = \Gamma(\operatorname{diag}(c_k^{-1}))$, $E_+ = e^{\Lambda(-\tau/2,0)}$ and $T = \Lambda(-\tau/2,0)$. Since $[a_j,T] = -\tau_j a_j^\dagger$ commutes with T, $a_jT^k = T^ka_j - k\tau_jT^{k-1}a_j^\dagger$, and at each coefficient $a_jE_+ = E_+(a_j - \tau_ja_j^\dagger)$; and $a_j^\dagger$ commutes with E_+. On coefficients, $a_jC = c_j^{-1}Ca_j$ and $a_j^\dagger C = c_jCa_j^\dagger$. The lowering operator $B = \frac12\sum_k\bar\tau_ka_k^2$ commutes with a_j, and $a_j^\dagger B^k = B^ka_j^\dagger - k\bar\tau_jB^{k-1}a_j$; since ψ, $a_j\psi$ and $a_j^\dagger\psi$ lie in $\jmath(\Gamma_s(\iota))$, all the series below converge absolutely (Lemma 10.8(b)) and may be split termwise:

$$a_jE_-\jmath\psi = E_-\jmath\xi, \qquad a_j^\dagger E_-\jmath\psi = E_-\jmath\xi' - \bar\tau_jE_-\jmath\xi.$$

Therefore, with $(\prod_k c_k)^{-1/2}$ omitted on both sides,

$$a_jE_+CE_-\jmath\psi = E_+C(c_j^{-1}a_j - \tau_jc_ja_j^\dagger)E_-\jmath\psi = E_+CE_-\jmath((c_j^{-1}+|\tau_j|^2c_j)\xi - \tau_jc_j\xi'),$$

and $c^{-1} + |\tau|^2c = c^{-1}(1+s^2) = c$, $\tau c = e^{i\phi}s$. Likewise $a_j^\dagger E_+CE_-\jmath\psi = E_+Cc_ja_j^\dagger E_-\jmath\psi = E_+CE_-\jmath(c_j\xi' - c_j\bar\tau_j\xi)$, and $c\bar\tau = e^{-i\phi}s$.

If $\psi \in \operatorname{dom}\bar a_j$, then also $a_j^\dagger\jmath\psi \in \jmath(\Gamma_s(\iota))$, and the first identity shows $\mathsf{S}(\zeta)\psi \in \operatorname{dom}\bar a_j$; the converse is the same statement for $\mathsf{S}(-\zeta) = \mathsf{S}(\zeta)^{-1}$, and applying $\mathsf{S}(\zeta)^*$ gives the operator form. □

Theorem 10.24 (One-parameter groups of squeezers). Lean ✓ *Fix* $\zeta_0 \in \mathbb{C}^\iota$*.*

(a) $\mathsf{S}(t\zeta_0)\mathsf{S}(t'\zeta_0) = \mathsf{S}((t+t')\zeta_0)$ *for all real* t, t'.

(b) *For every* $\psi \in \Gamma_s(\iota)$, $\zeta \mapsto \mathsf{S}(\zeta)\psi$ *is continuous from* $\mathbb{C}^\iota$ *to* $\Gamma_s(\iota)$.

(c) *Let* $\varphi \in \Gamma_s(\iota)$ *be finitely supported, and* $G \in \Gamma_s(\iota)$ *with* $\jmath G = \sum_j (\frac{1}{2}\bar\zeta_{0j} a_j^2 - \frac{1}{2}\zeta_{0j} a_j^{\dagger 2})\jmath\varphi$. *Then* $t \mapsto \mathsf{S}(t\zeta_0)\varphi$ *is differentiable at* $t = 0$ *with derivative* G.

Proof. Write $\rho_j = |\zeta_{0j}|$ and $\zeta_{0j} = \rho_j e^{i\phi_j}$. For real t, $\tau(t\zeta_{0j}) = e^{i\phi_j}\tanh(t\rho_j)$ and $\cosh|t\zeta_{0j}| = \cosh(t\rho_j)$, since tanh is odd.

(a) Write $C_t = \cosh(t\rho_j)$ and $S_t = \sinh(t\rho_j)$. In mode j, $L_{t\zeta_0}L_{t'\zeta_0}\beta = (C_tC_{t'} + S_tS_{t'})\beta - e^{i\phi_j}(C_tS_{t'} + S_tC_{t'})\bar\beta = L_{(t+t')\zeta_0}\beta$ by the addition formulas, so by Theorem 10.21(c) the unitaries $\mathsf{S}(t\zeta_0)\mathsf{S}(t'\zeta_0)$ and $\mathsf{S}((t+t')\zeta_0)$ conjugate every $D(\alpha)$ in the same way. Their vacuum expectations are, by the last formula of Theorem 10.21 with $\bar\tau(-t\zeta_{0j})\tau(t'\zeta_{0j})/4 = -\tanh(t\rho_j)\tanh(t'\rho_j)/4$, a real number of modulus below $\frac{1}{4}$, and by Lemma 10.18,

$$\prod_j \bigl(C_tC_{t'}(1 + \tanh(t\rho_j)\tanh(t'\rho_j))\bigr)^{-1/2} = \prod_j \cosh((t+t')\rho_j)^{-1/2} \neq 0,$$

which is $\langle 0|\mathsf{S}((t+t')\zeta_0)|0\rangle$. Corollary 10.17(a) concludes.

(b) By Theorem 10.21(a), $\langle \varepsilon(w), \mathsf{S}(\zeta)\varepsilon(z)\rangle$ is continuous in ζ, since τ and c are; τ is continuous at 0 because $|\tau(\zeta)| \le |\zeta|$. The operators $\mathsf{S}(\zeta)$ have norm 1, so by Theorem 10.5 $\zeta \mapsto \langle x, \mathsf{S}(\zeta)y\rangle$ is continuous for all $x, y \in \Gamma_s(\iota)$. Then $\|\mathsf{S}(\zeta)\psi - \mathsf{S}(\zeta')\psi\|^2 = 2\|\psi\|^2 - 2\,\mathrm{Re}\langle \mathsf{S}(\zeta)\psi, \mathsf{S}(\zeta')\psi\rangle$ tends to 0 as $\zeta' \to \zeta$.

(c) By linearity take $\varphi = |m\rangle$. With $\tau_j(t) = \tau(t\zeta_{0j})$ and $c_j(t) = \cosh(t\rho_j)$,

$$\jmath\mathsf{S}(t\zeta_0)|m\rangle = \Bigl(\prod_j c_j(t)\Bigr)^{-1/2} \sum_{m'} p_{m'}(t)\, e^{\Lambda(-\tau(t)/2,0)}|m'\rangle,$$

a finite sum over the m' in the support of $e^{\frac{1}{2}\sum_j \bar\tau_j(t)a_j^2}|m\rangle$, where $p_{m'}(t)$ is the coefficient at m' of $\Gamma(\mathrm{diag}(c_j(t)^{-1}))e^{\frac{1}{2}\sum_j \bar\tau_j(t)a_j^2}|m\rangle$, a polynomial in the $\bar\tau_j(t)$ times $\prod_j c_j(t)^{-m'_j}$. Since $\tau_j(0) = 0$, $\tau_j'(0) = \zeta_{0j}$ and $c_j'(0) = 0$, $p_{m'}(0) = \delta_{mm'}$ and $p'_{m'}(0)$ is the coefficient at m' of $\frac{1}{2}\sum_j \bar\zeta_{0j}a_j^2|m\rangle$, and the scalar prefactor has derivative 0. Each $e^{\Lambda(-\tau(t)/2,0)}|m'\rangle$ is square-summable, since $|\tau_j(t)/2| < \frac{1}{2}$ (proof of Lemma 10.8), and by Lemma 10.14 with $\kappa(t) = -\tau(t)/2$, $\ell = 0$ it is differentiable at 0 with derivative $\Lambda(-\zeta_0/2, 0)|m'\rangle = -\frac{1}{2}\sum_j \zeta_{0j}a_j^{\dagger 2}|m'\rangle$. By the product rule the derivative at 0 is $-\frac{1}{2}\sum_j \zeta_{0j}a_j^{\dagger 2}|m\rangle + \frac{1}{2}\sum_j \bar\zeta_{0j}a_j^2|m\rangle$. □

Notes and references

The exponential vectors and their totality are standard in the Fock-space calculus [14]; the proof of Lemma 10.4 avoids the Bargmann integral by rotating the argument. The construction of the displacement operators from their action on exponential vectors, by isometric extension, is Parthasarathy's definition of the Weyl operators [14, Ch. II, Secs. 19–20]; see also Guichardet [28]. The irreducibility of the Fock representation is classical; see [6, Sec. 5.2]. The normal-ordered forms of Definitions 10.11 and 10.20 are the standard disentangling of the exponentials in [4, (5)] [15]; for the squeezing operator see Truax [31]. Squeezed states go back to Stoler [29] and Yuen [30]; for the covariance of Theorem 10.21(c) see [30]. Fisher, Nieto and Sandberg [32] show that the construction does not extend naively beyond quadratic exponents. Tjoa and Cirac write the anti-normal-ordered forms in their (C17), $D(\alpha) = e^{|\alpha|^2/2}e^{-\bar\alpha a}e^{\alpha a^\dagger}$ and $\mathsf{S}(\zeta) = \sqrt{c}\, e^{\bar\tau a^2/2}c^{\hat n}e^{-\tau a^{\dagger 2}/2}$. On $\Gamma_s(\iota)$ the first holds on the exponential vectors, but the vacuum coefficient of the second, $\sqrt{c}\sum_k \binom{2k}{k}(-s^2/4)^k$, converges absolutely when $s < 1$, to $c^{-1/2} = \langle 0|\mathsf{S}(\zeta)|0\rangle$, and diverges for $s > 1$, while the normal-ordered forms need only $|\tau| < 1$, which holds for every ζ.

That $D(\alpha) = e^{-iA_\alpha}$ and $\mathsf{S}(\zeta) = e^{-iB_\zeta}$, where A_α and B_ζ are the self-adjoint closures of $i(\alpha \cdot a^\dagger - \bar{\alpha} \cdot a)$ and $\frac{i}{2}\sum_j(\bar{\zeta}_j a_j^2 - \zeta_j a_j^{\dagger 2})$ on the finitely supported vectors, is Corollary 13.8(b), which uses Theorems 10.15 and 10.24 and needs neither Stone's nor Nelson's theorem. The formal proof of Lemma 10.19 specializes Mehler's formula for commuting matrix arguments, proved by an ordinary differential equation in the companion notes on Gaussian-augmented matrix product states; we give the classical route through Bargmann's integral [7] and the complex Gaussian integral [16]. Likewise the formal proof of Lemma 10.18 for real y pairs the terms of the alternating series, where we cite Newton's binomial series, and the formal proofs of Theorem 10.21(e) and Theorem 10.23 are those given here: the product of squeezers by Corollary 10.17(a), and the ladder operators by a computation in all modes at once.

11 Gaussian unitaries

A Gaussian unitary U acts on the quadratures $\hat{R}$ by an affine symplectic map, $U^\dagger \hat{R} U = M\hat{R} + d$ [4, (A3)]. Since the quadratures are unbounded, we take the exponentiated form of this relation, an identity between bounded operators: conjugation by U sends each Weyl operator $W(u) = e^{-i\sigma(u,\hat{R})}$ to a phase times $W(M^{-1}u)$. The displacement, squeezing and passive unitaries of Sections 8 and 10 are Gaussian in this sense, and irreducibility (Theorem 10.16) turns the Bloch–Messiah decomposition of M into a factorization of U, Proposition 5 of [4].

Notation for this section.

$\iota = \{1, \dots, N\}$, $\Gamma_s(\iota)$	$N \geq 1$ modes, except where a statement allows any finite ι; the bosonic Fock space
$u = (q, p)$, $\sigma(u, v)$	a vector of $\mathbb{R}^{2N}$, positions first; $u^T \Omega v$ (Definition 11.1)
$\alpha(u)$, $\delta(\alpha)$	$(q + ip)/\sqrt{2} \in \mathbb{C}^N$ and its inverse (Definition 11.1)
$\hat{R} = (\hat{q}, \hat{p})$, $\mathcal{Q}$	quadratures on $\mathcal{F}$; $\bigcap_j \operatorname{dom} \bar{a}_j$ (Definition 11.1)
M_ζ	the symplectic matrix of $\mathsf{S}(\zeta)$ (Definition 11.1)
$W(u)$; (M, d)	Weyl operator; data of a Gaussian unitary (Definition 11.3)
$O_\mathcal{U}$, $\Sigma(r)$	orthogonal symplectic matrix of $\mathcal{U} \in \mathrm{U}(N)$, squeezing matrix (Definitions 9.2, 9.5)
$\mathsf{U}_\mathcal{U}$, $D(\alpha)$, $\mathsf{S}(\zeta)$, $R(\theta)$	passive, displacement, squeezing and rotation operators (Sections 8, 10)

The symplectic matrix of a Gaussian unitary is written M, since S is the squeezing operator; U is a unitary operator on $\Gamma_s(\iota)$ and $\mathcal{U}, \mathcal{V}$ are unitary matrices.

11.1 Phase-space coordinates

Definition 11.1 (Phase-space coordinates and quadratures)**.** (a) For $u = (q, p) \in \mathbb{R}^{2N}$, $\alpha(u) = (q + ip)/\sqrt{2} \in \mathbb{C}^N$; for $\alpha \in \mathbb{C}^N$, $\delta(\alpha) = \sqrt{2}\,(\operatorname{Re}\alpha, \operatorname{Im}\alpha) \in \mathbb{R}^{2N}$, the paper's (A33). The *symplectic form* is $\sigma(u, v) = u^T \Omega v = \sum_j (q_j p'_j - p_j q'_j)$ for $v = (q', p')$, with $\Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$.

(b) The *quadratures* on $\mathcal{F}$ are $\hat{R} = (\hat{q}, \hat{p})$, $\hat{q}_j = (a_j + a_j^\dagger)/\sqrt{2}$ and $\hat{p}_j = i(a_j^\dagger - a_j)/\sqrt{2}$, and $\mathcal{Q} = \bigcap_j \operatorname{dom} \bar{a}_j \subseteq \Gamma_s(\iota)$.

(c) For $\zeta \in \mathbb{C}^N$, with c_j, s_j, ϕ_j as in Definition 10.20, M_ζ is the real $2N \times 2N$ matrix with diagonal $N \times N$ blocks

$$M_\zeta = \begin{pmatrix} \operatorname{diag}(c_j - s_j \cos\phi_j) & \operatorname{diag}(-s_j \sin\phi_j) \\ \operatorname{diag}(-s_j \sin\phi_j) & \operatorname{diag}(c_j + s_j \cos\phi_j) \end{pmatrix},$$

where $\cos\phi_j$ and $\sin\phi_j$ are read as 0 when $\zeta_j = 0$ (then $s_j = 0$).

In the mode-pair order $(q_1, p_1, \dots, q_N, p_N)$, M_ζ is the direct sum of the 2×2 matrices $c_j 1 - s_j \left(\begin{smallmatrix} \cos\phi_j & \sin\phi_j \\ \sin\phi_j & -\cos\phi_j \end{smallmatrix}\right)$. Formally $\alpha(u)\cdot a^\dagger - \overline{\alpha(u)}\cdot a = -i\sigma(u,\hat R)$, so $D(\alpha(u))$ is $e^{-i\sigma(u,\hat R)}$, and the quadratures satisfy $[\hat R_k, \hat R_l] = i\Omega_{kl}$ on $\mathcal{F}$.

Lemma 11.2 (Phase-space coordinates). Lean ✓

(a) *$\alpha(\delta(\alpha)) = \alpha$ and $\delta(\alpha(u)) = u$; α is additive, odd and injective, and $|\alpha(u)|^2 = |u|^2/2$.*

(b) *σ is bilinear, $\sum_j (\alpha(u)_j\overline{\alpha(v)_j} - \overline{\alpha(u)_j}\alpha(v)_j) = -i\sigma(u,v)$, and if $\sigma(u,w) = 0$ for every u, then $w = 0$. For $M \in \mathrm{Sp}(2N,\mathbb{R})$, $\sigma(Mu, Mv) = \sigma(u,v)$ and $\sigma(M^{-1}u, v) = \sigma(u, Mv)$.*

(c) *$\alpha(O_{\mathcal{U}}u) = \mathcal{U}\alpha(u)$ for $\mathcal{U} \in \mathrm{U}(N)$.*

(d) *$\alpha(M_\zeta u) = L_\zeta \alpha(u)$, with L_ζ as in Theorem 10.21(c); $M_\zeta \in \mathrm{Sp}(2N,\mathbb{R})$, $M_{-\zeta}M_\zeta = 1$ and $M_\zeta^{-1} = M_{-\zeta}$.*

Proof. (a) The maps are mutually inverse real-linear bijections, and $|q_j + ip_j|^2/2 = (q_j^2 + p_j^2)/2$.

(b) $\alpha(u)_j\overline{\alpha(v)_j} = (q_j + ip_j)(q_j' - ip_j')/2$ has imaginary part $(p_jq_j' - q_jp_j')/2$, and the sum in question is $2i$ times the sum of these imaginary parts. With u the coordinate vectors, $\sigma(u,w)$ runs through w_{p_j} and $-w_{q_j}$. By Lemma 9.1, $M^T\Omega M = \Omega$, which is the first identity; the second is the first applied to $M^{-1}u$ and v.

(c) With $\mathcal{U} = X + iY$, Definition 9.2 gives $(Xq - Yp) + i(Yq + Xp) = \mathcal{U}(q + ip)$.

(d) In mode j, with $e^{i\phi}(q - ip) = (q\cos\phi + p\sin\phi) + i(q\sin\phi - p\cos\phi)$,

$$c(q + ip) - e^{i\phi}s(q - ip) = \big((c - s\cos\phi)q - s\sin\phi\, p\big) + i\big(-s\sin\phi\, q + (c + s\cos\phi)p\big),$$

which is $\sqrt{2}\,\alpha(M_\zeta u)_j$. Each 2×2 block has determinant $c^2 - s^2 = 1$, so it is symplectic, and so is the direct sum. Since $L_{-\zeta}L_\zeta = 1$ (Theorem 10.21(c), with ζ and $-\zeta$ exchanged) and α is injective, $M_{-\zeta}M_\zeta = 1$. □

11.2 Weyl operators and Gaussian unitaries

Definition 11.3 (Weyl operator; Gaussian unitary). The *Weyl operator* is $W(u) = D(\alpha(u))$, $u \in \mathbb{R}^{2N}$. A unitary operator U on $\Gamma_s(\iota)$ is a *Gaussian unitary with symplectic matrix M and displacement vector $d \in \mathbb{R}^{2N}$*, or with *data* (M, d), if $M \in \mathrm{Sp}(2N,\mathbb{R})$ and

$$U^*W(u)U = e^{-i\sigma(u,d)}\,W(M^{-1}u) \qquad \text{for every } u \in \mathbb{R}^{2N}.$$

Formally, $U^*e^{-i\sigma(u,\hat R)}U = e^{-i\sigma(u,M\hat R + d)}$ and $\sigma(u, M\hat R) = \sigma(M^{-1}u, \hat R)$, so the definition is the exponentiated form of $U^\dagger \hat R U = M\hat R + d$. The class contains products: $\mathsf{U}_{-I}\mathsf{S}(r)$ with $r > 0$ has the symplectic matrix $-\Sigma(r)$ (Theorem 11.5), of trace $-2\cosh r < -2$, while the symplectic matrix e^X of a single exponential e^{-iH} with H quadratic, X real and traceless, has $\operatorname{tr} e^X \geq -2$ (for $X^2 = -\det X\cdot 1$, the trace is $2\cosh\sqrt{-\det X}$ or $2\cos\sqrt{\det X}$); that e^{-iH} has a symplectic matrix of this form is Theorem 13.7. So the Gaussian unitaries are not all of the single-exponential form $e^{i\chi}e^{-iH_G}$ of [4, (1)]; they form the group generated by these exponentials, as the paper uses when it composes Gaussian unitaries in (A34) and (11). This group is the standard class, generated by phases, Weyl operators and the metaplectic operators [16, Ch. 4]; by Theorem 11.6 every Gaussian unitary is a phase times a product of displacement, squeezing and passive unitaries.

Proposition 11.4 (Weyl relation and data). Lean ✓

(a) *$W(u)W(v) = e^{-i\sigma(u,v)/2}W(u+v)$, $W(u)^* = W(-u)$, and $\langle 0|W(u)|0\rangle = e^{-|u|^2/4}$.*

(b) *If U_1, U_2 are Gaussian with data (M_1, d_1), (M_2, d_2), then U_1U_2 is Gaussian with data $(M_1M_2, d_1 + M_1d_2)$; if U is Gaussian with data (M, d), then U^* is Gaussian with data $(M^{-1}, -M^{-1}d)$; and a unitary $U = \lambda 1$ is Gaussian with data $(1, 0)$.*

(c) *If $\lambda W(u) = \lambda' W(v)$ with $|\lambda| = |\lambda'| = 1$, then $u = v$ and $\lambda = \lambda'$. If U is Gaussian with data (M, d) and with data (M', d'), then $M = M'$ and $d = d'$. If U and V are Gaussian with the same data, then $U = \lambda V$ with $|\lambda| = 1$.*

Proof. (a) By Theorem 10.12(c), Lemma 11.2(a),(b) and additivity of α; $W(u)^* = D(-\alpha(u)) = W(-u)$; and $\langle 0|W(u)|0\rangle = e^{-|\alpha(u)|^2/2} = e^{-|u|^2/4}$ by Theorem 10.12(d).

(b) $(U_1U_2)^* W(u) U_1U_2 = e^{-i\sigma(u,d_1)} U_2^* W(M_1^{-1}u) U_2 = e^{-i\sigma(u,d_1)-i\sigma(M_1^{-1}u,d_2)} W(M_2^{-1}M_1^{-1}u)$, and $\sigma(M_1^{-1}u, d_2) = \sigma(u, M_1d_2)$. For U^*, conjugate the defining relation at $u = Mv$ by U: $UW(v)U^* = e^{i\sigma(Mv,d)} W(Mv)$, and $\sigma(Mv, d) = \sigma(v, M^{-1}d)$. A scalar unitary commutes with $W(u)$.

(c) $W(-v)W(u) = e^{i\sigma(v,u)/2} W(u - v)$ by (a), so $\lambda W(u) = \lambda' W(v)$ makes $W(u - v)$ a scalar of modulus 1; then $e^{-|u-v|^2/4} = |\langle 0|W(u - v)|0\rangle| = 1$ gives $u = v$, and $\lambda = \lambda'$. For two data of U, this gives $M^{-1}u = M'^{-1}u$ for every u, so $M = M'$, and $e^{-i\sigma(u,d-d')} = 1$ for every u. If $\sigma(u, d - d') = t \neq 0$ for some u, then at $\pi u/t$ the phase is $e^{-i\pi} = -1$; so $\sigma(\cdot, d - d') = 0$ and $d = d'$ by Lemma 11.2(b). If U, V have the same data, then $U^*D(\alpha)U = V^*D(\alpha)V$ for every $\alpha = \alpha(\delta(\alpha))$, and Corollary 10.17(a) applies. □

Theorem 11.5 (Gaussian unitaries on Fock space; [4, (30), (31), (A17)–(A33)]). Lean ✓

(a) *$D(\alpha)$ is Gaussian with data $(1, \delta(\alpha))$.*

(b) *For every finite set ι, $\mathcal{U} \in \mathrm{U}(\iota)$ and $\beta, z \in \mathbb{C}^\iota$: $\mathsf{U}_\mathcal{U}^* D(\beta) \mathsf{U}_\mathcal{U} = D(\mathcal{U}^*\beta)$, the paper's (30); $\mathsf{U}_\mathcal{U} D(\beta) = D(\mathcal{U}\beta)\mathsf{U}_\mathcal{U}$; $\mathsf{U}_\mathcal{U}\varepsilon(z) = \varepsilon(\mathcal{U}z)$ and $\mathsf{U}_\mathcal{U}|0\rangle = |0\rangle$. For $\mathcal{U} \in \mathrm{U}(N)$, $\mathsf{U}_\mathcal{U}$ is Gaussian with data $(O_\mathcal{U}, 0)$, the paper's (A17)–(A19).*

(c) *$\mathsf{S}(\zeta)$ is Gaussian with data $(M_\zeta, 0)$, and $M_r = \Sigma(r)$ for every $r \in \mathbb{R}^N$, the paper's (A27)–(A29).*

(d) *For $\mathcal{U} \in \mathrm{U}(N)$, a mode j and $\zeta \in \mathbb{C}$, $\mathsf{U}_\mathcal{U}^* \mathsf{S}(\zeta e_j) \mathsf{U}_\mathcal{U}$ is Gaussian with data $(O_\mathcal{U}^T M_{\zeta e_j} O_\mathcal{U}, 0)$. With $b_j = \sum_k \mathcal{U}_{jk} a_k$ and $b_j^\dagger = \sum_k \overline{\mathcal{U}_{jk}} a_k^\dagger$: if $\varphi \in \Gamma_s(\iota)$ is finitely supported and $G \in \Gamma_s(\iota)$ satisfies $\jmath G = (\frac{1}{2}\bar\zeta b_j^2 - \frac{1}{2}\zeta b_j^{\dagger 2})\jmath\varphi$, then $t \mapsto \mathsf{U}_\mathcal{U}^* \mathsf{S}(t\zeta e_j)\mathsf{U}_\mathcal{U}\varphi$ is differentiable at $t = 0$ with derivative G: the paper's (31).*

Proof. (a) Applying the Weyl relation twice, $D(-\alpha)D(\beta)D(\alpha) = e^{\langle\alpha,\beta\rangle - \langle\beta,\alpha\rangle} D(\beta)$. For $\beta = \alpha(u)$ and $\alpha = \alpha(d)$, $d = \delta(\alpha)$, Lemma 11.2(b) gives $\langle\alpha(d), \alpha(u)\rangle - \langle\alpha(u), \alpha(d)\rangle = -i\sigma(u, d)$.

(b) By Lemma 10.3(b) and Theorem 10.12(b), $\mathsf{U}_{\mathcal{U}^*} D(\beta) \mathsf{U}_\mathcal{U} \varepsilon(z) = e^{-|\beta|^2/2 - \langle\beta, \mathcal{U}z\rangle} \varepsilon(z + \mathcal{U}^*\beta)$, which is $D(\mathcal{U}^*\beta)\varepsilon(z)$ because $|\mathcal{U}^*\beta| = |\beta|$ and $\langle\beta, \mathcal{U}z\rangle = \langle\mathcal{U}^*\beta, z\rangle$; Theorem 10.5 and $\mathsf{U}_\mathcal{U}^* = \mathsf{U}_{\mathcal{U}^*}$ (Theorem 8.7(b)) give (30). Replacing β by $\mathcal{U}\beta$ and multiplying by $\mathsf{U}_\mathcal{U}$ gives the second identity, and $\mathsf{U}_\mathcal{U}|0\rangle = |0\rangle$ is $z = 0$. For $\mathcal{U} \in \mathrm{U}(N)$, $\mathcal{U}^*\alpha(u) = \alpha(O_{\mathcal{U}^*}u) = \alpha(O_\mathcal{U}^{-1}u)$ by Lemma 11.2(c) and Theorem 9.4(a), and $O_\mathcal{U} \in \mathrm{Sp}(2N, \mathbb{R})$ by Theorem 9.4(b).

(c) By Theorem 10.21(c) and Lemma 11.2(d), $\mathsf{S}(\zeta)^* W(u) \mathsf{S}(\zeta) = D(L_{-\zeta}\alpha(u)) = W(M_{-\zeta}u) = W(M_\zeta^{-1}u)$. For real $r_j > 0$, $\phi_j = 0$ and the blocks are $c_j \mp s_j = e^{\mp r_j}$; for $r_j < 0$, $\phi_j = \pi$, $s_j = \sinh|r_j|$ and the blocks are $c_j \pm s_j = e^{\mp r_j}$; for $r_j = 0$ they are 1.

(d) The data follow from (b), (c) and Proposition 11.4(b), with $O_\mathcal{U}^{-1} = O_\mathcal{U}^T$. The vector $\mathsf{U}_\mathcal{U}\varphi$ is finitely supported (Lemma 8.3(c)), so by Theorem 10.24(c) $t \mapsto \mathsf{S}(t\zeta e_j)\mathsf{U}_\mathcal{U}\varphi$ has at 0 the derivative G_0 with $\jmath G_0 = (\frac{1}{2}\bar\zeta a_j^2 - \frac{1}{2}\zeta a_j^{\dagger 2})\Gamma(\mathcal{U})\jmath\varphi$; G_0 exists by Lemma 10.7. The bounded operator $\mathsf{U}_\mathcal{U}^*$ maps it to the derivative of the conjugated group, and $\jmath \mathsf{U}_\mathcal{U}^* G_0 = \Gamma(\mathcal{U}^*)(\frac{1}{2}\bar\zeta a_j^2 - \frac{1}{2}\zeta a_j^{\dagger 2})\Gamma(\mathcal{U})\jmath\varphi = \jmath G$ by Theorems 8.7(c) and 8.13(a). □

By (b), the product form of the paper's (30), $\mathsf{U}_{\mathcal{U}}^* D(\beta_0 e_j)\mathsf{U}_{\mathcal{U}} = \prod_k D(\overline{\mathcal{U}_{jk}}\beta_0 e_k)$, is the product formula of Theorem 10.12(d) for $\mathcal{U}^*(\beta_0 e_j)$, and $R(\theta)$ is Gaussian with data $(O_{\mathrm{diag}(e^{-i\theta_j})}, 0)$. Conjugating the group law and the strong continuity of Theorem 10.24(a),(b) by the fixed unitary $\mathsf{U}_{\mathcal{U}}$ shows that $t \mapsto \mathsf{U}_{\mathcal{U}}^* \mathsf{S}(t\zeta e_j)\mathsf{U}_{\mathcal{U}}$ is a strongly continuous one-parameter unitary group; by (d) its generator acts on finitely supported vectors as the squeezing generator in the modes b_j.

11.3 The Fock-space form of a Gaussian unitary

Theorem 11.6 ([4, Prop. 5, (A12)], for Gaussian unitaries in the sense of Definition 11.3)**.** Lean ✓ *Let U be a Gaussian unitary with data (M, d).*

(a) *If $M = O_{\mathcal{U}}\Sigma(r)O_{\mathcal{V}}$ with $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$ and $r \in \mathbb{R}^N$, then $U = \lambda\, \mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)\mathsf{U}_{\mathcal{V}}$ for some λ with $|\lambda| = 1$, where $\alpha = \alpha(O_{\mathcal{U}}^T d)$, that is $\delta(\alpha) = O_{\mathcal{U}}^T d$. In particular there are λ with $|\lambda| = 1$, $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$, $\alpha \in \mathbb{C}^N$ and $r \in \mathbb{R}^N$ with $r \geq 0$ such that $U = \lambda\, \mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)\mathsf{U}_{\mathcal{V}}$.*

(b) *For $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$ and $\alpha, \zeta \in \mathbb{C}^N$, the unitary $\mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(\zeta)\mathsf{U}_{\mathcal{V}}$ is Gaussian with the data $(O_{\mathcal{U}} M_\zeta O_{\mathcal{V}},\, O_{\mathcal{U}}\delta(\alpha))$, the paper's (A35).*

(c) *Let ι be a finite set, $\zeta \in \mathbb{C}^\iota$ and $\theta \in \mathbb{R}^\iota$ with $|\zeta_j| e^{-2i\theta_j} = \zeta_j$ for every j, and write $\Theta_\theta = \mathrm{diag}(e^{-i\theta_j})$, so that $R(\theta) = \mathsf{U}_{\Theta_\theta}$. Then $\mathsf{S}(\zeta) = \mathsf{U}_{\Theta_\theta}\, \mathsf{S}(|\zeta|)\, \mathsf{U}_{\Theta_\theta}^*$, where $|\zeta| = (|\zeta_j|)_j$. For $\iota = \{1, \dots, N\}$, $\mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(\zeta)\mathsf{U}_{\mathcal{V}} = \mathsf{U}_{\mathcal{U}\Theta_\theta} D(\Theta_\theta^* \alpha)\mathsf{S}(|\zeta|)\mathsf{U}_{\Theta_\theta^* \mathcal{V}}$.*

The admissible θ in (c) are $\theta_j = -\phi_j/2 + k_j\pi$, $k_j \in \mathbb{Z}$, where $\zeta_j \neq 0$, and arbitrary where $\zeta_j = 0$; the choice $\theta_j = -\phi_j/2$ gives $\Theta_\theta = \mathrm{diag}(e^{i\phi_j/2})$, the paper's absorption of the squeezing phases into the passive factors (A38). The numbers $r_j \geq 0$ in (a) are determined by M up to order (Proposition 9.12(b)), and λ is determined by U and the factors, since the product of the factors is unitary.

Proof. (b) By Theorem 11.5 the factors have data $(O_{\mathcal{U}}, 0)$, $(1, \delta(\alpha))$, $(M_\zeta, 0)$, $(O_{\mathcal{V}}, 0)$, and by Proposition 11.4(b) the product has data $(O_{\mathcal{U}} M_\zeta O_{\mathcal{V}},\, O_{\mathcal{U}}\delta(\alpha))$.

(a) By (b) with $\zeta = r$, $M_r = \Sigma(r)$, the product $\mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)\mathsf{U}_{\mathcal{V}}$ has data $(M, O_{\mathcal{U}}\delta(\alpha)) = (M, d)$, the paper's (A36); Proposition 11.4(c) gives $U = \lambda\, \mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)\mathsf{U}_{\mathcal{V}}$, the paper's (A37). For the existence form, Theorem 9.11 writes $M = O\Sigma(r)O'$ with O, O' orthogonal symplectic and $r \geq 0$, and Theorem 9.4(b) writes $O = O_{\mathcal{U}}$, $O' = O_{\mathcal{V}}$.

(c) By Theorem 11.5(b) and Theorem 10.21(c), the unitary $\mathsf{U}_\Theta \mathsf{S}(|\zeta|)\mathsf{U}_\Theta^*$, $\Theta = \Theta_\theta$, conjugates $D(\alpha)$ to $D(\Theta L_{-|\zeta|}\Theta^* \alpha)$, and in mode j $e^{-i\theta_j}(c_j e^{i\theta_j}\alpha_j + s_j \overline{e^{i\theta_j}\alpha_j}) = c_j\alpha_j + e^{-2i\theta_j} s_j \bar{\alpha}_j = (L_{-\zeta}\alpha)_j$, since $e^{-2i\theta_j} = e^{i\phi_j}$ when $\zeta_j \neq 0$ and $s_j = 0$ otherwise. So it has the conjugation action of $\mathsf{S}(\zeta)$. Its vacuum expectation is $\langle \mathsf{U}_\Theta^* |0\rangle, \mathsf{S}(|\zeta|)\mathsf{U}_\Theta^* |0\rangle\rangle = \langle 0|\mathsf{S}(|\zeta|)|0\rangle = (\prod_j c_j)^{-1/2}$, which is $\langle 0|\mathsf{S}(\zeta)|0\rangle \neq 0$ (Theorem 10.21(a)). Corollary 10.17(a) gives the first identity. The second follows from it, from $D(\alpha)\mathsf{U}_\Theta = \mathsf{U}_\Theta D(\Theta^* \alpha)$ (Theorem 11.5(b)) and from $\mathsf{U}_{\mathcal{U}}\mathsf{U}_\Theta = \mathsf{U}_{\mathcal{U}\Theta}$. □

Corollary 11.7 (The vacuum, one mode, and the quadratures)**.** Lean ✓

(a) *For one mode and $\alpha, \zeta \in \mathbb{C}$, $D(\alpha)\mathsf{S}(\zeta)|0\rangle = c^{-1/2} e^{-|\alpha|^2/2 - \tau\bar{\alpha}^2/2} |\kappa, \ell\rangle$ with $\kappa = -\tau/2$ and $\ell = \alpha + \tau\bar{\alpha} = \alpha - 2\kappa\bar{\alpha}$, the paper's (8) with its constant. For N modes and $\alpha, \zeta \in \mathbb{C}^N$,*

$$D(\alpha)\mathsf{S}(\zeta)|0\rangle = \Big(\prod_j c_j\Big)^{-1/2} e^{-|\alpha|^2/2 - \sum_j \tau_j \bar{\alpha}_j^2/2} \bigotimes_j |-\tau_j/2,\, \alpha_j + \tau_j \bar{\alpha}_j\rangle.$$

If $U = \lambda\, \mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)\mathsf{U}_{\mathcal{V}}$ with $\lambda \in \mathbb{C}$, $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$, $\alpha \in \mathbb{C}^N$ and $r \in \mathbb{R}^N$, then $U|0\rangle = \lambda\, \mathsf{U}_{\mathcal{U}} \bigotimes_j (D(\alpha_j)\mathsf{S}(r_j)|0\rangle)$, the paper's (11) and (A40)–(A42). Every Gaussian unitary U satisfies this for some λ with $|\lambda| = 1$, $\mathcal{U}$, $\mathcal{V}$, α and $r \geq 0$.

(b) *For one mode, every Gaussian unitary is* $\lambda D(\alpha)\mathsf{S}(\zeta)R(\theta)$ *for some* $\lambda,\alpha,\zeta\in\mathbb{C}$ *and* $\theta\in\mathbb{R}$ *with* $|\lambda|=1$*, the paper's (6). Moreover* $R(\theta)R(\theta')=R(\theta+\theta')$ *on any number of modes, and every unitary* 1×1 *matrix is* $e^{-i\theta}$ *for some real* θ*.*

(c) *Let* U *be Gaussian with data* (M,d)*. Then* $U\psi\in\mathcal{Q}$ *if and only if* $\psi\in\mathcal{Q}$*, so* U *maps* $\mathcal{Q}$ *onto itself. If* $\psi\in\mathcal{Q}$ *and* $\xi\in\Gamma_s(\iota)$ *satisfies* $\jmath\xi=\sum_l M_{kl}\hat{R}_l\jmath\psi+d_k\jmath\psi$*, then* $\hat{R}_k\jmath U\psi=\jmath U\xi$*; that is,* $U^*\hat{R}_kU=\sum_l M_{kl}\hat{R}_l+d_k$ *on* $\mathcal{Q}$*, the paper's (A3). Such a* ξ *exists for every* $\psi\in\mathcal{Q}$*, all real* M*,* d *and every* k*.*

Proof. (a) We compare inner products with $\varepsilon(w)$, $w\in\mathbb{C}^N$. By Theorem 10.12(b), $D(-\alpha)\varepsilon(w)=e^{-|\alpha|^2/2+\langle\alpha,w\rangle}\varepsilon(w-\alpha)$, so by Theorem 10.21(a) at $z=0$,

$$\langle\varepsilon(w),D(\alpha)\mathsf{S}(\zeta)|0\rangle\rangle=\Big(\prod_j c_j\Big)^{-1/2}\exp\sum_j\Big(-\tfrac12|\alpha_j|^2+\alpha_j\bar{w}_j-\tfrac12\tau_j(\bar{w}_j-\bar{\alpha}_j)^2\Big).$$

By the remark after Lemma 10.19, the right side of the claim gives $(\prod_j c_j)^{-1/2}$ times $\exp\sum_j(-\frac12|\alpha_j|^2-\frac12\tau_j\bar{\alpha}_j^2-\frac12\tau_j\bar{w}_j^2+(\alpha_j+\tau_j\bar{\alpha}_j)\bar{w}_j)$, which is equal. Both vectors lie in $\jmath(\Gamma_s(\iota))$, the product vector because $|\tau_j/2|<\frac12$ (proof of Lemma 10.8), so they are equal by Theorem 10.5; one mode is the case $N=1$. For the given U, $\mathsf{U}_\mathcal{V}|0\rangle=|0\rangle$, and the several-mode formula is the product vector of the one-mode formulas, since $\bigotimes_j(x_j\chi_j)=(\prod_j x_j)\bigotimes_j\chi_j$ for numbers x_j. The existence form is Theorem 11.6(a).

(b) A unitary 1×1 matrix is a number of modulus 1, so $e^{-i\theta}$, and $R(\theta)R(\theta')=R(\theta+\theta')$ because $\Theta_\theta\Theta_{\theta'}=\Theta_{\theta+\theta'}$ and $\mathcal{U}\mapsto\mathsf{U}_\mathcal{U}$ is a homomorphism. By Theorem 11.6(a), $U=\lambda R(\theta_1)D(\alpha)\mathsf{S}(r)R(\theta_2)$ with $r\ge0$. By Theorem 11.5(b), $R(\theta_1)D(\alpha)=D(e^{-i\theta_1}\alpha)R(\theta_1)$. By Theorem 11.6(c) with $\zeta=re^{-2i\theta_1}$, which satisfies $|\zeta|e^{-2i\theta_1}=\zeta$ for every real θ_1 because $r\ge0$, $R(\theta_1)\mathsf{S}(r)=\mathsf{S}(re^{-2i\theta_1})R(\theta_1)$. Hence $U=\lambda D(e^{-i\theta_1}\alpha)\mathsf{S}(re^{-2i\theta_1})R(\theta_1+\theta_2)$.

(c) Say that a unitary V *implements* (M,d) on $\mathcal{Q}$ if V maps $\mathcal{Q}$ onto $\mathcal{Q}$ and $\hat{R}_k\jmath V\psi=\jmath V\xi$ whenever $\psi\in\mathcal{Q}$ and $\jmath\xi=\sum_l M_{kl}\hat{R}_l\jmath\psi+d_k\jmath\psi$. A phase implements $(1,0)$. By Theorem 10.23, $\mathcal{Q}=\bigcap_j\operatorname{dom}\overline{a_j^\dagger}$ as well, so for $\psi\in\mathcal{Q}$ every $\hat{R}_l\jmath\psi$ lies in $\jmath(\Gamma_s(\iota))$, and ξ exists. By Theorem 10.13, with $(\alpha_j+\bar{\alpha}_j)/\sqrt{2}=\sqrt{2}\operatorname{Re}\alpha_j$ and $i(\bar{\alpha}_j-\alpha_j)/\sqrt{2}=\sqrt{2}\operatorname{Im}\alpha_j$, $D(\alpha)$ implements $(1,\delta(\alpha))$. By Theorem 10.23, $\mathsf{S}(\zeta)$ implements the matrix of $a\mapsto c\,a-e^{i\phi}s\,a^\dagger$ on $(\hat{q},\hat{p})$, which is M_ζ by the computation in the proof of Lemma 11.2(d). The passive unitary maps $\mathcal{Q}$ into $\mathcal{Q}$, since $a_j\Gamma(\mathcal{U})\jmath\psi=\Gamma(\mathcal{U})\sum_k\mathcal{U}_{jk}a_k\jmath\psi$ (Theorem 8.5) and $\Gamma(\mathcal{U})$ preserves $\jmath(\Gamma_s(\iota))$; with $\mathsf{U}_{\mathcal{U}^*}$ it maps onto. By Theorem 8.7(c), $\Gamma(\mathcal{U}^*)\hat{R}\Gamma(\mathcal{U})=O_\mathcal{U}\hat{R}$ on $\mathcal{F}$, since $(X+iY)(\hat{q}+i\hat{p})=(X\hat{q}-Y\hat{p})+i(Y\hat{q}+X\hat{p})$; so $\mathsf{U}_\mathcal{U}$ implements $(O_\mathcal{U},0)$. If V_1 implements (M_1,d_1) and V_2 implements (M_2,d_2), then V_1V_2 implements $(M_1M_2,d_1+M_1d_2)$: with $\jmath\xi_l=\sum_m(M_2)_{lm}\hat{R}_m\jmath\psi+(d_2)_l\jmath\psi$,

$$\hat{R}_k\jmath V_1V_2\psi=\jmath V_1\Big(\sum_l(M_1)_{kl}V_2\xi_l+(d_1)_kV_2\psi\Big)=\jmath V_1V_2\Big(\sum_l(M_1)_{kl}\xi_l+(d_1)_k\psi\Big),$$

and $\jmath$ of the last bracket is $\sum_m(M_1M_2)_{km}\hat{R}_m\jmath\psi+(d_1+M_1d_2)_k\jmath\psi$. By Theorem 11.6(a), U is a phase times $\mathsf{U}_\mathcal{U}D(\alpha)\mathsf{S}(r)\mathsf{U}_\mathcal{V}$, which implements $(O_\mathcal{U}\Sigma(r)O_\mathcal{V},O_\mathcal{U}\delta(\alpha))=(M,d)$, the order of (A35). □

In (b) the sign of $e^{-i\theta_1}$ needs no choice: Theorem 11.6(c) holds for every admissible θ, not only for $\theta_j=-\phi_j/2$. Equivalently, U_{-I} commutes with $\mathsf{S}(r)$, since both preserve the parity of the particle number.

Notes and references

Tjoa and Cirac [4], Appendix A, lift the factors of the Bloch–Messiah decomposition $S = O_U \Sigma O_W$ (A11) to Fock space in (A17)–(A33), compute the action of $\tilde{U}_G = \mathsf{U} D(\alpha) S(r) \mathsf{W}$ on the quadratures (A35), choose $\delta = O_U^T d$ (A36), and conclude $U_G = e^{i\chi} \tilde{U}_G$ (A37) because $U_G \tilde{U}_G^\dagger$ commutes with every canonical operator and the Fock representation is irreducible; they absorb the squeezing phases into the passive factors (A38) and obtain (11) in (A40)–(A42). We prove their Proposition 5 for Gaussian unitaries in the Weyl form of Definition 11.3, with irreducibility proved in Theorem 10.16. These unitaries satisfy (A3) on $\mathcal{Q}$ (Corollary 11.7(c)); the converse, from (A3) with the paper's unspecified domains to the Weyl form, needs input on self-adjointness and domains and is not proved here. The overlaps of one-mode Gaussian vectors in Corollary 11.7(a) come from Lemma 10.19, the one-mode case of Mehler's formula, which the companion notes on Gaussian-augmented matrix product states use for matrix arguments. The metaplectic representation and the group of Gaussian unitaries are treated in [16].

12 Characteristic functions and pure Gaussian states

Tjoa and Cirac define a pure Gaussian state as $U|0\rangle$ for a Gaussian unitary U [4, (4)]; the textbook definition is a unit vector whose characteristic function $\langle \psi, W(u)\psi \rangle$ is a Gaussian function of u. We prove that the two agree. Two facts carry the proof, and both rest on irreducibility (Theorem 10.16): a vector is determined up to a phase by its characteristic function, and a unit vector with a Gaussian characteristic function has a symplectic covariance matrix, so that it cannot be a nontrivial Gaussian mixture.

Notation for this section.

$\Gamma_s(N)$	the bosonic Fock space of the modes $\{1, \dots, N\}$, $N \geq 1$; N' another number of modes
$W(u)$, σ, $\alpha(u)$, $\delta(\alpha)$, $\hat{R}$, $\mathcal{Q}$	as in Section 11
(M, d)	data of a Gaussian unitary (Definition 11.3)
χ_ψ	characteristic function of ψ (Definition 12.1)
$\chi^{\mathrm{G}}_{d,\gamma}$	Gaussian function of displacement d and covariance γ (Definition 12.5)
γ, $A \succeq 0$	a real $2N \times 2N$ matrix, symmetric where stated; A is Hermitian and positive semidefinite (Section 1)
γ_κ	the covariance matrix of the normalized vector $\lvert \kappa, \ell \rangle$ (Lemma 12.17)

Inner products are antilinear in the first argument. A Gaussian unitary has some data (M, d), unique by Proposition 11.4(c).

12.1 Characteristic functions

Definition 12.1 (Characteristic function)**.** For $\psi \in \Gamma_s(N)$ and $u \in \mathbb{R}^{2N}$, the *characteristic function* of ψ is $\chi_\psi(u) = \langle \psi, W(u)\psi \rangle$.

For a unit vector this is the symmetrically ordered characteristic function [33] $\mathrm{Tr}(\rho D(\xi))$ of $\rho = |\psi\rangle\langle\psi|$ at $\xi = \alpha(u)$; formally $W(u) = e^{-i\sigma(u, \hat{R})}$.

Lemma 12.2 (Elementary properties)**.** Lean ✓ *Let $\psi \in \Gamma_s(N)$, $u \in \mathbb{R}^{2N}$ and $a \in \mathbb{C}$. Then $\chi_\psi(0) = \|\psi\|^2$, $\chi_\psi(-u) = \overline{\chi_\psi(u)}$, $|\chi_\psi(u)| \leq \|\psi\|^2$ and $\chi_{a\psi} = |a|^2 \chi_\psi$. If U is a Gaussian unitary with data (M, d), then*

$$\chi_{U\psi}(u) = e^{-i\sigma(u,d)}\, \chi_\psi(M^{-1}u).$$

Proof. $W(0) = D(0) = 1$; $\langle\psi, W(-u)\psi\rangle = \langle\psi, W(u)^*\psi\rangle = \langle W(u)\psi, \psi\rangle$ by Proposition 11.4(a); the bound is the Cauchy–Schwarz inequality, $W(u)$ being unitary; the scaling is sesquilinearity. Finally $\langle U\psi, W(u)U\psi\rangle = \langle\psi, U^*W(u)U\psi\rangle$, and Definition 11.3 applies. □

The inner products of the vectors $W(u)\psi$ are values of χ_ψ, so the characteristic function fixes the Gram matrix of the orbit of ψ; since the orbit spans a dense subspace, Lemma 10.9 turns equal characteristic functions into an isometry.

Lemma 12.3 (The intertwiner). Lean ✓

(a) *For $\psi \in \Gamma_s(N)$ and $u, v \in \mathbb{R}^{2N}$, $\langle W(u)\psi, W(v)\psi\rangle = e^{i\sigma(u,v)/2}\chi_\psi(v-u)$.*

(b) *If $\psi \neq 0$, the vectors $W(u)\psi$, $u \in \mathbb{R}^{2N}$, span a dense subspace of $\Gamma_s(N)$.*

(c) *Let σ' be the symplectic form of $\mathbb{R}^{2N'}$ and $E : \mathbb{R}^{2N} \to \mathbb{R}^{2N'}$ linear with $\sigma'(Eu, Ev) = \sigma(u, v)$ for all u, v. Let $\psi \in \Gamma_s(N)$, $\psi \neq 0$, and $\psi' \in \Gamma_s(N')$ with $\chi_\psi(u) = \chi_{\psi'}(Eu)$ for every u. Then there is exactly one linear isometry $T : \Gamma_s(N) \to \Gamma_s(N')$ with $T\psi = \psi'$ and $TW(u)x = W(Eu)Tx$ for all $u \in \mathbb{R}^{2N}$ and $x \in \Gamma_s(N)$.*

Proof. (a) $\langle W(u)\psi, W(v)\psi\rangle = \langle\psi, W(-u)W(v)\psi\rangle = e^{-i\sigma(-u,v)/2}\chi_\psi(v-u)$ by Proposition 11.4(a).

(b) Let K be the closure of the span. Since $W(v)W(u)\psi = e^{-i\sigma(v,u)/2}W(v+u)\psi$ and $W(v)$ is continuous, $W(v)K \subseteq K$ for every v, so $D(\alpha)K = W(\delta(\alpha))K \subseteq K$ for every α. As $\psi \in K$ is nonzero, Corollary 10.17(b) gives $K = \Gamma_s(N)$.

(c) By (a) for ψ and for ψ', the linearity of E and the two hypotheses, $\langle W(Eu)\psi', W(Ev)\psi'\rangle = e^{i\sigma'(Eu,Ev)/2}\chi_{\psi'}(E(v-u)) = \langle W(u)\psi, W(v)\psi\rangle$. By (b) and Lemma 10.9 there is exactly one linear isometry T with $TW(u)\psi = W(Eu)\psi'$ for every u; at $u = 0$, $T\psi = \psi'$. For fixed v, the bounded operators $TW(v)$ and $W(Ev)T$ agree on every $W(u)\psi$:

$$\begin{aligned} TW(v)W(u)\psi &= e^{-i\sigma(v,u)/2}W(E(v+u))\psi' \\ &= e^{-i\sigma'(Ev,Eu)/2}W(Ev+Eu)\psi' = W(Ev)TW(u)\psi, \end{aligned}$$

hence everywhere by (b). Conversely, an isometry with $T\psi = \psi'$ and $TW(u) = W(Eu)T$ sends $W(u)\psi$ to $W(Eu)\psi'$, so it is the one of Lemma 10.9. □

Theorem 12.4 (A vector is determined up to a phase by its characteristic function). Lean ✓
Let $\psi, \psi' \in \Gamma_s(N)$ with $\psi \neq 0$ and $\chi_{\psi'} = \chi_\psi$. Then $\psi' = \lambda\psi$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$.

Proof. Lemma 12.3(c) with $N' = N$ and $E = 1$ gives an isometry T with $T\psi = \psi'$ that commutes with every $W(u)$, hence with every $D(\alpha) = W(\delta(\alpha))$. By Theorem 10.16, $T = \lambda 1$, and $|\lambda| = 1$ since T is an isometry. So $\psi' = T\psi = \lambda\psi$. □

Conversely $\chi_{\lambda\psi} = \chi_\psi$ when $|\lambda| = 1$, by Lemma 12.2.

12.2 Gaussian characteristic functions and the state condition

For a state with mean $d = \langle\hat{R}\rangle$ and covariance matrix $\gamma_{kl} = \langle\{\hat{R}_k - d_k, \hat{R}_l - d_l\}\rangle$, twice the symmetrized covariance as in Definition 2.3, the formal identity $W(u) = e^{-i\sigma(u,\hat{R})}$ and the moment formula for Gaussian states give $\langle W(u)\rangle = \exp(-i\sigma(u,d) - \frac{1}{4}(\Omega^T u)^T\gamma(\Omega^T u))$. This motivates the definition; its identification with the moments is Proposition 12.10.

Definition 12.5 (Gaussian characteristic function). For $d \in \mathbb{R}^{2N}$ and a real $2N \times 2N$ matrix γ, the *Gaussian function* is

$$\chi^{\mathrm{G}}_{d,\gamma}(u) = \exp\Big(-i\sigma(u,d) - \tfrac{1}{4}\,u^T\Omega\gamma\Omega^T u\Big), \qquad u \in \mathbb{R}^{2N}.$$

A vector $\psi \in \Gamma_s(N)$ *has the Gaussian characteristic function with displacement vector d and covariance matrix γ* if $\gamma^T = \gamma$ and $\chi_\psi = \chi^{\mathrm{G}}_{d,\gamma}$.

$\chi^{\mathrm{G}}_{d,\gamma}$ has the form $\exp(i\,\mathrm{disp}(u) - \mathrm{cov}(u,u)/4)$ with the linear form $\mathrm{disp}(u) = -\sigma(u,d)$ and the bilinear form $\mathrm{cov}(u,v) = u^T\Omega\gamma\Omega^T v$, symmetric when γ is. The vacuum has $\chi_{|0\rangle}(u) = e^{-|u|^2/4} = \chi^{\mathrm{G}}_{0,1}(u)$ (Proposition 11.4(a)), so it has displacement 0 and covariance 1, the convention of the reading guide. Only unit vectors have a Gaussian characteristic function, since $\chi^{\mathrm{G}}_{d,\gamma}(0) = 1$.

The admissibility of γ follows from a positivity property that every characteristic function has: by Lemma 12.3(a), the matrix $[e^{i\sigma(u_j,u_k)/2}\chi_\psi(u_k - u_j)]_{jk}$ is the Gram matrix of the vectors $W(u_j)\psi$. We state the second-order argument that extracts the state condition from it in general. For a real vector space V with a bilinear form σ and a function $\chi : \mathsf{V} \to \mathbb{C}$, the *twisted kernel* of a finite family (u_j) in V is $[e^{i\sigma(u_j,u_k)/2}\chi(u_k - u_j)]_{jk}$; σ is *alternating* if $\sigma(u,u) = 0$ for every u.

Lemma 12.6 (The state condition from positive semidefinite kernels). Lean ✓ *Let* cov *be a symmetric bilinear form and* disp *a linear form on* V*, both real, and* $\chi(u) = \exp(i\,\mathrm{disp}(u) - \mathrm{cov}(u,u)/4)$*. For a finite family* (u_j) *put* $H_{jk} = (\mathrm{cov}(u_j,u_k) + i\sigma(u_j,u_k))/2$*.*

(a) *For every family* (u_j) *and* $s \in \mathbb{R}$*, the twisted kernel of* (su_j) *has the entries* $\overline{\chi(su_j)}\,\chi(su_k)\,e^{s^2 H_{jk}}$*.*

Let moreover every twisted kernel of χ *be positive semidefinite. Then for every finite family* (u_j) *and all* $b_j \in \mathbb{C}$*:*

(b) *if* $\sum_j b_j = 0$*, then* $\sum_{j,k} \bar{b}_j b_k (e^{s^2 H_{jk}} - 1)$ *is a real number* ≥ 0 *for every* $s \in \mathbb{R}$*;*

(c) *the same holds with no condition on* $\sum_j b_j$*;*

(d) $\mathrm{Re}\sum_{j,k} \bar{b}_j b_k (\mathrm{cov}(u_j,u_k) + i\sigma(u_j,u_k)) \geq 0$*;*

(e) *if* σ *is alternating, the* state condition *holds:* $\big[\mathrm{cov}(u_j,u_k) + i\sigma(u_j,u_k)\big]_{jk} \succeq 0$ *for every finite family.*

Proof. (a) Expanding $\mathrm{cov}(s(u_k - u_j), s(u_k - u_j))$ and using the linearity of disp, $\chi(s(u_k - u_j)) = \overline{\chi(su_j)}\chi(su_k)e^{s^2\mathrm{cov}(u_j,u_k)/2}$, since disp and cov are real; multiply by $e^{is^2\sigma(u_j,u_k)/2}$.

(b) Apply the positivity of the twisted kernel of (su_j) to the coefficients $b_k/\chi(su_k)$, which is allowed since χ has no zeros: by (a), $\sum_{j,k} \bar{b}_j b_k e^{s^2 H_{jk}}$ is real and ≥ 0. Subtract $\sum_{j,k} \bar{b}_j b_k = |\sum_j b_j|^2 = 0$.

(c) Apply (b) to the family $(0, u_1, \ldots, u_m)$ with the coefficients $(-\sum_j b_j, b_1, \ldots, b_m)$, which sum to 0. The entries of H in the row and column of 0 vanish, so their terms $e^0 - 1$ contribute nothing.

(d) Divide (c) by $s^2 > 0$ and let $s \to 0$: $(e^{s^2 H_{jk}} - 1)/s^2 \to H_{jk}$, so $\sum_{j,k} \bar{b}_j b_k H_{jk}$ is a limit of real numbers ≥ 0, and its real part is ≥ 0.

(e) If σ is alternating, it is antisymmetric, so the matrix is Hermitian and its quadratic form is real; by (d) it is ≥ 0. □

For a Gaussian state on the algebra of the canonical commutation relations over (V, σ), the same five statements hold with χ the expectation of the Weyl operators and cov, disp its covariance and displacement, since the twisted kernels of a state are positive semidefinite.

Lemma 12.7 (Gaussian characteristic functions). Lean ✓ *Let* $\psi \in \Gamma_s(N)$ *have the Gaussian characteristic function with* (d, γ)*.*

(a) $\|\psi\| = 1$*.*

(b) *If* ψ *also has it with* (d', γ')*, then* $d = d'$ *and* $\gamma = \gamma'$*.*

(c) γ *is admissible:* $\gamma - i\Omega \succeq 0$.

Proof. (a) $\|\psi\|^2 = \chi_\psi(0) = 1$.

(b) Taking moduli, $u^T\Omega\gamma\Omega^T u = u^T\Omega\gamma'\Omega^T u$ for every u; both matrices are symmetric, so by polarization $\Omega\gamma\Omega^T = \Omega\gamma'\Omega^T$, and $\gamma = \gamma'$ since Ω is orthogonal. Then $e^{-i\sigma(u,d-d')} = 1$ for every u, and the rescaling argument in the proof of Proposition 11.4(c) gives $d = d'$.

(c) By Lemma 12.3(a) every twisted kernel of χ_ψ is a Gram matrix, hence positive semidefinite, and χ_ψ has the form of Lemma 12.6 with $\mathrm{disp}(u) = -\sigma(u,d)$ and $\mathrm{cov}(u,v) = u^T\Omega\gamma\Omega^T v$. The form σ is alternating, so by Lemma 12.6(e) for the coordinate vectors, $\Omega\gamma\Omega^T + i\Omega \succeq 0$. Congruence by Ω gives $\gamma + i\Omega = \Omega^T(\Omega\gamma\Omega^T + i\Omega)\Omega \succeq 0$, and entrywise complex conjugation, which preserves $\succeq 0$, gives $\gamma - i\Omega \succeq 0$. □

Theorem 12.13 below compares γ with a symplectic matrix below it, which the next lemma supplies; the matrix it constructs is explicit.

Lemma 12.8 (A symplectic matrix below an admissible one). Lean ✓ *If γ is admissible, there is a positive definite $\gamma_0 \in \mathrm{Sp}(2N,\mathbb{R})$ with $\gamma - \gamma_0 \succeq 0$.*

Proof. By Theorem 3.1, γ is positive definite; let $\gamma^{1/2}$ be its positive square root and $A = \gamma^{-1/2}\Omega\gamma^{-1/2}$, a real, antisymmetric, invertible matrix. Congruence by $\gamma^{-1/2}$ turns $\gamma - i\Omega \succeq 0$ into $1 - iA \succeq 0$. For real vectors x, y and $v = x + iy$, $v^*(1 - iA)v = |x|^2 + |y|^2 + 2x^TAy$, using $y^TAx = -x^TAy$. With $y = Ax$ this is $|x|^2 + |Ax|^2 - 2|Ax|^2 \geq 0$, since $x^TA^2x = -|Ax|^2$; so $A^TA \preceq 1$. Let $|A| = (A^TA)^{1/2}$, symmetric and positive definite. For every real x, by the Cauchy–Schwarz inequality, $x^T|A|x \leq |x|\,||A|x| = |x|(x^TA^TAx)^{1/2} \leq |x|^2$, so $|A| \preceq 1$.

Put $\gamma_0 = \gamma^{1/2}|A|\gamma^{1/2}$, symmetric and positive definite. Then $\gamma - \gamma_0 = \gamma^{1/2}(1 - |A|)\gamma^{1/2} \succeq 0$. A commutes with $A^TA = -A^2$, hence with its square root $|A|$, and $|A|^2 = -A^2$. From $\Omega^{-1} = -\Omega$, $\gamma^{1/2}\Omega\gamma^{1/2} = -A^{-1}$, and

$$\gamma_0\Omega\gamma_0 = \gamma^{1/2}|A|(-A^{-1})|A|\gamma^{1/2} = -\gamma^{1/2}|A|^2A^{-1}\gamma^{1/2} = \gamma^{1/2}A\gamma^{1/2} = \Omega.$$

Since γ_0 is symmetric, $\gamma_0\Omega\gamma_0^T = \Omega$. □

If γ is itself symplectic, then $|A| = 1$ and $\gamma_0 = \gamma$. Physically, $\gamma - \gamma_0 \succeq 0$ says that the Gaussian state of γ is a Gaussian mixture of displaced states with covariance γ_0.

Theorem 12.9 (Gaussian unitaries on the vacuum). Lean ✓ *If U is a Gaussian unitary with data (M,d), then $U|0\rangle$ has the Gaussian characteristic function with displacement vector d and covariance matrix MM^T.*

Proof. By Lemma 12.2 and Proposition 11.4(a), $\chi_{U|0\rangle}(u) = e^{-i\sigma(u,d)}e^{-|M^{-1}u|^2/4}$. From $M\Omega M^T = \Omega$, $M^{-1} = \Omega M^T\Omega^T$, so $(M^{-1})^TM^{-1} = \Omega M\Omega^T\Omega M^T\Omega^T = \Omega MM^T\Omega^T$. The matrix MM^T is symmetric. □

MM^T is positive definite and symplectic by Lemma 9.7. The displacement vector and the covariance matrix are the first and second moments of the quadratures, in the following sense.

Proposition 12.10 (Moments; [4, (13)]). Lean ✓ *Let U be a Gaussian unitary with data (M,d), $\psi = U|0\rangle$ and $\gamma = MM^T$. There are $\phi_1, \dots, \phi_{2N} \in \Gamma_s(N)$ with $\hat{R}_k\jmath\psi = \jmath\phi_k$ and $\langle\psi,\phi_k\rangle = d_k$ for every k, and*

$$\langle\phi_k - d_k\psi,\ \phi_l - d_l\psi\rangle = \tfrac{1}{2}(\gamma_{kl} + i\Omega_{kl}) \qquad \text{for all } k,l.$$

Proof. $|0\rangle \in \mathcal{Q}$, so by Corollary 11.7(c), with the finitely supported $\xi_k = \sum_m M_{km}\hat{R}_m|0\rangle + d_k|0\rangle$, the vectors $\phi_k = U\xi_k$ satisfy $\hat{R}_k\jmath\psi = \jmath\phi_k$. From $\hat{q}_j|0\rangle = |e_j\rangle/\sqrt{2}$ and $\hat{p}_j|0\rangle = i|e_j\rangle/\sqrt{2}$, $\langle 0|\hat{R}_m|0\rangle = 0$ and $\langle\hat{R}_m|0\rangle, \hat{R}_n|0\rangle\rangle = \frac{1}{2}(\delta_{mn} + i\Omega_{mn})$. Since U is unitary, $\langle\psi,\phi_k\rangle = \langle 0,\xi_k\rangle = d_k$, and

$$\langle\phi_k - d_k\psi, \phi_l - d_l\psi\rangle = \sum_{m,n} M_{km}M_{ln}\langle\hat{R}_m|0\rangle, \hat{R}_n|0\rangle\rangle = \tfrac{1}{2}\left(MM^T + iM\Omega M^T\right)_{kl},$$

and $M\Omega M^T = \Omega$. □

In particular $\psi \in \mathcal{Q}$, since $a_j = (\hat{q}_j + i\hat{p}_j)/\sqrt{2}$. In the paper's notation $r_k = \langle \hat{R}_k \rangle = d_k$ and $\Gamma_{kl} = \frac{1}{2}\langle\{\hat{R}_k - r_k, \hat{R}_l - r_l\}\rangle = \gamma_{kl}/2$, the anticommutator expectation being read as $2\,\mathrm{Re}\langle(\hat{R}_k - r_k)\psi, (\hat{R}_l - r_l)\psi\rangle$.

12.3 Pure Gaussian states

Definition 12.11 (Pure Gaussian state; [4, (4)])**.** A vector $\psi \in \Gamma_s(N)$ is a *pure Gaussian state* if $\psi = U|0\rangle$ for some Gaussian unitary U.

The global phase in [4, (4)] is absorbed: λU with $|\lambda| = 1$ is Gaussian with the same data (Proposition 11.4(b)). The vacuum is pure Gaussian, with $U = 1$, and 0 is not, since $\|U|0\rangle\| = 1$.

Lemma 12.12 (Existence)**.** Lean ✓

(a) *For every $M \in \mathrm{Sp}(2N, \mathbb{R})$ and $d \in \mathbb{R}^{2N}$ there is a Gaussian unitary with data (M, d).*

(b) *Every positive definite $\gamma \in \mathrm{Sp}(2N, \mathbb{R})$ is MM^T for some $M \in \mathrm{Sp}(2N, \mathbb{R})$.*

Proof. (a) By Theorems 9.11 and 9.4(b), $M = O_{\mathcal{U}}\Sigma(r)O_{\mathcal{V}}$ with $\mathcal{U}, \mathcal{V} \in \mathrm{U}(N)$; by Theorem 11.6(b) with $\zeta = r$, $\mathsf{U}_{\mathcal{U}} D(\alpha) \mathsf{S}(r) \mathsf{U}_{\mathcal{V}}$ has data $(M, O_{\mathcal{U}}\delta(\alpha))$, and $\delta(\alpha) = O_{\mathcal{U}}^T d$ gives displacement d.

(b) By Theorem 9.10(b), $\gamma = O\Sigma(r)^2 O^T$ with O orthogonal symplectic; take $M = O\Sigma(r)$, which is symplectic, with $MM^T = O\Sigma(r)^2 O^T$ since $\Sigma(r)$ is symmetric. □

The next theorem is the step from a Gaussian characteristic function to a pure state. If γ lay strictly above the symplectic γ_0 of Lemma 12.8, the state would be a Gaussian mixture; the proof realizes the mixture with $2N$ ancilla modes and lets irreducibility exclude it.

Theorem 12.13 (The covariance matrix is symplectic)**.** Lean ✓ *If $\psi \in \Gamma_s(N)$ has the Gaussian characteristic function with (d, γ), then $\gamma \in \mathrm{Sp}(2N, \mathbb{R})$; as γ is symmetric, $\gamma\Omega\gamma = \Omega$.*

Proof. A symplectic part. By Lemma 12.7(c) γ is admissible, and Lemma 12.8 gives a positive definite $\gamma_0 \in \mathrm{Sp}(2N, \mathbb{R})$ with $\Delta = \gamma - \gamma_0 \succeq 0$. Let $C = (\Omega\Delta\Omega^T)^{1/2}$, symmetric and positive semidefinite, so that $u^T\Omega\gamma\Omega^T u = u^T\Omega\gamma_0\Omega^T u + |Cu|^2$.

Ancillas. Take $N' = 3N$ modes, the N modes of ψ followed by $2N$ ancilla modes. Reordering the coordinates of $\mathbb{R}^{6N}$ (positions first) by subsystem, we write $w = (w_s; x, y)$ with $w_s \in \mathbb{R}^{2N}$ for the first N modes and $x, y \in \mathbb{R}^{2N}$ for the positions and the momenta of the ancillas. Under this reordering Ω_{3N} becomes $\Omega_N \oplus \Omega_{2N}$ (the proof of Theorem 1.5), so $\sigma'(w, w') = \sigma(w_s, w_s') + x \cdot y' - y \cdot x'$. The map $E(u) = (u; Cu, 0)$ is linear with $\sigma'(Eu, Ev) = \sigma(u, v)$. The matrix $\gamma_0 \boxplus 1$ (Definition 1.4) is positive definite and symplectic, so by Lemmas 12.12(b) and (a) and Theorem 12.9 there is $\psi' \in \Gamma_s(3N)$ with the Gaussian characteristic function of $((d; 0, 0), \gamma_0 \boxplus 1)$:

$$\chi_{\psi'}(w) = \exp\Big(-i\sigma(w_s, d) - \tfrac{1}{4} w_s^T \Omega\gamma_0\Omega^T w_s - \tfrac{1}{4}\big(|x|^2 + |y|^2\big)\Big).$$

At $w = Eu$ this is $\exp(-i\sigma(u, d) - \frac{1}{4}u^T\Omega\gamma_0\Omega^T u - \frac{1}{4}|Cu|^2) = \chi_\psi(u)$.

The intertwiner. Since $\|\psi\| = 1$, Lemma 12.3(c) gives a linear isometry $T : \Gamma_s(N) \to \Gamma_s(3N)$ with $T\psi = \psi'$ and $TW(v) = W(Ev)T$. Taking adjoints at $-v$, $T^*W(Ev) = W(v)T^*$.

Commuting Weyl operators. For $h \in \mathbb{R}^{2N}$ let $w_h = (0; h, 0)$. Then $\sigma'(w_h, Ev) = h \cdot 0 - 0 \cdot Cv = 0$, so $W(w_h)$ commutes with every $W(Ev)$, and $X_h = T^*W(w_h)T$ satisfies

$$X_h W(v) = T^*W(w_h)W(Ev)T = T^*W(Ev)W(w_h)T = W(v)X_h.$$

By Theorem 10.16, $X_h = f_h 1$ for a number f_h.

$C = 0$. For every u, since $\sigma'(w_h, Eu) = 0$,

$$f_h\,\chi_\psi(u) = \langle\psi, X_h W(u)\psi\rangle = \langle T\psi, W(w_h)TW(u)\psi\rangle = \langle\psi', W(w_h)W(Eu)\psi'\rangle = \chi_{\psi'}(w_h + Eu),$$

and $w_h + Eu = (u; Cu + h, 0)$, so the right side is $\chi_\psi(u)\exp(-\frac{1}{4}(|Cu+h|^2 - |Cu|^2))$. At $u = 0$, $f_h = e^{-|h|^2/4}$. As $\chi_\psi(u) \neq 0$, $e^{-|h|^2/4} = e^{-(2Cu\cdot h + |h|^2)/4}$ with real exponents, so $Cu \cdot h = 0$ for all u, h, and $h = Cu$ gives $Cu = 0$. Hence $C = 0$, $\Omega\Delta\Omega^T = 0$, $\Delta = 0$, and $\gamma = \gamma_0 \in \mathrm{Sp}(2N, \mathbb{R})$. □

Theorem 12.14 (The two definitions agree)**.** Lean ✓

(a) *$\psi \in \Gamma_s(N)$ is a pure Gaussian state if and only if it has the Gaussian characteristic function with some (d, γ).*

(b) *If ψ has the Gaussian characteristic function with (d, γ) and $M \in \mathrm{Sp}(2N, \mathbb{R})$ satisfies $\gamma = MM^T$, then $\psi = U|0\rangle$ for some Gaussian unitary U with data (M, d).*

Proof. (a) If $\psi = U|0\rangle$, Theorem 12.9 applies. Conversely, by Theorem 12.13 γ is symplectic, and by Lemma 12.7(c) and Theorem 3.1 it is positive definite; Lemma 12.12(b) gives $M \in \mathrm{Sp}(2N, \mathbb{R})$ with $MM^T = \gamma$, and (b) applies.

(b) By Lemma 12.12(a) there is a Gaussian unitary U' with data (M, d), and by Theorem 12.9 $\chi_{U'|0\rangle} = \chi^{\mathrm{G}}_{d,MM^T} = \chi_\psi$. Theorem 12.4, with $U'|0\rangle \neq 0$, gives $\psi = \lambda U'|0\rangle$ with $|\lambda| = 1$, and $U = \lambda U'$ has data (M, d) by Proposition 11.4(b). □

Corollary 12.15 (Uniqueness)**.** Lean ✓ *If ψ and ψ' both have the Gaussian characteristic function with (d, γ), then $\psi' = \lambda\psi$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$: a pure Gaussian state is determined, up to a phase, by its displacement vector and covariance matrix.*

Proof. $\psi \neq 0$ by Lemma 12.7(a), and $\chi_{\psi'} = \chi_\psi$; Theorem 12.4. □

By Lemma 12.7(c) and Theorem 12.13, the covariance matrix of a pure Gaussian state is admissible and symplectic. Every such matrix occurs.

Corollary 12.16 (Every admissible symplectic γ occurs)**.** Lean ✓ *Let γ be admissible and symplectic, and $d \in \mathbb{R}^{2N}$. There is a Gaussian unitary U such that $U|0\rangle$ has the Gaussian characteristic function with (d, γ). Hence a vector with the Gaussian characteristic function with (d, γ) exists if and only if γ is admissible and symplectic.*

Proof. γ is positive definite by Theorem 3.1, so Lemma 12.12(b) gives $M \in \mathrm{Sp}(2N, \mathbb{R})$ with $\gamma = MM^T$, and Lemma 12.12(a) a Gaussian unitary U with data (M, d). By Theorem 12.9, $U|0\rangle$ has the Gaussian characteristic function with $(d, MM^T) = (d, \gamma)$. The converse is the sentence before the corollary. □

12.4 The parametrization of squeezed coherent states

Tjoa and Cirac parametrize single-mode squeezed coherent states by the unnormalized vectors $|\kappa, \ell\rangle \propto D(\alpha)\mathsf{S}(\zeta)|0\rangle$ with $\kappa = -\frac{1}{2}e^{i\phi}\tanh r$ and $\ell = \alpha - 2\kappa\bar{\alpha}$ [4, (8)]. The next lemma inverts this parametrization and computes the covariance matrix.

Lemma 12.17 (Inverting the paper's (8))**.** Lean ✓

(a) *For every $\tau_0 \in \mathbb{C}$ with $|\tau_0| < 1$ there is $\zeta \in \mathbb{C}$ with $\tau(\zeta) = \tau_0$. For every $\zeta \in \mathbb{C}$, $\cosh|\zeta| = (1 - |\tau(\zeta)|^2)^{-1/2}$.*

(b) *Let ι be a finite set and $\kappa, \ell, \zeta \in \mathbb{C}^\iota$ with $\tau(\zeta_j) = -2\kappa_j$ for every j, and put $\alpha_j = (\ell_j + 2\kappa_j\bar{\ell_j})/(1 - 4|\kappa_j|^2)$. Then*

$$\bigotimes_j |\kappa_j, \ell_j\rangle = \prod_j \Big((1 - 4|\kappa_j|^2)^{-1/4} e^{|\alpha_j|^2/2 - \kappa_j\bar{\alpha}_j^2}\Big)_J\, D(\alpha)\mathsf{S}(\zeta)|0\rangle.$$

(c) *For N modes and $\tau(\zeta_j) = -2\kappa_j$ for every j, $M_\zeta^2 = \gamma_\kappa$, where γ_κ has no entries between different modes and its 2×2 matrix on (q_j, p_j) is*

$$\frac{1}{1-4|\kappa_j|^2}\begin{pmatrix} 1+4|\kappa_j|^2+4\operatorname{Re}\kappa_j & 4\operatorname{Im}\kappa_j \\ 4\operatorname{Im}\kappa_j & 1+4|\kappa_j|^2-4\operatorname{Re}\kappa_j \end{pmatrix}.$$

The condition $\tau(\zeta_j) = -2\kappa_j$ forces $|\kappa_j| < \frac{1}{2}$, and by (a) it can be met exactly when $|\kappa_j| < \frac{1}{2}$; for $|\kappa_j| \geq \frac{1}{2}$ the vector $|\kappa_j, \ell_j\rangle$ is not square-summable (Lemma 10.8(b) and its proof).

Proof. (a) For $\tau_0 \neq 0$ take $\zeta = \operatorname{artanh}(|\tau_0|)\,\tau_0/|\tau_0|$, so that $\tau(\zeta) = (\tau_0/|\tau_0|)\tanh\operatorname{artanh}|\tau_0| = \tau_0$; for $\tau_0 = 0$ take $\zeta = 0$. And $|\tau(\zeta)| = \tanh|\zeta|$, $1-\tanh^2 = \cosh^{-2}$.

(b) With $D_j = 1-4|\kappa_j|^2 > 0$, $\alpha_j - 2\kappa_j\bar{\alpha}_j = (\ell_j + 2\kappa_j\bar{\ell}_j - 2\kappa_j\bar{\ell}_j - 4|\kappa_j|^2\ell_j)/D_j = \ell_j$. Corollary 11.7(a), which holds on any finite set of modes, with $\tau_j = -2\kappa_j$ reads $D(\alpha)\mathsf{S}(\zeta)|0\rangle = \prod_j c_j^{-1/2} e^{-|\alpha_j|^2/2+\kappa_j\bar{\alpha}_j^2} \bigotimes_j |\kappa_j, \ell_j\rangle$, and $c_j^{-1/2} = (1-|\tau_j|^2)^{1/4} = D_j^{1/4}$ by (a). Dividing by the nonzero scalar gives the identity.

(c) Per mode, $M_\zeta = c\,1 - s\,R$ with $R = \left(\begin{smallmatrix}\cos\phi & \sin\phi \\ \sin\phi & -\cos\phi\end{smallmatrix}\right)$, $R^2 = 1$, so $M_\zeta^2 = (c^2+s^2)1 - 2cs\,R$. Entry by entry, $c^2+s^2 = c^2(1+\tanh^2 r)$ and $2cs = 2c^2\tanh r$ with $c^2 = (1-\tanh^2 r)^{-1}$ and $\tanh r\, e^{i\phi} = \tau = -2\kappa$: so $c^2+s^2 = (1+4|\kappa|^2)/D$, $2cs\cos\phi = -4\operatorname{Re}\kappa/D$ and $2cs\sin\phi = -4\operatorname{Im}\kappa/D$. For $\kappa = 0$, $\zeta = 0$ and both sides are 1. □

Corollary 12.18 (The norm of $|\kappa, \ell\rangle$; [4, (9)])**.** *For $\kappa, \ell \in \mathbb{C}$ with $|\kappa| < \frac{1}{2}$ and $\alpha = (\ell + 2\kappa\bar{\ell})/(1-4|\kappa|^2)$,*

$$\||\kappa, \ell\rangle\|^2 = \frac{e^{|\alpha|^2 - 2\operatorname{Re}(\kappa\bar{\alpha}^2)}}{\sqrt{1-4|\kappa|^2}} = \frac{1}{\sqrt{1-4|\kappa|^2}} \exp\frac{|\ell|^2 + \kappa\bar{\ell}^2 + \bar{\kappa}\ell^2}{1-4|\kappa|^2}.$$

Proof. Choose ζ with $\tau(\zeta) = -2\kappa$ (Lemma 12.17(a)). By Lemma 12.17(b) for one mode and the unitarity of $D(\alpha)\mathsf{S}(\zeta)$, the squared norm is $|(1-4|\kappa|^2)^{-1/4} e^{|\alpha|^2/2 - \kappa\bar{\alpha}^2}|^2$, the first expression. With $D = 1-4|\kappa|^2$, expanding $|\ell+2\kappa\bar{\ell}|^2$ and $\kappa(\bar{\ell}+2\bar{\kappa}\ell)^2$ gives $D^2(|\alpha|^2 - 2\operatorname{Re}(\kappa\bar{\alpha}^2)) = D(|\ell|^2 + \kappa\bar{\ell}^2 + \bar{\kappa}\ell^2)$, the second. It is also Lemma 10.19 at $(\kappa', \ell') = (\kappa, \ell)$, with $\mathcal{S}(|\kappa|^2) = (1-4|\kappa|^2)^{-1/2}$. □

For one mode, Theorem 11.6(b) gives $D(\alpha)\mathsf{S}(\zeta)$ the data $(M_\zeta, \delta(\alpha))$, so by Theorem 12.9 and Lemma 12.17 the normalized vector $|\kappa, \ell\rangle/\||\kappa, \ell\rangle\|$ has displacement $\delta(\alpha)$ and covariance γ_κ. At $\kappa = 0$ it is the coherent state with $\gamma = 1$, and at $\ell = 0$ the squeezed vacuum with $d = 0$; for real $\kappa > 0$, $\gamma_{qq} = (1+2\kappa)/(1-2\kappa)$ exceeds the vacuum value 1 and $\gamma_{pp} = (1-2\kappa)/(1+2\kappa)$ lies below it.

12.5 The vacuum component

A photon-number measurement on a pure Gaussian state ψ finds no photon with the probability $|\langle 0, \psi\rangle|^2$, the squared modulus of its *vacuum component* $\langle 0, \psi\rangle$, and this probability is never zero. By Theorem 11.6 it is a passive unitary applied to a product of displaced squeezed vacua; the passive unitary leaves the vacuum component unchanged, and each one-mode factor is a nonzero multiple of a vector $|\kappa, \ell\rangle$ with vacuum coefficient 1 (Corollary 11.7(a)). So no nonzero vector orthogonal to the vacuum, in particular no number state other than the vacuum, is a multiple of a pure Gaussian state.

Theorem 12.19 (The vacuum component of a pure Gaussian state)**.** Lean ✓ *Every pure Gaussian state $\psi \in \Gamma_s(N)$ satisfies $\langle 0, \psi\rangle = \psi(0) \neq 0$.*

Proof. The inner product with $|0\rangle$ picks the coefficient at the multi-index 0, so $\langle 0, \varphi\rangle = \varphi(0)$ for every $\varphi \in \Gamma_s(N)$. For $\mathcal{U} \in \mathrm{U}(N)$ the passive unitary preserves this coefficient: $\mathsf{U}_\mathcal{U}^* = \mathsf{U}_{\mathcal{U}^*}$ by Theorem 8.7(b), and $\Gamma(\mathcal{U}^*)|0\rangle = |0\rangle$ by Lemma 8.3(b), so $\langle 0, \mathsf{U}_\mathcal{U}\varphi\rangle = \langle \mathsf{U}_{\mathcal{U}^*}|0\rangle, \varphi\rangle = \langle 0, \varphi\rangle$.

Let $\psi = U|0\rangle$ with U a Gaussian unitary. By Corollary 11.7(a), $U|0\rangle = \lambda\, \mathsf{U}_{\mathcal{U}} D(\alpha)\mathsf{S}(r)|0\rangle$ for some λ with $|\lambda| = 1$, $\mathcal{U} \in \mathrm{U}(N)$, $\alpha \in \mathbb{C}^N$ and $r \in \mathbb{R}^N$ with $r \geq 0$, and

$$D(\alpha)\mathsf{S}(r)|0\rangle = \Big(\prod_j c_j\Big)^{-1/2} e^{-|\alpha|^2/2 - \sum_j \tau_j \bar{\alpha}_j^2/2} \bigotimes_j |-\tau_j/2,\, \alpha_j + \tau_j \bar{\alpha}_j\rangle,$$

with $c_j = \cosh r_j \geq 1$ and $\tau_j = \tanh r_j$. The coefficient of a product vector at 0 is the product of the coefficients of its factors at 0 (Definition 7.4), and each factor $e^{\kappa a^{\dagger 2} + \ell a^\dagger}|0\rangle$ has the coefficient 1 at 0, because its exponent raises the particle number, so that the finite sum of Definition 6.6 at the multi-index 0 has the single term $|0\rangle(0) = 1$. With the first paragraph,

$$\langle 0, \psi\rangle = \lambda \Big(\prod_j c_j\Big)^{-1/2} e^{-|\alpha|^2/2 - \sum_j \tau_j \bar{\alpha}_j^2/2} \neq 0. \qquad \square$$

Corollary 12.20 (Vectors orthogonal to the vacuum)**.** Lean ✓ *Let $\varphi \in \Gamma_s(N)$ be nonzero with $\langle 0, \varphi\rangle = 0$. Then $\varphi \neq c\,\psi$ for every pure Gaussian state $\psi \in \Gamma_s(N)$ and every $c \in \mathbb{C}$.*

Proof. If $\varphi = c\,\psi$, then $0 = \langle 0, \varphi\rangle = c\,\langle 0, \psi\rangle$, so $c = 0$ by Theorem 12.19, and $\varphi = 0$. $\square$

Corollary 12.21 (Number states are not Gaussian)**.** Lean ✓ *For every multi-index $m \neq 0$, the number state $|m\rangle \in \Gamma_s(N)$ is not a multiple of a pure Gaussian state: $|m\rangle \neq c\,\psi$ for every pure Gaussian state ψ and every $c \in \mathbb{C}$.*

Proof. $|m\rangle \neq 0$, and $\langle 0, |m\rangle\rangle$ is the coefficient of $|m\rangle$ at $0 \neq m$, which is 0; apply Corollary 12.20. $\square$

Notes and references

Tjoa and Cirac [4, Sec. II A] define pure Gaussian states by their (4), give the parametrization (8) with its norm (9), the multimode form (11) and the moments (13), and state that by Wick's theorem a Gaussian state is completely characterized by its displacement vector and covariance matrix. For pure states this characterization is Corollary 12.15; Wick's theorem for higher moments is not treated here. The paper gives the norm (9) but not the covariance matrix of $|\kappa, \ell\rangle$, which is γ_κ of Lemma 12.17(c).

Theorem 12.14 is, for finitely many modes, the classical result of Manuceau and Verbeure [35] that the pure quasi-free states of the algebra of the canonical commutation relations are the images of the Fock state under Bogoliubov transformations; see also Holevo [36, Ch. V] and, for pure Gaussian states and their Wigner functions, Arvind, Dutta, Mukunda and Simon [12, Sec. XII]. The standard proofs that the covariance matrix of a pure Gaussian state is symplectic [16, 17] use Williamson's normal form or the Moyal identity; Theorem 12.13 uses neither, only the explicit symplectic matrix of Lemma 12.8 and irreducibility. Lemma 12.8 uses the device of the quasi-free literature [35], [37], the polar part $|A|$ of $A = \gamma^{-1/2}\Omega\gamma^{-1/2}$. We have not checked whether the formula $\gamma_0 = \gamma^{1/2}|A|\gamma^{1/2}$ appears in the literature. Once the Fock representation of the algebra of the canonical commutation relations is constructed, every vector of Definition 12.5 defines a Gaussian state of that algebra with $\mathrm{disp}(u) = -\sigma(u, d)$ and $\mathrm{cov}(u, v) = u^T\Omega\gamma\Omega^T v$; this construction is not part of these notes. The derivation of the state condition from the positivity of the twisted kernels, Lemma 12.6, is modelled on the characterization of quasi-free states in [35] and [36, Ch. V]. Lemma 12.3(c) and Theorem 12.4 are modelled on the uniqueness of the cyclic representation of a state up to unitary equivalence [38, Thm. 2.3.16], combined with irreducibility. The formal proof of Lemma 12.8 obtains $A^T A \preceq 1$ from the pair form of the state condition and $|A| \preceq 1$ by quadratic forms, as above; that of Lemma 12.17(c) compares the entries directly, through $\cosh^2 r\,(1 - \tanh^2 r) = 1$ and $\sinh r = \tanh r \cosh r$.

For one mode, the vacuum coefficient of a squeezed coherent state is the first term of its number-state expansion, which is classical [30]. Theorem 12.19 reaches N modes through the factorization of Theorem 11.6; this argument is ours. Corollary 12.21 is the remark after (10) in [4] that the number states are Gaussian only for $n = 0$.

13 Quadratic Hamiltonians

Gaussian unitaries are defined by their action on the Weyl operators (Definition 11.3); they are often written as e^{-iH} with H quadratic in the quadratures [4, (1)]. For $H = \frac{1}{2}\hat{R}^T K \hat{R} + f^T \hat{R}$ with K real symmetric and f real, H is essentially self-adjoint on the finitely supported vectors, its closure generates exactly one unitary group e^{-itH}, and e^{-itH} is a Gaussian unitary whose data are the classical flow of the Hamilton function $\frac{1}{2}x^T K x + f^T x$. The construction does not use the spectral theorem: the group is built from the exponential series, which on finitely supported vectors converges with a radius independent of the vector.

Notation for this section.

E; A, $\operatorname{dom} A$, A^*, $\overline{A}$	a complex Hilbert space; an operator on E, its domain, adjoint and closure
U, U_t	a one-parameter unitary group
$\mathcal{D}$	the finitely supported vectors of $\Gamma_s(\iota)$, $\iota = \{1, \dots, N\}$, $N \geq 1$ modes
K, f, H, C_H	a real symmetric $2N \times 2N$ matrix, a vector of $\mathbb{R}^{2N}$, $\frac{1}{2}\hat{R}^T K \hat{R} + f^T \hat{R}$, Definition 13.4
$M(t)$, $d(t)$	$e^{t\Omega K}$ and $\int_0^t e^{s\Omega K}\Omega f \, ds$ (not to be confused with the matrix M_ζ of $\mathsf{S}(\zeta)$)
Φ_v	the field $\sigma(v, \hat{R})$; $W(v) = e^{-i\bar{\Phi}_v}$ (proof of Theorem 13.7)

An *operator* is a linear map defined on a subspace; $\bar{T}$ with a short bar is still the maximal realization of a map T of $\mathcal{F}$ (Definition 7.6), and $\overline{A}$ with a long bar is a closure. Inner products are linear in the second argument.

13.1 One-parameter unitary groups

Definition 13.1 (One-parameter unitary group; generator)**.** A *strongly continuous one-parameter unitary group* on E is a map $t \mapsto U_t$ from $\mathbb{R}$ to the unitary operators with $U_0 = 1$, $U_{s+t} = U_s U_t$ and $t \mapsto U_t\psi$ continuous for every ψ. An operator A is its *generator*, and U is *generated* by A, written $U_t = e^{-itA}$, if $\operatorname{dom} A$ consists of the ψ at which $t \mapsto U_t\psi$ is differentiable at 0, and $\frac{d}{dt}U_t\psi|_{t=0} = -iA\psi$ there.

These ψ form a subspace, on which $\psi \mapsto i\frac{d}{dt}U_t\psi|_{t=0}$ is linear; so every such group has exactly one generator. Reed and Simon write $U(t) = e^{itA}$ [9, Sec. VIII.4]; we keep the physicist's sign. The group e^{-itA} of the functional calculus of a self-adjoint A has generator A in this sense [9], so by Theorem 13.2(c) below the two meanings of e^{-itA} agree. The zero operator generates only $U_t = 1$.

Theorem 13.2 (Generators)**.** Lean ✓ *Let A generate U.*

(a) *For $\psi \in \operatorname{dom} A$ and every t: $U_t\psi \in \operatorname{dom} A$, $AU_t\psi = U_t A\psi$, and $\frac{d}{dt}U_t\psi = -iU_t A\psi$.*

(b) *$\operatorname{dom} A$ is dense, and A is self-adjoint.*

(c) *If A also generates V, then $U = V$.*

(d) *Let B be self-adjoint and $E_0 \subseteq \operatorname{dom} B$ a subspace with $\overline{B \restriction E_0} = B$ (E_0 is a* core *for B). If V is a strongly continuous one-parameter unitary group with $\frac{d}{dt}V_t\psi|_{t=0} = -iB\psi$ for every $\psi \in E_0$, then B generates V.*

(e) *For a unitary Z, the operator Z^*AZ on $\{\psi : Z\psi \in \operatorname{dom} A\}$ generates $t \mapsto Z^*U_tZ$; for real c, $A + c$ generates $t \mapsto e^{-itc}U_t$.*

(f) *A bounded self-adjoint T generates $t \mapsto \exp(-itT)$.*

Proof. (a) The quotient $(U_\tau U_t\psi - U_t\psi)/\tau = U_t(U_\tau\psi - \psi)/\tau$ tends to $-iU_tA\psi$, and it is also the difference quotient of $s \mapsto U_s\psi$ at t.

(b) For $\varepsilon > 0$ put $\psi_\varepsilon = \varepsilon^{-1}\int_0^\varepsilon U_s\psi\,ds$. Then $(U_\tau\psi_\varepsilon - \psi_\varepsilon)/\tau = \varepsilon^{-1}\tau^{-1}(\int_\varepsilon^{\varepsilon+\tau} - \int_0^\tau)U_s\psi\,ds$ tends to $\varepsilon^{-1}(U_\varepsilon\psi - \psi)$, so $\psi_\varepsilon \in \operatorname{dom} A$, and $\psi_\varepsilon \to \psi$: the domain is dense. Differentiating $\langle U_\tau\psi, U_\tau\varphi\rangle = \langle\psi,\varphi\rangle$ at $\tau = 0$ shows that A is symmetric. If $\psi_n \to \psi$ and $A\psi_n \to \xi$, then $U_\tau\psi_n - \psi_n = -i\int_0^\tau U_sA\psi_n\,ds$ by (a), and in the limit $U_\tau\psi - \psi = -i\int_0^\tau U_s\xi\,ds$, whose quotient by τ tends to $-i\xi$; so A is closed. Finally let $A^*\varphi = \pm i\varphi$. For $\psi \in \operatorname{dom} A$, $g(t) = \langle\varphi, U_t\psi\rangle$ satisfies $g' = \mp g$, so $g(t) = e^{\mp t}g(0)$; as g is bounded, $g(0) = 0$, and $\varphi = 0$ because $\operatorname{dom} A$ is dense. A closed symmetric operator with $\ker(A^* \mp i) = 0$ has $\operatorname{ran}(A \pm i) = E$ and is self-adjoint [9].

(c) For $\psi, \varphi \in \operatorname{dom} A$ the function $s \mapsto \langle V_s\varphi, U_s\psi\rangle$ has derivative $i\langle AV_s\varphi, U_s\psi\rangle - i\langle V_s\varphi, AU_s\psi\rangle = 0$ by (a) and symmetry, so $\langle V_t\varphi, U_t\psi\rangle = \langle\varphi,\psi\rangle = \langle V_t\varphi, V_t\psi\rangle$. Since V_t maps $\operatorname{dom} A$ onto itself, $U_t\psi = V_t\psi$ on the dense $\operatorname{dom} A$, and by continuity on E.

(d) The generator C of V extends $B \restriction E_0$ and is closed by (b), so $C \supseteq B$. As C is symmetric, $C \subseteq C^* \subseteq B^* = B$.

(e) The quotient for $Z^*U_\tau Z\psi$ converges exactly when that for $U_\tau Z\psi$ does; the shift is the product rule.

(f) The norm-continuous group $\exp(-itT)$ has derivative $-iT\psi$ at 0 for every ψ; so its generator extends the everywhere defined T and equals it. □

Theorem 13.3 (Analytic vectors with a common radius)**.** Lean ✓ *Let $E_0 \subseteq E$ be a dense subspace, $A : E_0 \to E_0$ linear and symmetric, and $\rho > 0$ with $\sum_n \rho^n\|A^nx\|/n! < \infty$ for every $x \in E_0$. Then $\overline{A}$ is self-adjoint, it generates exactly one strongly continuous one-parameter unitary group U, and $U_tx = \sum_n (-it)^nA^nx/n!$ for $|t| < \rho$ and $x \in E_0$.*

Proof. For $|t| \le \rho$ the series $V_tx = \sum_n(-it)^nA^nx/n!$ converges absolutely on E_0. Expanding $\langle V_tx, V_ty\rangle$ as a double series and using $\langle A^mx, A^ny\rangle = \langle x, A^{m+n}y\rangle$, the terms with $m + n = k$ sum to $\langle x, A^ky\rangle(it - it)^k/k!$; so V_t preserves inner products and extends to an isometry of E. For $|s| + |t| \le \rho$ the double series of V_sV_tx converges absolutely, and summing along $m + n = k$ with the binomial theorem gives $V_sV_t = V_{s+t}$; in particular V_t is unitary for $|t| \le \rho/2$. The group is $U_t = V_{t/m}^m$ for any $m \ge 1$ with $|t|/m \le \rho/2$, which does not depend on m (by the local law both $V_{t/m}^m$ and $V_{t/m'}^{m'}$ equal $V_{t/(mm')}^{mm'}$); the local law gives $U_{s+t} = U_sU_t$ and $U_t = V_t$ for $|t| < \rho$. On E_0, $t \mapsto V_tx$ is a convergent power series with derivative $-iAx$ at 0; uniform boundedness and density give strong continuity.

$\overline{A}$ is closed and symmetric. If $\langle\varphi, (A - i)\xi\rangle = 0$ for all $\xi \in E_0$, then, since A maps E_0 into itself, $\langle\varphi, A^nx\rangle = i^n\langle\varphi, x\rangle$, so $\langle\varphi, U_tx\rangle = e^t\langle\varphi, x\rangle$ for $|t| < \rho$, then for every $x \in E$, and by the group law for every t. Boundedness forces $\varphi = 0$; likewise for $A + i$. So $\operatorname{ran}(\overline{A} \pm i)$ is dense; it is also closed, because $\|(\overline{A} \pm i)x\|^2 = \|\overline{A}x\|^2 + \|x\|^2$ and $\overline{A}$ is closed. Hence $\operatorname{ran}(\overline{A} \pm i) = E$ and $\overline{A}$ is self-adjoint [9]. Theorem 13.2(d), with core E_0, shows that $\overline{A}$ generates U, and (c) gives uniqueness. □

The hypothesis is stronger than that of Nelson's analytic vector theorem [13, Thm. X.39], which allows the radius to depend on the vector and needs no invariant domain. Quadratic Hamiltonians satisfy it (Lemma 13.5); polynomials of higher degree in the quadratures in general do not: for one mode and $A = \hat{q}^3$ on $\mathcal{D}$, $\|A^m|0\rangle\|^2 = \|\hat{q}^{3m}|0\rangle\|^2 = (6m-1)!!/8^m$ grows like $(m!)^3$ up to exponential factors, so $\sum_m \rho^m\|A^m|0\rangle\|/m!$ diverges for every $\rho > 0$.

13.2 The quadratic Hamiltonian

Definition 13.4 (Quadratic Hamiltonian)**.** For a real symmetric $2N\times 2N$ matrix K and $f\in\mathbb{R}^{2N}$, the *quadratic Hamiltonian* is the linear map of $\mathcal{F}$

$$H=\sum_{k,l}\tfrac{1}{2}K_{kl}\hat{R}_k\hat{R}_l+\sum_k f_k\hat{R}_k,$$

and $\bar{H}$ is its maximal realization. We write

$$C_H=\sum_{k,l}|K_{kl}|+\sqrt{2}\sum_k|f_k|,\qquad M(t)=e^{t\Omega K},\qquad d(t)=\int_0^t M(s)\Omega f\,ds.$$

The pair $(M(t),d(t))$ is the flow of the classical equations of motion $\dot{x}=\Omega Kx+\Omega f$ of $\frac{1}{2}x^TKx+f^Tx$: $x(t)=M(t)x(0)+d(t)$. Hence $M(s+t)=M(s)M(t)$ and $d(s+t)=d(s)+M(s)d(t)$, and $M(t)\in\mathrm{Sp}(2N,\mathbb{R})$ because ΩK is a Hamiltonian matrix, $\Omega^{-1}(\Omega K)^T\Omega=-\Omega K$. For one mode, $K=1$ and $f=0$, $H=a^\dagger a+\frac{1}{2}$ and $M(t)=O_{e^{-it}}$, the symplectic matrix of $\mathsf{U}_{e^{-it}}$ in Theorem 11.5(b); for $K=0$, $d(t)=t\Omega f$.

Lemma 13.5 (Bounds, symmetry and commutators)**.** Lean ✓

(a) *If $\psi\in\Gamma_s(\iota)$ vanishes above ν particles, then $H\jmath\psi=\jmath\varphi$ for a $\varphi\in\Gamma_s(\iota)$ vanishing above $\nu+2$, with $\|\varphi\|\le C_H(\nu+2)\|\psi\|$. H is symmetric on $\mathcal{D}$, and for $\rho\ge 0$ with $2C_H\rho<1$ and $\psi\in\mathcal{D}$, $\sum_m\rho^m\|H^m\psi\|/m!<\infty$.*

(b) *On $\mathcal{F}$: $[\hat{R}_k,\hat{R}_l]=i\Omega_{kl}$, $[\hat{R}_k,H]=i(\sum_m(\Omega K)_{km}\hat{R}_m+(\Omega f)_k)$, and for the field $\Phi_v=\sigma(v,\hat{R})=\sum_k(\Omega^Tv)_k\hat{R}_k$, $[\Phi_v,H]=-i(\Phi_{\Omega Kv}+f^Tv)$.*

Proof. (a) On vectors vanishing above μ, $\|a_j\omega\|^2\le\mu\|\omega\|^2$ and $\|a_j^\dagger\omega\|^2\le(\mu+1)\|\omega\|^2$ by $\|a_j\omega\|^2=\sum_m m_j|\omega(m)|^2$ and $\|a_j^\dagger\omega\|^2=\|a_j\omega\|^2+\|\omega\|^2$; so $\|\hat{R}_k\omega\|\le\sqrt{2}\sqrt{\mu+1}\|\omega\|$ and $\|\hat{R}_k\hat{R}_l\omega\|\le 2(\mu+2)\|\omega\|$, which gives the bound on H. Iterating, for $\psi\in\mathcal{D}$ vanishing above ν,

$$\|H^m\psi\|\le C_H^m\prod_{j<m}(\nu+2j+2)\,\|\psi\|\le(2C_H)^m\,m!\binom{m+\nu}{\nu}\|\psi\|,$$

and $\sum_m\binom{m+\nu}{\nu}x^m$ converges for $x<1$. H is a polynomial in the a_j and $a_j^\dagger$, hence coefficient-finite (Lemma 7.5(a)). Symmetry follows from the identity

$$\textstyle\sum_n\overline{(Hx)(n)}\,(\jmath\varphi)(n)=\sum_n\overline{x(n)}\,(H\jmath\varphi)(n)\qquad(x\in\mathcal{F},\ \varphi\in\mathcal{D}),\tag{13.1}$$

both sums being finite. It holds for a_j in place of H with $a_j^\dagger$ on the right, and for $a_j^\dagger$ with a_j, since $(a_jx)(n)=\sqrt{n_j+1}\,x(n+e_j)$ and the sums run over the finitely many n at which $\jmath\varphi$, $a_j^\dagger\jmath\varphi$ or $a_j\jmath\varphi$ is nonzero. It follows for each $\hat{R}_k$, and for H because K is symmetric and K, f are real.

(b) The first identity follows from $[a_j,a_k^\dagger]=\delta_{jk}$. With $K=K^T$, $[\hat{R}_k,\hat{R}_l\hat{R}_m]=i\Omega_{kl}\hat{R}_m+i\Omega_{km}\hat{R}_l$ gives the second. For the third, $[\Phi_v,\hat{R}_l]=-iv_l$ since $\Omega^T\Omega^T=-1$, so $[\Phi_v,H]=-i(v^TK\hat{R}+f^Tv)$, and $v^TK\hat{R}=\Phi_{\Omega Kv}$. □

Theorem 13.6 (Essential self-adjointness and the unitary group)**.** Lean ✓ *Let H be a quadratic Hamiltonian.*

(a) *$\bar{H}$ is self-adjoint and is the closure of $H\restriction\mathcal{D}$; moreover $(H\restriction\mathcal{D})^*=\bar{H}$.*

(b) *$\bar{H}$ generates exactly one strongly continuous one-parameter unitary group, $t\mapsto e^{-it\bar{H}}$.*

(c) *For $\psi \in \mathcal{D}$ and $2C_H|t| < 1$, $e^{-it\bar{H}}\psi = \sum_m (-it)^m H^m \psi/m!$.*

Proof. Theorem 13.3 applies to $H \restriction \mathcal{D}$ with every $\rho > 0$ such that $2C_H\rho < 1$, by Lemma 13.5(a); $\mathcal{D}$ is dense (Lemma 7.2(c)). So $\overline{H \restriction \mathcal{D}}$ is self-adjoint. It is contained in $\bar{H}$, which is closed (Theorem 7.7, since H is coefficient-finite). Conversely $\bar{H} \subseteq (H \restriction \mathcal{D})^*$, because for $\psi \in \operatorname{dom}\bar{H}$ and $\varphi \in \mathcal{D}$ the identity (13.1) with $x = \jmath\psi$ gives $\langle \bar{H}\psi, \varphi\rangle = \langle \psi, H\varphi\rangle$; and $(H \restriction \mathcal{D})^*$ is contained in the adjoint of the closure, which is the closure itself. The three operators therefore coincide. □

13.3 The affine symplectic map

Theorem 13.7 (e^{-itH} is a Gaussian unitary)**.** Lean ✓ *Let H be a quadratic Hamiltonian. For every real t, $e^{-it\bar{H}}$ is a Gaussian unitary with data $(M(t), d(t))$:*

$$e^{it\bar{H}}\, W(u)\, e^{-it\bar{H}} = e^{-i\sigma(u,d(t))}\, W(M(t)^{-1}u) \qquad (u \in \mathbb{R}^{2N}).$$

In particular $e^{-i\bar{H}}$ has data $(e^{\Omega K}, d(1))$, and by Corollary 11.7(c) $e^{i\bar{H}}\hat{R}e^{-i\bar{H}} = e^{\Omega K}\hat{R} + d(1)$ on $\bigcap_j \operatorname{dom}\bar{a}_j$, the paper's (3).

Proof. Write $U_t = e^{-it\bar{H}}$. The field Φ_v is H for $K = 0$ and $f = \Omega^T v$, and $D(\alpha(v)) = e^{-i\Phi_v}$ formally. By Theorem 10.15 the map $s \mapsto W(sv)$ is a strongly continuous unitary group whose derivative at 0 on $\mathcal{D}$ is $-i\Phi_v$, so by Theorems 13.6 and 13.2(d) it is generated by $\bar{\Phi}_v$: $W(sv) = e^{-is\bar{\Phi}_v}$.

Let $2C_H|t| < 1$, $\psi \in \mathcal{D}$ and $u \in \mathbb{R}^{2N}$. We show $\bar{\Phi}_u U_t\psi = U_t\xi$ with $\jmath\xi = (\Phi_{M(t)^{-1}u} + \sigma(u, d(t)))\jmath\psi$. Write $\operatorname{ad} X = XH - HX$. Then $\Phi_u H^m = \sum_k \binom{m}{k} H^{m-k} \operatorname{ad}^k \Phi_u$, and by Lemma 13.5(b) $\operatorname{ad}^k \Phi_u = (-i)^k(\Phi_{(\Omega K)^k u} + \lambda_k)$ with $\lambda_0 = 0$ and $\lambda_k = f^T(\Omega K)^{k-1}u$. Substituting into the series of Theorem 13.6(c) and summing first over k uses

$$\sum_k \frac{(-t)^k}{k!}(\Omega K)^k u = M(t)^{-1}u, \qquad \sum_{k\geq 1} \frac{(-t)^k}{k!}\lambda_k = \sigma(u, d(t)).$$

The double series reduces to finitely many products of absolutely convergent series, since Φ_w is a finite combination of the $\hat{R}_k$, and the series of $\bar{H}$ on the finitely supported vectors $\hat{R}_k\psi$ converge by Lemma 13.5(a). Comparing coefficients, each of which is a continuous function on $\Gamma_s(\iota)$ by Lemma 7.5(c), gives the identity.

Now $s \mapsto e^{is\sigma(u,d(t))}U_t^* W(su)U_t$ is a strongly continuous unitary group; by Theorem 13.2(e) and the identity just proved, its derivative at 0 on $\mathcal{D}$ is $-i\Phi_{M(t)^{-1}u}$. By Theorem 13.2(d) and (c) it equals $s \mapsto W(sM(t)^{-1}u)$; at $s = 1$ this is the claim for small t. For arbitrary t, $U_t = U_{t/m}^m$ with $2C_H|t|/m < 1$, and Proposition 11.4(b) with the flow laws gives the data $(M(t), d(t))$. □

Corollary 13.8 (The paper's (1) and (5))**.** Lean ✓

(a) *Let h and Δ be complex $N \times N$ matrices, h Hermitian and Δ symmetric, let $\eta \in \mathbb{C}^N$, and let*

$$H_G = \sum_{ij} h_{ij}a_i^\dagger a_j + \tfrac{1}{2}\sum_{ij}(\Delta_{ij}a_i^\dagger a_j^\dagger + \bar{\Delta}_{ij}a_i a_j) + \sum_j (\eta_j a_j^\dagger + \bar{\eta}_j a_j),$$

the paper's (1). Then $H_G = H - \frac{1}{2}\operatorname{tr} h$ on $\mathcal{F}$ for $f = \delta(\eta)$ and the symmetric matrix

$$K = \begin{pmatrix} \operatorname{Re} h + \operatorname{Re}\Delta & \operatorname{Im}\Delta - \operatorname{Im} h \\ \operatorname{Im}\Delta + \operatorname{Im} h & \operatorname{Re} h - \operatorname{Re}\Delta \end{pmatrix}.$$

So $\bar{H}_G$ is self-adjoint and generates exactly one unitary group, and $e^{ic}e^{-i\bar{H}_G}$ is a Gaussian unitary with data $(e^{\Omega K}, d(1))$ for every real c.

(b) *For $\alpha, \zeta \in \mathbb{C}^N$ let A_α and B_ζ be the maximal realizations of $i\sum_j(\alpha_j a_j^\dagger - \bar{\alpha}_j a_j)$ and $\frac{i}{2}\sum_j(\bar{\zeta}_j a_j^2 - \zeta_j a_j^{\dagger 2})$. Then A_α generates $s \mapsto D(s\alpha)$ and B_ζ generates $s \mapsto \mathsf{S}(s\zeta)$: $D(\alpha) = e^{-iA_\alpha}$ and $\mathsf{S}(\zeta) = e^{-iB_\zeta}$, which the paper's (5) writes as $e^{\alpha\cdot a^\dagger - \bar{\alpha}\cdot a}$ and $e^{\frac{1}{2}\sum_j(\bar{\zeta}_j a_j^2 - \zeta_j a_j^{\dagger 2})}$.*

Proof. (a) Substitute $a_j = (\hat{q}_j + i\hat{p}_j)/\sqrt{2}$; in each mode pair the coefficients of $a_i^\dagger a_j^\dagger$, $a_i a_j$, $a_i^\dagger a_j$ and $a_i a_j^\dagger$ in $\frac{1}{2}\hat{R}^T K \hat{R}$ are $\frac{1}{2}\Delta_{ij}$, $\frac{1}{2}\bar{\Delta}_{ij}$, $\frac{1}{2}h_{ij}$ and $\frac{1}{2}\bar{h}_{ij}$, and normal ordering $a_i a_j^\dagger = a_j^\dagger a_i + \delta_{ij}$ with $\bar{h}_{ij} = h_{ji}$ produces $\frac{1}{2}\operatorname{tr} h$. The linear terms give $f = \delta(\eta)$. A real constant multiplies the group by a phase (Theorem 13.2(e)), and phases are Gaussian with data $(1, 0)$.

(b) Both operators are of the form (a) with $h = 0$; Theorems 10.15 and 10.24 supply the groups, their continuity and their derivatives on $\mathcal{D}$, and Theorem 13.2(d) applies. □

Notes and references

Tjoa and Cirac [4, (1), (3), (5)] write Gaussian unitaries as e^{-iH_G} up to a phase and state that they act on the quadratures by affine symplectic maps; Theorem 13.7 and Corollary 13.8 supply the domains and the data $(M(t), d(t))$. The standard rigorous routes use Nelson's theorem with Stone's theorem, or the metaplectic representation [16, Ch. 4]; Theorem 13.3 replaces both.

Index of terms

Each term is defined where it first appears; the number is that of the definition, or of the statement or section where the term is introduced. Terms marked * are introduced in these notes or used in a sense or normalization different from the usual one.

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Lean declarations

The linked Lean ✓ tag marks a result whose Lean proof has been checked and whose correspondence with the statement here has been reviewed. Only marked statements are listed, each with the declarations of BosonicLean that prove it.

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1.3 `Gaussian.isAdmissible_subsystem`
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`BosonicLean.Fock.FockSpace.secondQuantUnitary`
`BosonicLean.Fock.FockSpace.toFormal_secondQuantUnitary`
`BosonicLean.Fock.FockSpace.secondQuantUnitary_symm`
`BosonicLean.Fock.FockSpace.secondQuantUnitary_mul`
`BosonicLean.Fock.FockSpace.secondQuantRep`
`BosonicLean.Fock.FockSpace.secondQuantRep_apply`
`BosonicLean.Fock.FockSpace.star_secondQuantRep`
`BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_annihilate_mul`
`BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_create_mul`
`BosonicLean.Fock.FormalFock.secondQuant_mul_create_mul_conjTranspose`
`BosonicLean.Fock.FormalFock.secondQuant_mul_totalNumber_mul_conjTranspose`
`BosonicLean.Fock.FockSpace.continuous_secondQuantRep_apply`
BosonicLean/Fock/SecondQuantization.lean

8.9 BosonicLean.Fock.FormalFock.toBargmann_dSecondQuant
BosonicLean.Fock.FormalFock.toBargmann_dSecondQuant_single_one
BosonicLean.Fock.FormalFock.toBargmann_dSecondQuant_pow_single_one
BosonicLean.Fock.FormalFock.dSecondQuant_sector
BosonicLean.Fock.FormalFock.dSecondQuant_single_one
BosonicLean.Fock.FormalFock.dSecondQuant_smul
Derivation.iterate_apply_mul
BosonicLean.Fock.FormalFock.norm_coeff_toBargmann_dSecondQuant_pow_le
BosonicLean.Fock.FormalFock.norm_coeff_toBargmann_dSecondQuant_pow_le_sup'
BosonicLean.Fock.FormalFock.summable_norm_coeff_toBargmann_dSecondQuant_pow
BosonicLean.Fock.FormalFock.summable_norm_coeff_dSecondQuant_pow
BosonicLean.Fock.FormalFock.isLinearMap_coeffExp_dSecondQuant
BosonicLean.Fock.FormalFock.coeffExp_dSecondQuant_sector
BosonicLean/Fock/SecondQuantization.lean

8.10 BosonicLean.Fock.FormalFock.hasSum_coeff_toBargmann_dSecondQuant_pow
BosonicLean.Fock.FormalFock.hasSum_coeff_dSecondQuant_pow
BosonicLean.Fock.FormalFock.secondQuant_exp
BosonicLean.Fock.FormalFock.secondQuant_exp_I_smul
BosonicLean/Fock/SecondQuantization.lean

8.11 BosonicLean.exists_isHermitian_exp_eq
BosonicLean.exp_I_smul_mem_unitaryGroup
BosonicLean/LinearAlgebra/UnitaryExp.lean

8.13 BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_mul_mul
BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_add_mul
BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_smul_mul
BosonicLean.Fock.FormalFock.secondQuant_conjTranspose_mul_lift_mul
BosonicLean.Fock.FockSpace.mem_domain_maxRealization_conj
BosonicLean.Fock.FockSpace.domain_maxRealization_conj
BosonicLean.Fock.FockSpace.maxRealization_conj_apply
BosonicLean.Fock.FockSpace.inner_maxRealization_conj
BosonicLean.Fock.FockSpace.toFormal_star_mul_mul_secondQuantRep
BosonicLean.Fock.FockSpace.domain_maxRealization_conj_eq_top
BosonicLean.Fock.FockSpace.inner_secondQuantRep_apply
BosonicLean/Fock/SecondQuantization.lean

9.1 Gaussian.mem_symplecticGroup_iff
Gaussian.mem_symplecticGroup_iff'
Gaussian.symplecticForm_eq_fromBlocks
SymplecticGroup.symplectic_det
BosonicLean/PhaseSpace/BlochMessiah.lean
Mathlib/LinearAlgebra/SymplecticGroup.lean (Mathlib)

9.3 Gaussian.mem_symplecticGroup_iff_commute
Gaussian.commute_symplecticForm_iff
BosonicLean/PhaseSpace/BlochMessiah.lean

9.4 Gaussian.orthogonalSymplecticOfUnitary_mul
Gaussian.orthogonalSymplecticOfUnitary_one
Gaussian.orthogonalSymplecticOfUnitary_star
Gaussian.orthogonalSymplecticOfUnitary_injective
Gaussian.orthogonalSymplecticOfUnitary_neg
Gaussian.orthogonalSymplecticOfUnitary_eq_neg_symplecticForm
Gaussian.orthogonalSymplecticOfUnitary_mem
Gaussian.mem_orthogonalSymplectic_iff
Gaussian.mem_orthogonalSymplectic_iff_fromBlocks
Gaussian.coe_unitaryOfOrthogonalSymplectic
Gaussian.unitaryOfOrthogonalSymplectic_orthogonalSymplecticOfUnitary
Gaussian.orthogonalSymplecticOfUnitary_unitaryOfOrthogonalSymplectic
Gaussian.orthogonalSymplecticEquivUnitary

Gaussian.coe_orthogonalSymplecticEquivUnitary
Gaussian.coe_orthogonalSymplecticEquivUnitary_symm
Gaussian.orthogonalSymplecticEquivUnitary_apply
Gaussian.orthogonalSymplecticEquivUnitary_symm_apply
BosonicLean/PhaseSpace/BlochMessiah.lean

9.6 Gaussian.squeezingMatrix_mem
Gaussian.squeezingMatrix_add
Gaussian.squeezingMatrix_zero
Gaussian.squeezingMatrix_transpose
Gaussian.posDef_squeezingMatrix
Gaussian.squeezingMatrix_neg_mul
Gaussian.squeezingMatrix_mul_neg
BosonicLean/PhaseSpace/BlochMessiah.lean

9.7 Gaussian.posDef_mul_transpose
Gaussian.mul_transpose_mem
Gaussian.mul_symplecticForm_eq
BosonicLean/PhaseSpace/BlochMessiah.lean

9.8 Gaussian.ne_zero_of_hasEigenvalue
Gaussian.mulVec_symplecticForm_mulVec
Gaussian.map_eigenspace_symplecticForm
Gaussian.dotProduct_symplecticForm_eq_zero
Gaussian.isotropic_eigenspace
Gaussian.symplecticForm_mulVec_mem_eigenspace_one
BosonicLean/PhaseSpace/BlochMessiah.lean

9.9 Gaussian.orthonormal_sumElim_neg_symplecticForm
Gaussian.finrank_orthogonal_span_sumElim_neg_symplecticForm
Gaussian.mapsTo_orthogonal_span_sumElim_neg_symplecticForm
Gaussian.columns_sumElim_neg_symplecticForm
Gaussian.exists_unit_eigenvector_mem
BosonicLean/PhaseSpace/BlochMessiah.lean

9.10 Gaussian.exists_orthonormal_isotropic_eigenvectors
Gaussian.exists_orthogonalSymplectic_diagonalization_columns
Gaussian.exists_orthogonalSymplectic_diagonalization
BosonicLean/PhaseSpace/BlochMessiah.lean

9.11 Gaussian.exists_blochMessiah
Gaussian.blochMessiah_right_mem
BosonicLean/PhaseSpace/BlochMessiah.lean

9.12 Gaussian.charpoly_mul_transpose_of_eq
Gaussian.charpoly_transpose_mul_of_eq
Gaussian.singularValues_of_eq
Gaussian.roots_map_abs_log
Gaussian.map_univ_eq_of_blochMessiah
Gaussian.exists_perm_of_blochMessiah
BosonicLean/PhaseSpace/BlochMessiah.lean

9.13 Gaussian.neg_one_mem_orthogonalSymplectic
Gaussian.neg_mem_orthogonalSymplectic
Gaussian.neg_ne_self_of_mem_orthogonalGroup
Gaussian.eq_neg_mul_squeezingMatrix_mul_neg
BosonicLean/PhaseSpace/BlochMessiah.lean

9.14 Gaussian.polar_of_blochMessiah
Gaussian.cfcSqrt_mul_transpose_eq
Gaussian.polar_decomposition
Gaussian.polar_unique
BosonicLean/PhaseSpace/BlochMessiah.lean

10.2 `BosonicLean.hasSum_prod_apply_of_summable_norm`
`BosonicLean/Analysis/ProductSeries.lean`

10.3 `BosonicLean.Fock.FormalFock.coeff_expVecFormal`
`BosonicLean.Fock.FormalFock.expVecFormal_eq_exp`
`BosonicLean.Fock.FormalFock.raiseDisp_op`
`BosonicLean.Fock.FockSpace.toFormal_expVec`
`BosonicLean.Fock.FockSpace.expVec_apply`
`BosonicLean.Fock.FockSpace.toFormal_vacuum`
`BosonicLean.Fock.FormalFock.memâĎŞp_expVecFormal`
`BosonicLean.Fock.FormalFock.hasSum_conj_mul_expVecFormal`
`BosonicLean.Fock.FockSpace.inner_expVec`
`BosonicLean.Fock.FockSpace.norm_expVec_sq`
`BosonicLean.Fock.FockSpace.continuous_expVec`
`BosonicLean.Fock.FormalFock.annihilate_expVecFormal`
`BosonicLean.Fock.FormalFock.secondQuant_expVecFormal`
`BosonicLean.Fock.FormalFock.expVecFormal_eq_tensorVec`
`BosonicLean.Fock.FockSpace.expVec_zero`
`BosonicLean.Fock.FormalFock.expVecFormal_zero`
`BosonicLean/Fock/ExpVec.lean`

10.4 `BosonicLean.Fock.FockSpace.eq_zero_of_hasSum_pow`
`BosonicLean.Fock.FockSpace.hasSum_inner_expVec_smul`
`BosonicLean.Fock.FockSpace.coeff_eq_zero_of_inner_expVec_eq`
`BosonicLean.Fock.FockSpace.coeff_zero_of_inner_expVec_eq`
`BosonicLean/Fock/ExpVec.lean`

10.5 `BosonicLean.Fock.FockSpace.dense_span_expVec`
`BosonicLean.Fock.FockSpace.ext_expVec`
`BosonicLean.Fock.FockSpace.ext_inner_expVec`
`BosonicLean/Fock/ExpVec.lean`

10.7 `BosonicLean.Fock.FormalFock.gaussRaise_op`
`BosonicLean.Fock.FormalFock.gaussRaise_exp_tensorVec`
`BosonicLean.Fock.FormalFock.gaussRaise_exp_vacuum`
`BosonicLean.Fock.FormalFock.VanishesAbove.mono`
`BosonicLean.Fock.FormalFock.VanishesAbove.add`
`BosonicLean.Fock.FormalFock.VanishesAbove.smul`
`BosonicLean.Fock.FormalFock.vanishesAbove_sum`
`BosonicLean.Fock.FormalFock.vanishesAbove_single`
`BosonicLean.Fock.FormalFock.VanishesAbove.create`
`BosonicLean.Fock.FormalFock.VanishesAbove.gaussRaise_op`
`BosonicLean.Fock.FormalFock.VanishesAbove.secondQuant`
`BosonicLean.Fock.FormalFock.VanishesAbove.finite_support`
`BosonicLean.Fock.FockSpace.exists_toFormal_eq_of_vanishesAbove`
`BosonicLean.Fock.FockSpace.exists_vanishesAbove_of_finite`
`BosonicLean/Fock/ExpVec.lean`

10.8 `BosonicLean.Fock.FormalFock.lowerDisp`
`BosonicLean.Fock.FormalFock.lowerSqueeze`
`BosonicLean.Fock.FormalFock.raiseSqueeze`
`BosonicLean.Fock.FormalFock.raiseSqueeze_op`
`BosonicLean.Fock.FockSpace.summable_coeffExp_lowerDisp`
`BosonicLean.Fock.FockSpace.coeff_coeffExp_lowerDisp`
`BosonicLean.Fock.FockSpace.summable_coeffExp_lowerSqueeze`
`BosonicLean.Fock.FockSpace.coeff_coeffExp_lowerSqueeze`
`BosonicLean.Fock.FormalFock.sqNorm_gaussVec_eq_top`
`BosonicLean.Fock.FockSpace.continuous_coeff_coeffExp_lowerDisp`
`BosonicLean.Fock.FockSpace.continuous_coeff_coeffExp_lowerSqueeze`
`BosonicLean.Fock.FockSpace.not_memâĎŞp_raiseSqueeze`
`BosonicLean/Fock/ExpVec.lean`

10.9 `BosonicLean.exists_unique_linearIsometry_of_inner_eq`
`BosonicLean/Analysis/GramIsometry.lean`

10.10 `BosonicLean.Fock.FockSpace.exists_linearIsometry_of_expVec`
`BosonicLean.Fock.FockSpace.eq_of_eqOn_expVec`
`BosonicLean/Fock/ExpVec.lean`

10.12 `BosonicLean.Fock.FockSpace.orderedDisplacementLin`
`BosonicLean.Fock.FockSpace.orderedDisplacementLin_apply`
`BosonicLean.Fock.FockSpace.continuous_coeff_orderedDisplacementLin`
`BosonicLean.Fock.FockSpace.displacement`
`BosonicLean.Fock.FockSpace.toFormal_displacement`
`BosonicLean.Fock.FormalFock.orderedDisplacement_expVecFormal`
`BosonicLean.Fock.FockSpace.displacement_expVec`
`BosonicLean.Fock.FockSpace.displacement_mul_neg`
`BosonicLean.Fock.FockSpace.star_displacement`
`BosonicLean.Fock.FockSpace.displacement_zero`
`BosonicLean.Fock.FockSpace.displacement_mul`
`BosonicLean.Fock.FockSpace.displacement_mul_of_phase_eq_zero`
`BosonicLean.Fock.FockSpace.displacement_vacuum`
`BosonicLean.Fock.FockSpace.displacement_eq_prod`
`BosonicLean.Fock.FockSpace.fiberFock`
`BosonicLean.Fock.FockSpace.toFormal_fiberFock`
`BosonicLean.Fock.FockSpace.displacement_single_apply`
`BosonicLean/Fock/Displacement.lean`

10.13 `BosonicLean.Fock.FockSpace.annihilate_displacement`
`BosonicLean.Fock.FockSpace.create_displacement`
`BosonicLean.Fock.FockSpace.displacement_mem_maxDomain_iff`
`BosonicLean/Fock/Displacement.lean`

10.14 `BosonicLean.Fock.FockSpace.hasDerivAt_gaussRaise_exp`
`BosonicLean/Fock/ExpVec.lean`

10.15 `BosonicLean.Fock.FockSpace.displacement_smul_mul`
`BosonicLean.Fock.FockSpace.continuous_displacement_apply`
`BosonicLean.Fock.FockSpace.hasDerivAt_displacement_smul`
`BosonicLean/Fock/Displacement.lean`

10.16 `BosonicLean.Fock.FockSpace.inner_expVec_apply_vacuum`
`BosonicLean.Fock.FockSpace.apply_vacuum_of_commute`
`BosonicLean.Fock.FockSpace.eq_smul_one_of_commute_displacement`
`BosonicLean/Fock/Displacement.lean`

10.17 `BosonicLean.Fock.FockSpace.exists_eq_smul_of_conj_eq`
`BosonicLean.Fock.FockSpace.eq_of_conj_eq_of_inner_vacuum`
`BosonicLean.Fock.FockSpace.eq_bot_or_eq_top_of_invariant`
`BosonicLean.Fock.FockSpace.dense_span_displacement_vacuum`
`BosonicLean/Fock/Displacement.lean`

10.18 `BosonicLean.centralBinomSeries_ofReal_of_abs_lt`
`BosonicLean/Analysis/Mehler.lean`

10.19 `BosonicLean.Fock.FormalFock.hasSum_conj_mul_coeff_gaussVec`
`BosonicLean/Fock/Squeeze.lean`

10.21 `BosonicLean.Fock.FockSpace.orderedSqueezeLin`
`BosonicLean.Fock.FockSpace.orderedSqueezeLin_apply`
`BosonicLean.Fock.FockSpace.continuous_coeff_orderedSqueezeLin`
`BosonicLean.Fock.FockSpace.squeeze`
`BosonicLean.Fock.FockSpace.toFormal_squeeze`
`BosonicLean.Fock.FormalFock.orderedSqueeze_expVecFormal`
`BosonicLean.Fock.FockSpace.squeeze_expVec`
`BosonicLean.Fock.FockSpace.squeeze_vacuum`

BosonicLean.Fock.FockSpace.inner_expVec_squeeze_expVec
BosonicLean.Fock.FockSpace.squeezeMap
BosonicLean.Fock.FockSpace.squeeze_mul_displacement
BosonicLean.Fock.FockSpace.star_squeeze_mul_displacement_mul
BosonicLean.Fock.FockSpace.squeezeMap_squeezeMap_neg
BosonicLean.Fock.FockSpace.squeeze_mul_neg
BosonicLean.Fock.FockSpace.star_squeeze
BosonicLean.Fock.FockSpace.squeeze_zero
BosonicLean.Fock.FockSpace.squeeze_eq_prod
BosonicLean.Fock.FockSpace.squeeze_add_of_disjoint
BosonicLean.Fock.FockSpace.squeeze_single_apply
BosonicLean.Fock.FockSpace.inner_squeeze_vacuum
BosonicLean/Fock/Squeeze.lean

10.22 BosonicLean.Fock.FockSpace.displacement_matrixElement
BosonicLean/Fock/Displacement.lean
BosonicLean.Fock.FockSpace.squeeze_matrixElement
BosonicLean.Fock.FockSpace.rotation_matrixElement
BosonicLean.Fock.FockSpace.rotation_zero
BosonicLean.Fock.FockSpace.rotation_two_pi
BosonicLean/Fock/Squeeze.lean

10.23 BosonicLean.Fock.FockSpace.maxDomain_create
BosonicLean.Fock.FockSpace.annihilate_squeeze
BosonicLean.Fock.FockSpace.create_squeeze
BosonicLean.Fock.FockSpace.squeeze_mem_maxDomain_iff
BosonicLean/Fock/Squeeze.lean

10.24 BosonicLean.Fock.FockSpace.squeeze_smul_mul
BosonicLean.Fock.FockSpace.continuous_squeeze_apply
BosonicLean.Fock.FockSpace.hasDerivAt_squeeze_smul
BosonicLean/Fock/Squeeze.lean

11.2 BosonicLean.Fock.complexOfPhase_phaseOfComplex
BosonicLean.Fock.phaseOfComplex_complexOfPhase
BosonicLean.Fock.complexOfPhase_add
BosonicLean.Fock.complexOfPhase_neg
BosonicLean.Fock.complexOfPhase_injective
BosonicLean.Fock.sum_norm_complexOfPhase_sq
BosonicLean.Fock.symplecticBilin_apply
BosonicLean.Fock.sum_complexOfPhase_mul_conj_sub
BosonicLean.Fock.eq_zero_of_symplecticBilin_eq_zero
BosonicLean.Fock.symplecticBilin_mulVec
BosonicLean.Fock.symplecticBilin_inv_mulVec
BosonicLean.Fock.complexOfPhase_mulVec_orthogonalSymplecticOfUnitary
BosonicLean.Fock.complexOfPhase_squeezeSymplectic_mulVec
BosonicLean.Fock.squeezeSymplectic_mem
BosonicLean.Fock.squeezeSymplectic_neg_mul
BosonicLean.Fock.inv_squeezeSymplectic
BosonicLean/Weyl/GaussianUnitary.lean

11.4 BosonicLean.Fock.FockSpace.weylOp_mul
BosonicLean.Fock.FockSpace.star_weylOp
BosonicLean.Fock.FockSpace.inner_vacuum_weylOp
BosonicLean.Fock.FockSpace.IsGaussianUnitary.mul
BosonicLean.Fock.FockSpace.IsGaussianUnitary.star
BosonicLean.Fock.FockSpace.isGaussianUnitary_of_eq_smul_one
BosonicLean.Fock.FockSpace.IsGaussianUnitary.unique
BosonicLean.Fock.FockSpace.IsGaussianUnitary.exists_eq_smul
BosonicLean.Fock.FockSpace.eq_of_smul_weylOp_eq
BosonicLean/Weyl/GaussianUnitary.lean

11.5 BosonicLean.Fock.FockSpace.isGaussianUnitary_displacement
BosonicLean.Fock.FockSpace.isGaussianUnitary_secondQuantRep
BosonicLean.Fock.FockSpace.isGaussianUnitary_squeeze
BosonicLean.Fock.squeezeSymplectic_ofReal
BosonicLean.Fock.FockSpace.isGaussianUnitary_conj_squeeze_single
BosonicLean.Fock.FockSpace.hasDerivAt_conj_squeeze_single
BosonicLean/Weyl/GaussianUnitary.lean
BosonicLean.Fock.FockSpace.star_secondQuantRep_mul_displacement_mul
BosonicLean.Fock.FockSpace.secondQuantRep_mul_displacement
BosonicLean/Fock/Squeeze.lean
BosonicLean.Fock.FockSpace.secondQuantRep_expVec
BosonicLean.Fock.FockSpace.secondQuantRep_vacuum
BosonicLean/Fock/ExpVec.lean

11.6 BosonicLean.Fock.FockSpace.IsGaussianUnitary.eq_of_blochMessiah
BosonicLean.Fock.FockSpace.IsGaussianUnitary.exists_decomposition
BosonicLean.Fock.FockSpace.isGaussianUnitary_decomposition
BosonicLean.Fock.FockSpace.secondQuantRep_mul_displacement_mul_squeeze_mul_secondQuantRep
BosonicLean/Weyl/GaussianUnitary.lean
BosonicLean.Fock.FockSpace.squeeze_eq_rotation_mul_squeeze_norm_mul
BosonicLean.Fock.FockSpace.squeeze_eq_conj
BosonicLean/Fock/Squeeze.lean

11.7 BosonicLean.Fock.FormalFock.tensorVec_smul
BosonicLean.Fock.FockSpace.displacement_squeeze_vacuum_unit
BosonicLean.Fock.FockSpace.displacement_squeeze_vacuum
BosonicLean.Fock.FockSpace.apply_vacuum_of_eq_decomposition
BosonicLean.Fock.FockSpace.IsGaussianUnitary.apply_vacuum
BosonicLean.Fock.FockSpace.IsGaussianUnitary.exists_eq_oneMode
BosonicLean.Fock.FockSpace.rotation_mul
BosonicLean.Fock.FockSpace.phaseDiag_mul
BosonicLean.Fock.FockSpace.exists_phaseDiag_eq
BosonicLean.Fock.FockSpace.IsGaussianUnitary.apply_mem_maxDomain_iff
BosonicLean.Fock.FockSpace.IsGaussianUnitary.quadrature_apply
BosonicLean.Fock.FockSpace.exists_toFormal_eq_affine
BosonicLean/Weyl/GaussianUnitary.lean

12.2 BosonicLean.Fock.FockSpace.charFun_zero
BosonicLean.Fock.FockSpace.charFun_neg
BosonicLean.Fock.FockSpace.norm_charFun_le
BosonicLean.Fock.FockSpace.charFun_smul
BosonicLean.Fock.FockSpace.IsGaussianUnitary.charFun_apply
BosonicLean/Weyl/PureGaussian.lean

12.3 BosonicLean.Fock.FockSpace.inner_weylOp_apply
BosonicLean.Fock.FockSpace.dense_span_weylOp_apply
BosonicLean.Fock.FockSpace.exists_unique_linearIsometry_of_charFun_comp
BosonicLean/Weyl/PureGaussian.lean

12.4 BosonicLean.Fock.FockSpace.exists_eq_smul_of_charFun_eq
BosonicLean/Weyl/PureGaussian.lean

12.6 BosonicLean.Weyl.twistedKernel_smul_of_eq_exp
BosonicLean.Weyl.sum_exp_sub_one_nonneg_of_sum_eq_zero_of_posSemidef_twistedKernel
BosonicLean.Weyl.sum_exp_sub_one_nonneg_of_posSemidef_twistedKernel
BosonicLean.Weyl.re_sum_nonneg_of_posSemidef_twistedKernel
BosonicLean.Weyl.isStateConditionC_of_posSemidef_twistedKernel
BosonicLean/Weyl/State.lean

12.7 BosonicLean.Fock.FockSpace.HasGaussianCharFun.norm_eq_one
BosonicLean.Fock.FockSpace.HasGaussianCharFun.unique
BosonicLean.Fock.FockSpace.HasGaussianCharFun.isAdmissible

BosonicLean/Weyl/PureGaussian.lean

12.8 BosonicLean.Fock.FockSpace.exists_symplectic_le_of_isAdmissible
BosonicLean/Weyl/PureGaussian.lean

12.9 BosonicLean.Fock.FockSpace.IsGaussianUnitary.hasGaussianCharFun_apply_vacuum
BosonicLean/Weyl/PureGaussian.lean

12.10 BosonicLean.Fock.FockSpace.IsGaussianUnitary.moments
BosonicLean/Weyl/PureGaussian.lean

12.12 BosonicLean.Fock.FockSpace.exists_isGaussianUnitary
BosonicLean.Fock.FockSpace.exists_eq_mul_transpose
BosonicLean/Weyl/PureGaussian.lean

12.13 BosonicLean.Fock.FockSpace.HasGaussianCharFun.mem_symplecticGroup
BosonicLean/Weyl/PureGaussian.lean
Gaussian.phaseSplitEquiv
Gaussian.phaseSplitEquiv_coe
Gaussian.symplecticForm_directSum
BosonicLean/PhaseSpace/DirectSum.lean

12.14 BosonicLean.Fock.FockSpace.isPureGaussian_iff_exists_hasGaussianCharFun
BosonicLean.Fock.FockSpace.HasGaussianCharFun.exists_isGaussianUnitary
BosonicLean/Weyl/PureGaussian.lean

12.15 BosonicLean.Fock.FockSpace.HasGaussianCharFun.exists_eq_smul
BosonicLean/Weyl/PureGaussian.lean

12.16 BosonicLean.Fock.FockSpace.exists_isGaussianUnitary_hasGaussianCharFun
BosonicLean.Fock.FockSpace.exists_hasGaussianCharFun_iff
BosonicLean/Weyl/PureGaussian.lean

12.17 BosonicLean.Fock.exists_tanhPhase_eq
BosonicLean.Fock.cosh_norm_eq_of_tanhPhase
BosonicLean.Fock.FockSpace.tensorVec_gaussVec_eq
BosonicLean.Fock.squeezeSymplectic_mul_self
BosonicLean/Weyl/PureGaussian.lean

12.19 BosonicLean.Fock.FockSpace.inner_vacuum_left
BosonicLean.Fock.FockSpace.IsPureGaussian.inner_vacuum_ne_zero
BosonicLean/Weyl/PureGaussian.lean

12.20 BosonicLean.Fock.FockSpace.ne_smul_of_inner_vacuum_eq_zero
BosonicLean/Weyl/PureGaussian.lean

12.21 BosonicLean.Fock.FockSpace.numberState_ne_smul
BosonicLean/Weyl/PureGaussian.lean

13.2 BosonicLean.IsGenerator.hasDerivAt_apply
BosonicLean.IsGenerator.isSelfAdjoint
BosonicLean.IsGenerator.unique
BosonicLean.IsGenerator.of_core
BosonicLean.IsGenerator.unitaryConj
BosonicLean.IsGenerator.addConst
BosonicLean.isGenerator_exp
BosonicLean/Analysis/UnitaryGroup.lean

13.3 BosonicLean.isSelfAdjoint_closure_and_isGenerator_of_analytic
BosonicLean/Analysis/UnitaryGroup.lean

13.5 BosonicLean.Fock.FockSpace.exists_toFormal_eq_quadraticHamiltonian
BosonicLean.Fock.FockSpace.summable_quadraticHamiltonianOnFinite
BosonicLean.Fock.FockSpace.quadrature_mul_sub
BosonicLean.Fock.FockSpace.quadrature_mul_quadraticHamiltonian_sub
BosonicLean.Fock.FockSpace.field_mul_quadraticHamiltonian_sub
BosonicLean/Weyl/Quadratic.lean

BosonicLean/Weyl/QuadraticHamiltonian.lean

13.6 BosonicLean.Fock.FockSpace.quadraticHamiltonian_closure_eq
BosonicLean.Fock.FockSpace.existsUnique_isGenerator_quadraticHamiltonian
BosonicLean/Weyl/Quadratic.lean

13.7 BosonicLean.Fock.FockSpace.isGaussianUnitary_of_isGenerator_quadraticHamiltonian
BosonicLean.Fock.quadraticSymplectic_mem
BosonicLean/Weyl/QuadraticHamiltonian.lean

13.8 BosonicLean.Fock.FockSpace.isGaussianUnitary_ladderHamiltonian
BosonicLean.Fock.FockSpace.isGenerator_displacement
BosonicLean.Fock.FockSpace.isGenerator_squeeze
BosonicLean/Weyl/QuadraticHamiltonian.lean